F5 Certified! Administrator (F5-CA) Exam Blueprint Review

Presented by:

Sam Park (Sr. SE)
Anthony Graber (Principal SE)
Tyler Nelson (Sr. SE)
Ammar Jaghlit (Sr. SE)
Stephen Hellams (Sr. SE)
John Vernetti (Sr. SE)
Hachul Jun (Sr. SE)

April 8-9, 2026

The goal:

If you are almost ready, this is an opportunity for a final review and to ask questions. You should already be familiar with BIG-IP and TMOS Administration. We will be covering blueprint objectives – not the test. We do not have knowledge of the test questions.

 ©2026 F5

Certifications

A few important things to note:
•All certification exams will be delivered on YOUR laptop during the event. (See step 3 below)
•Anyone taking an exam during the event must be registered in the F5 Certified program.

1. Register for the F5 Certified program OR login to your certification account in the Education Services Portal:
Go to the Education Services Portal (https://education.f5.com) and click on the “Sign Up” link or login.

2. Schedule your exam at Symposium:
Certification exams are free during the event when you use this exam scheduling link. Important: You ***must*** be registered in the F5 Certified program before scheduling any exams. (See step 1.)

3. Check your laptop:
All certification exams are delivered on YOUR laptop, so make sure it’s compatible with the secure exam browser by taking the pre-flight test. Please take this test and verify access ***before*** arriving at the event.

If you need help logging into the Education Services Portal or have any questions about exams or scheduling, the Certification team is here for you! Please contact us directly at support@mail.education.f5.com

 ©2026 F5

F5 Certified Administrator (F5-CA, BIG-IP)

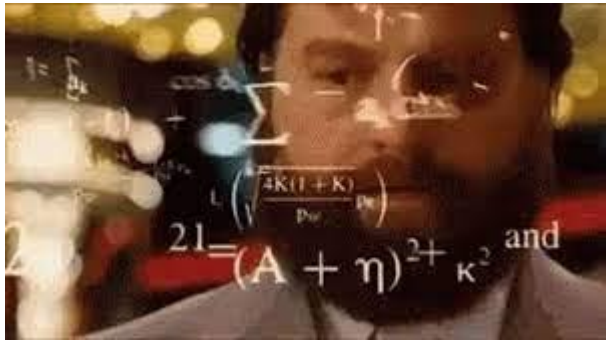
The F5 Certified Administrator certification is earned by passing these five exams, in any order:

- F5CAB1: BIG-IP Administration Install, Initial Configuration, and Upgrade
- F5CAB2: BIG-IP Administration Data Plane Concepts
- F5CAB3: BIG-IP Administration Data Plane Configuration
- F5CAB4: BIG-IP Administration Control Plane Administration
- F5CAB5: BIG-IP Administration Support and Troubleshooting

 ©2026 F5

Exam Structure

F5 Certified! Administrator Exam



- The questions are all multiple choice.
  - There are no true/false questions.
  - There are no “all of the above/none of the above” questions.
- Questions are not adaptive – do not increase/decrease difficulty based on how you are doing
- 30 questions
- Test duration is 30 minutes
- Passing score is 245 (70%) out of a range between 100 and 350
- Non-native English-speaking students have an additional 30 minutes!
- No command line engines (although you will have to know a few TMSH commands)

 ©2026 F5

Exam Structure - continued

F5 Certified! Administrator Exam

Advice:

- Flag long, complicated questions
- View whole exhibit before you close them (attachments)
- Manage Your Time!
- You can flag, review and re-answer questions (within the 30-minute test limit!)

 ©2026 F5

F5CAB2 : BIG-IP Administration Data Plane Concepts

Explain the relationship between interfaces, trunks, VLANs, self-IPs, routes and their status/statistics

- Explain the dependencies of interfaces/trunks, VLANs, self-IPs
- Compare Interface status (Up/Down)
- Illustrate the use of a trunk in a BIG-IP solution
- Demonstrate ability to assign VLAN to interface and/or trunk
- Distinguish between tagged vs untagged VLAN
- Identify, based on traffic, which VLAN/route/egress IP would be used

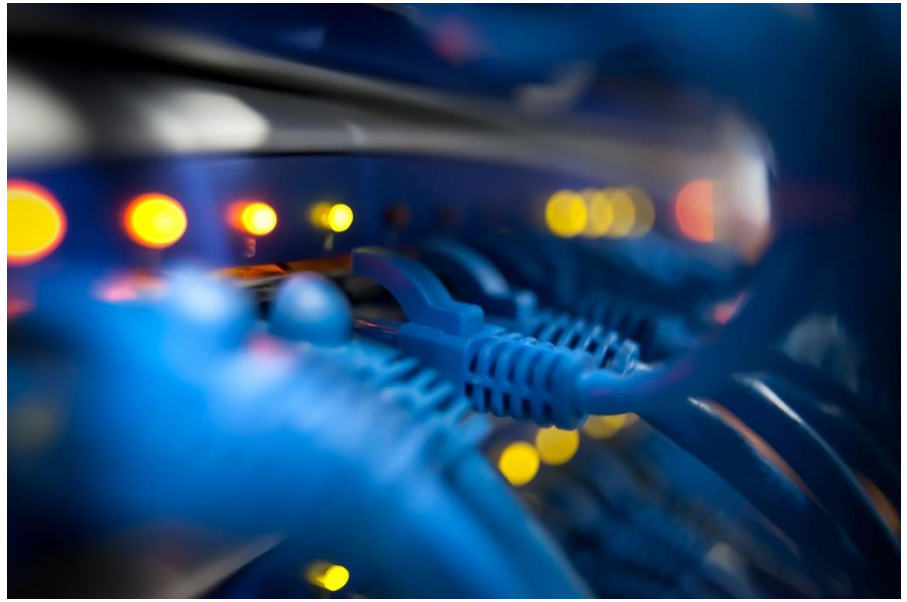
 ©2026 F5

Configuring the network

Overview of BIG-IP TMOS Routing

1. Configure the out-of-band management interface (eth0/mgmt) on the control plane
   - LCD panel (hardware)
   - config command
2. Set up Interfaces and Trunks (L1)
3. Assign interfaces and trunks to VLANs (L2)
4. Assign Self IPs to VLANs (L3)
5. Set up Default Gateway

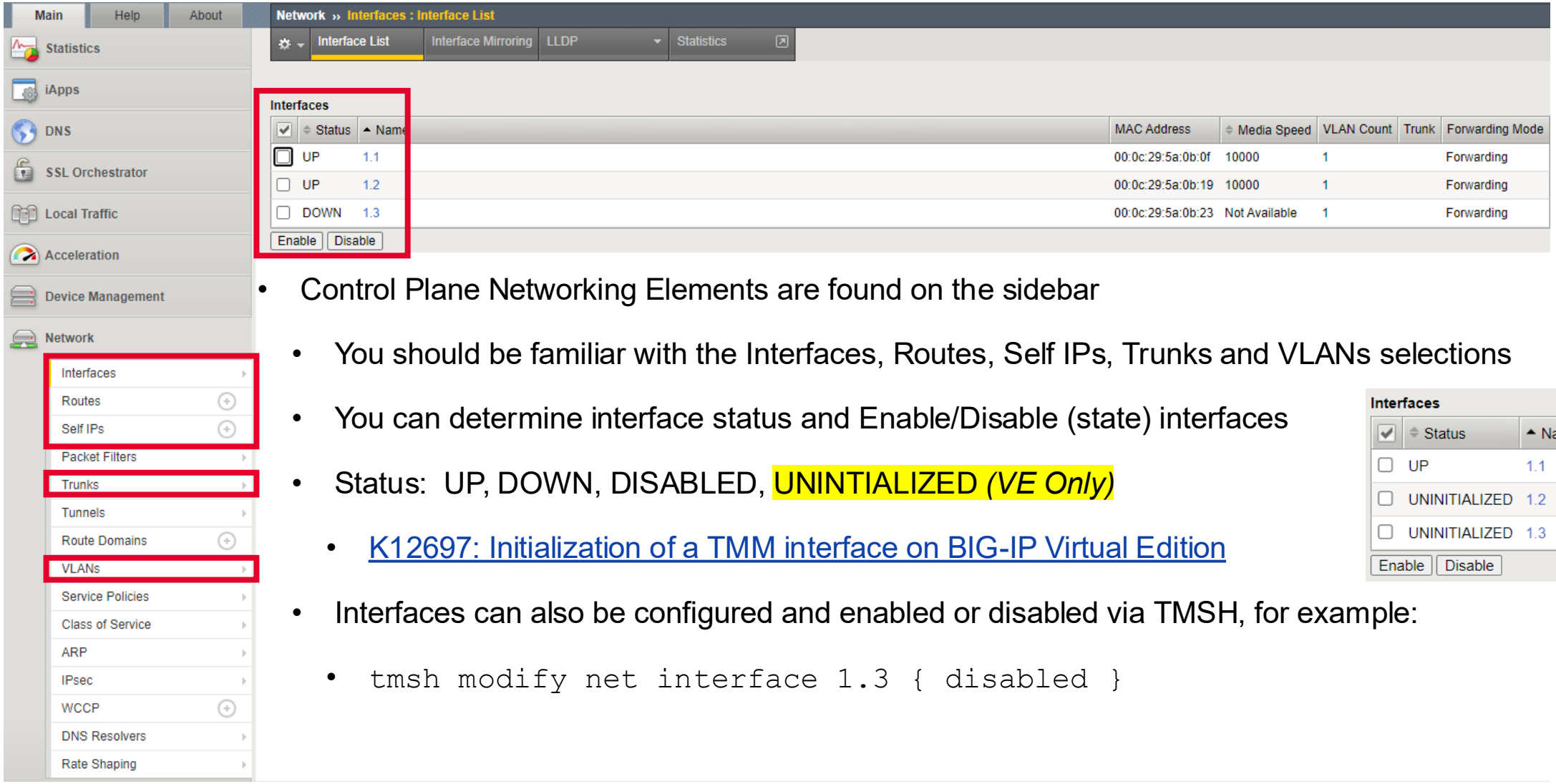
```
[root@bigip02:Standby:Changes Pending] config # ip route get 1.1.1.1
1.1.1.1 via 10.1.10.1 dev client_vlan  src 10.1.10.246
```



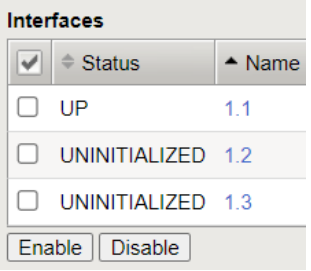
 ©2026 F5

Interfaces

Manual Chapter : Interfaces



- Control Plane Networking Elements are found on the sidebar
  - You should be familiar with the Interfaces, Routes, Self IPs, Trunks and VLANs selections
  - You can determine interface status and Enable/Disable (state) interfaces
  - Status: UP, DOWN, DISABLED, UNINTIALIZED *(VE Only)*
    - K12697: Initialization of a TMM interface on BIG-IP Virtual Edition
  - Interfaces can also be configured and enabled or disabled via TMSH, for example:
    - `tmsh modify net interface 1.3 { disabled }`



 ©2026 F5

Traffic Management Shell (TMSH)

https://clouddocs.f5.com/cli/tmsh-reference/latest/ with link to Full TMSH Reference Guide PDF

- When does the configuration get written to disk?
  - In the GUI the changes are made to the running configuration and written to disk immediately.
  - In TMSH configuration changes are made to the running configuration, but NOT written to disk
    - A TMSH command is required to save the configuration to disk, or a change made through the GUI will force a write to disk

```
(tmos)# save sys config
Saving running configuration...
  /config/bigip.conf
  /config/bigip_base.conf
  /config/bigip_user.conf
Saving Ethernet mapping...done
```

- Show vs List
  - **show** commands allow you to view runtime information, statistics and status
  - **list** commands allow you to view the running configuration and settings

 ©2026 F5

tmsh vlan examples

```
(tmos)# list net vlan

net vlan ha_vlan {
    fwd-mode l3
    interfaces {
        1.3 { }
    }
    tag 4092
}
net vlan new_vlan {
    fwd-mode l3
    interfaces {
        1.3 {
            tagged
        }
    }
    tag 30
}
```

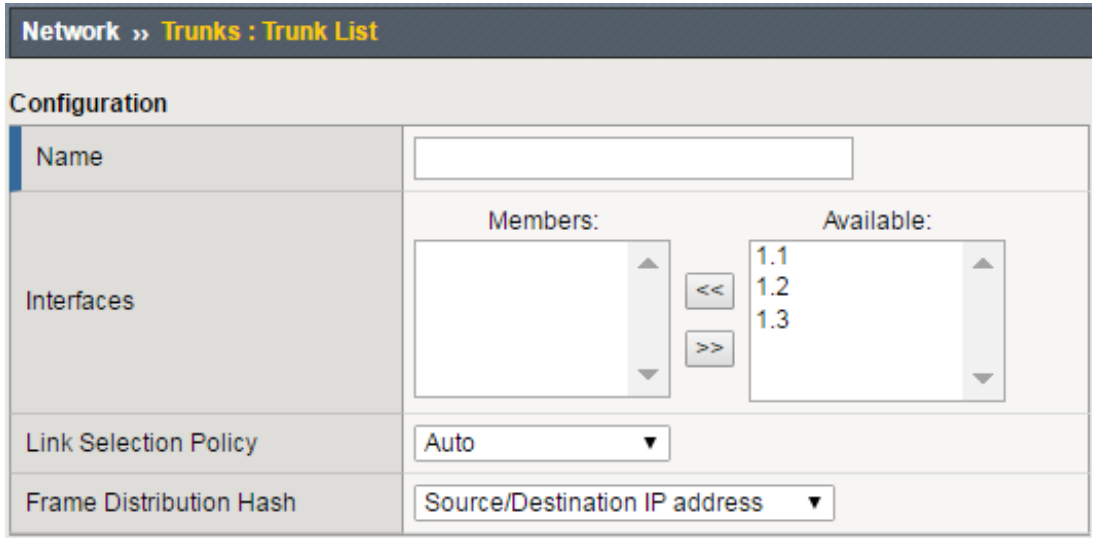
```
(tmos)# show net vlan new_vlan

-------------------------------------
Net::Vlan: new_vlan
-------------------------------------
Interface Name       new_vlan
Mac Address (True)   00:0c:29:5a:0b:23
MTU                  1500
Tag                  30

  -----------------------
  | Net::Vlan-Member: 1.3
  -----------------------
  | Tagged    yes

    ------------------------------------------------------------------
    | Net::Interface
    | Name   Status    Bits   Bits   Pkts   Pkts   Drops   Errs      Media
    |                    In    Out     In    Out
    ------------------------------------------------------------------
    | 1.3       up   867.1K  1.1M    652   3.3K       0      0  10000T-FD
```

 ©2026 F5

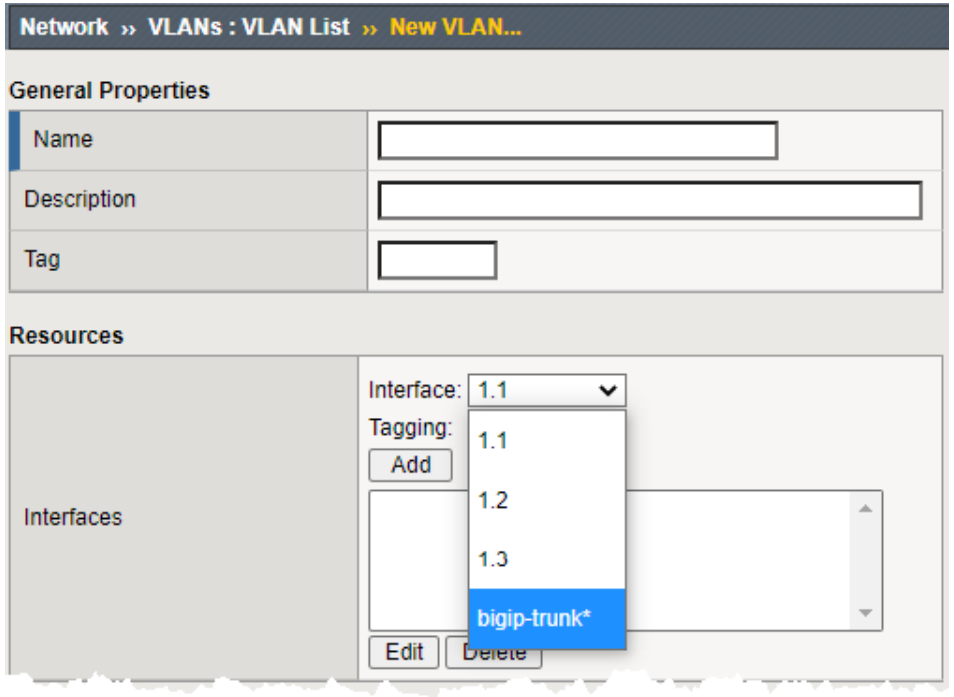
BIG-IP Trunks

- BIG-IP trunks can be set up as LACP (default) or Etherchannel (Cisco link aggregation)
  - IMPORTANT: A BIG-IP trunk (interface) is not equivalent to a Cisco trunk (VLAN tagging)
    - Cisco terminology uses Port Channel for link aggregation and trunk for 802.1q VLAN tagging

A trunk is created from the Network >> Trunks



Once created the trunk shows up as an interface

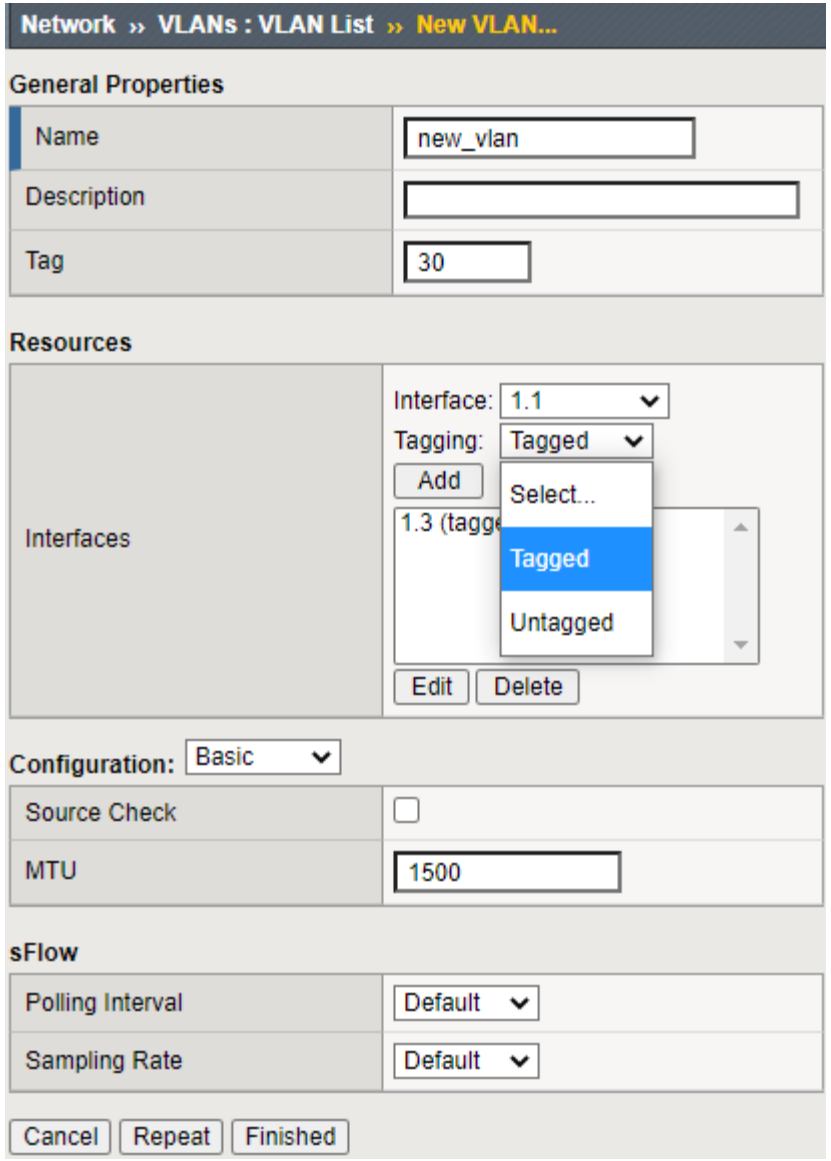


 ©2026 F5

Tagged vs Untagged VLANs

Manual Chapter : VLANs VLAN Groups and VXLAN

- If you wish to have more than one VLAN over the same physical interface or trunk
- Place interfaces and trunks into the Untagged or Tagged boxes
- **Untagged** interfaces do not require a Tag be entered
  - The BIG-IP will assign a Tag to logically separate internal traffic
- **Tagged** interfaces run 802.1q VLAN tagging
  - You need to manually enter the tag



 ©2026 F5

Types of Self IPs

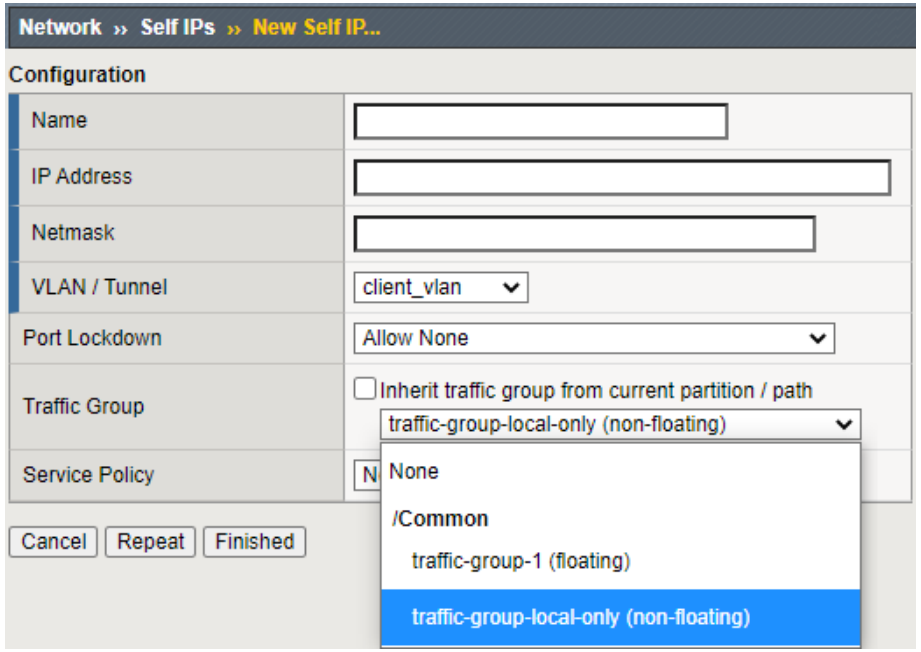
Manual Chapter : Self IP Addresses

- Self IPs have **Port Lockdown (allow none)** configured by default and only respond to ICMP traffic.
- You should understand the difference between floating and non-floating self IPs.
- There are two types of self IP addresses that you can create:
  - A **static (non-floating) self IP** address is an IP address that the BIG-IP system does not share with another BIG-IP system.
    - Any self IP address that you assign to the default traffic group **traffic-group-local-only** is a static self IP address.
    - If the BIG-IP goes down, the static self IPs go down with it.
    - Used for monitoring based on route table
  - A **floating self IP** address is an IP address that two (or more) BIG-IP systems share.
    - Any self IP address that you assign to the default traffic group **traffic-group-1** is a floating self IP address.
    - Or any traffic group that is **NOT** traffic-group-local-only (all other traffic groups are floating)
    - A floating self IP only responds on the Active BIG-IP, if the Active BIG-IP goes down the floating self IP is activated on another BIG-IP in the Device Service Cluster (DSC)

 ©2026 F5

Self IPs

[Manual Chapter : Self IP Addresses]



Name	Application	IP Address	Netmask	VLAN / Tunnel	Traffic Group	Partition / Path
client_ip		10.1.10.245	255.255.255.0	client_vlan	traffic-group-local-only	Common
floating-ip		10.1.20.240	255.255.255.0	server_vlan	traffic-group-1	Common
ha_ip		192.168.20.245	255.255.255.0	ha_vlan	traffic-group-local-only	Common
server_ip		10.1.20.245	255.255.255.0	server_vlan	traffic-group-local-only	Common

```
(tmos)# list net self
net self floating-ip {
    address 10.1.20.240/24
    floating enabled
    traffic-group traffic-group-1
    unit 1
    vlan server_vlan
}
net self ha_ip {
    address 192.168.20.245/24
    allow-service {
        default
    }
    traffic-group traffic-group-local-only
    vlan ha_vlan
}
net self server_ip {
    address 10.1.20.245/24
    traffic-group traffic-group-local-only
    vlan server_vlan
}
net self client_ip {
    address 10.1.10.245/24
    traffic-group traffic-group-local-only
    vlan client_vlan
}
```

 ©2026 F5

F5CAB2 : BIG-IP Administration Data Plane Concepts

Identify the different virtual server types

- Standard, Forwarding, Stateless, Reject
- Performance (Layer 4) and Performance (HTTP)

 ©2026 F5

Virtual Server Types

Virtual server type	Description of virtual server type
Standard	**A Standard virtual server directs client traffic to a load balancing pool and is the most basic type of virtual server. It is a general purpose virtual server that does everything not expressly provided by the other types of virtual servers.**
Forwarding (Layer 2)	A Forwarding (Layer 2) virtual server typically shares the same IP address as a node in an associated Virtual Local Area Network (VLAN). You use a Forwarding (Layer 2) virtual server in conjunction with a VLAN group.
Forwarding (IP)	*A Forwarding (IP) virtual server forwards packets directly to the destination IP address specified in the client request. A Forwarding (IP) virtual server has no pool members to load balance.*
Performance (Layer 4)	*A Performance (Layer 4) virtual server has a FastL4 profile associated with it. A Performance (Layer 4) virtual server increases the speed at which the virtual server processes packets.*
Performance (HTTP)	*A Performance (HTTP) virtual server has a FastHTTP profile associated with it. The Performance (HTTP) virtual server and related profile increase the speed at which the virtual server processes HTTP requests.*
Stateless	*A Stateless virtual server improves the performance of User Datagram Protocol (UDP) traffic in specific scenarios.*
Reject	*A Reject virtual server rejects any traffic destined for the virtual server IP address.*
DHCP Relay	A Dynamic Host Configuration Protocol (DHCP) relay virtual server relays DHCP client requests for an IP address to one or more DHCP servers, and provides DHCP server responses with an available IP address for the client.
Internal	An Internal virtual server enables usage of Internet Content Adaptation Protocol (ICAP) servers to modify HTTP requests and responses by creating and applying an ICAP profile and adding Request Adapt or Response Adapt profiles to the virtual server.
Message Routing	A Message Routing virtual server uses a Session Initiation Protocol (SIP) application protocol and functions in accordance with a SIP session profile and SIP router profile.

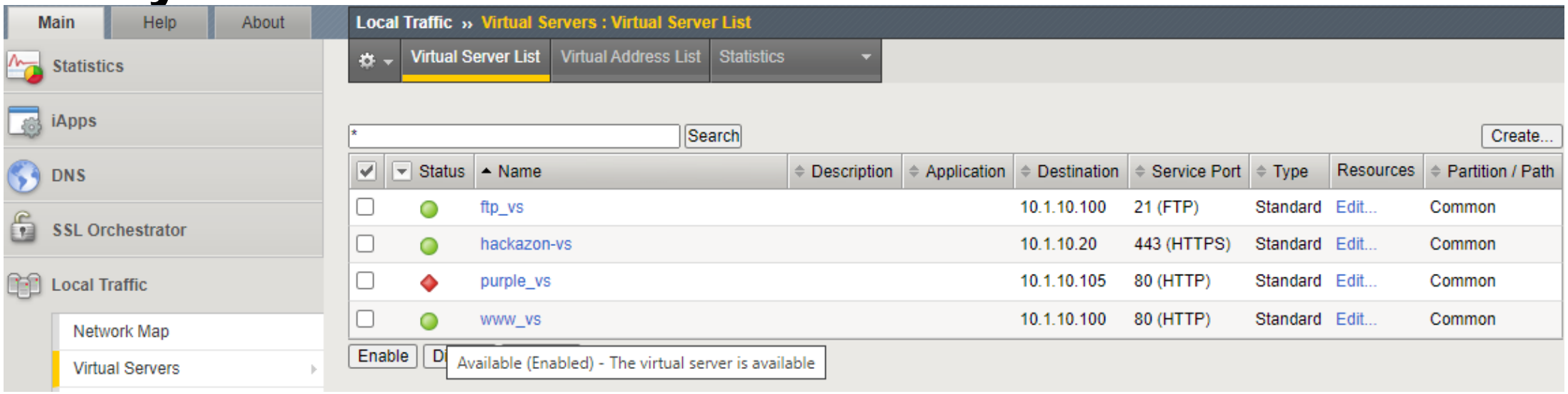
F5CAB3 : BIG-IP Administration Data Plane Configuration

Identify the reason a virtual server is not working as expected

- Identify the current configured state of the virtual server
- Identify the current availability status of the virtual server
- Identify misconfigured IP address and/or Port
- Identify conflicting/misconfigured profiles

 ©2026 F5

Identify the state and status of a virtual server




```
# show ltm virtual www_vs
-----------------------------------------------------------
Ltm::Virtual Server: www_vs
-----------------------------------------------------------
Status
  Availability     : available
  State            : enabled
  Reason           : The virtual server is available
  CMP              : enabled
  CMP Mode         : all-cpus
  Destination      : 10.1.10.100:80

Traffic                              ClientSide  Ephemeral  General
  Bits In                                577.1K          0        -
<cut>
```

 ©2026 F5

Virtual Server Statistics

[Manual Chapter: Virtual Servers]

Statistics » Module Statistics : Local Traffic » Virtual Servers

Traffic Summary | DNS | Local Traffic | Network | Memory

Display Options

Statistics Type	Virtual Servers
Data Format	Normalized
Auto Refresh	Disabled Refresh

* Search

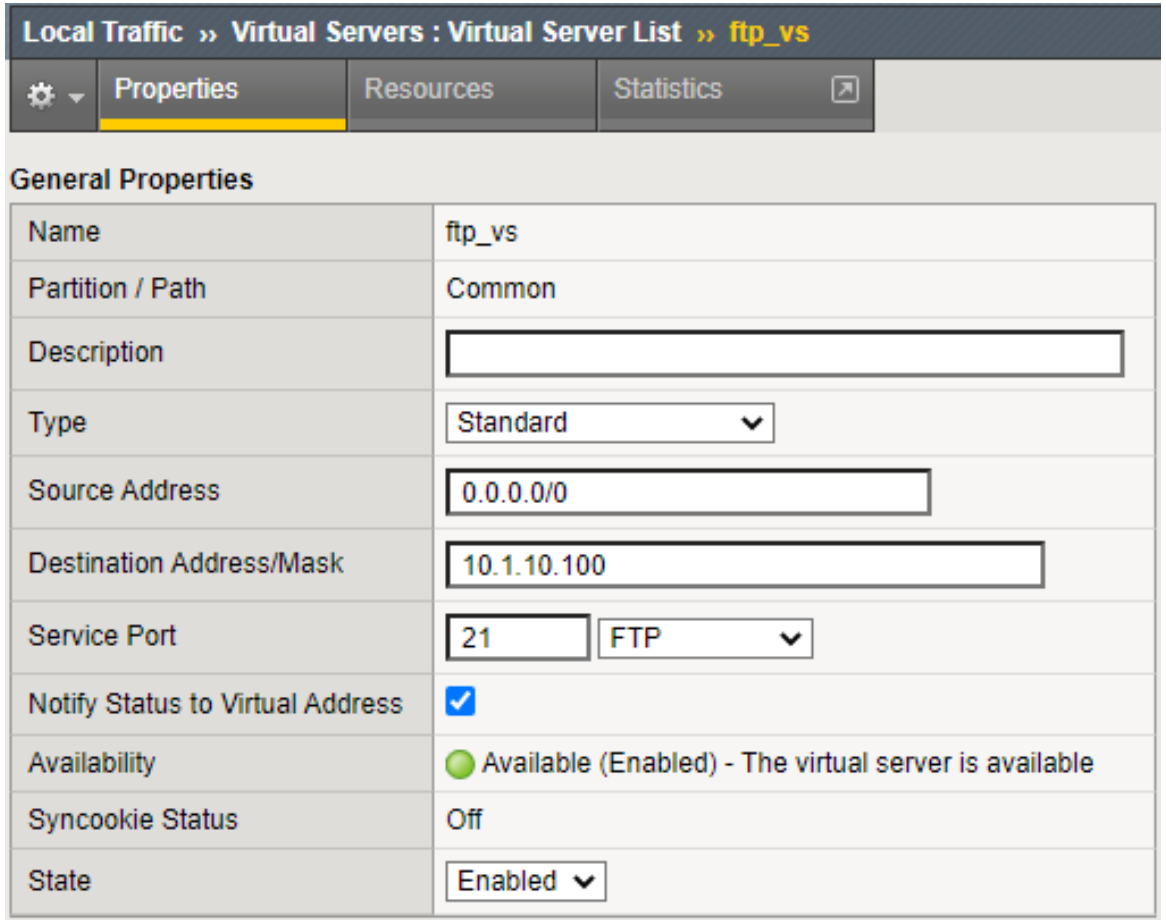
				Bits		Packets		Connections			Requests	CPU Utilization Avg.		
Status	Virtual Server	Partition / Path	Details	In	Out	In	Out	Current	Maximum	Total	Total	5 Sec.	1 Min.	5 Min.
	avr_virtual1	Common	View...	3.5M	34.7M	7.0K	7.1K	474	593	1.5K	1.4K	2%	0%	0%
	avr_virtual2	Common	View...	3.4M	34.6M	8.0K	6.8K	362	483	1.4K	1.3K	2%	0%	0%
	demo_iapp_redir_vs	Common/demo_iapp.app	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	demo_iapp_vs	Common/demo_iapp.app	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	secure_vs	Common	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	subnet_10_128_20_vs	Common	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	wildcard_vs	Common	View...	199.4K	0	368	0	0	56	368	0	0%	0%	0%
	www_vs	Common	View...	13.8M	153.9M	28.1K	32.1K	463	735	4.9K	4.9K	1%	0%	0%

Reset

 ©2026 F5

Identify misconfigured IP address and/or Port

Manual Chapter: Virtual Servers




```
(tmos)# list ltm virtual ftp_vs
ltm virtual ftp_vs {
    destination 10.1.10.100:ftp
    ip-protocol tcp
    mask 255.255.255.255
    pool ftp_pool
    profiles {
        ftp { }
        tcp { }
    }
    source 0.0.0.0/0
    source-address-translation {
        pool SNAT249_pool
        type snat
    }
    translate-address enabled
    translate-port enabled
    vs-index 2
}
```

 ©2026 F5

Profile Types

Manual : BIG-IP Local Traffic Management: Profiles Reference

K23843660: BIG-IP LTM-DNS operations guide | Chapter 5: BIG-IP LTM profiles

Profile Type	Description
Protocol profiles	
Fast L4	Defines the behavior of Layer 4 IP traffic.
Fast HTTP	Improves the speed at which a virtual server processes HTTP requests.
TCP	Defines the behavior of TCP traffic.
UDP	Defines the behavior of UDP traffic.
SSL profiles	
Client	Defines the behavior of client-side SSL traffic. See also Persistence Profiles.
Server	Defines the behavior of server-side SSL traffic. See also Persistence Profiles.

 ©2026 F5

Profile Types

Manual : BIG-IP Local Traffic Management: Profiles Reference

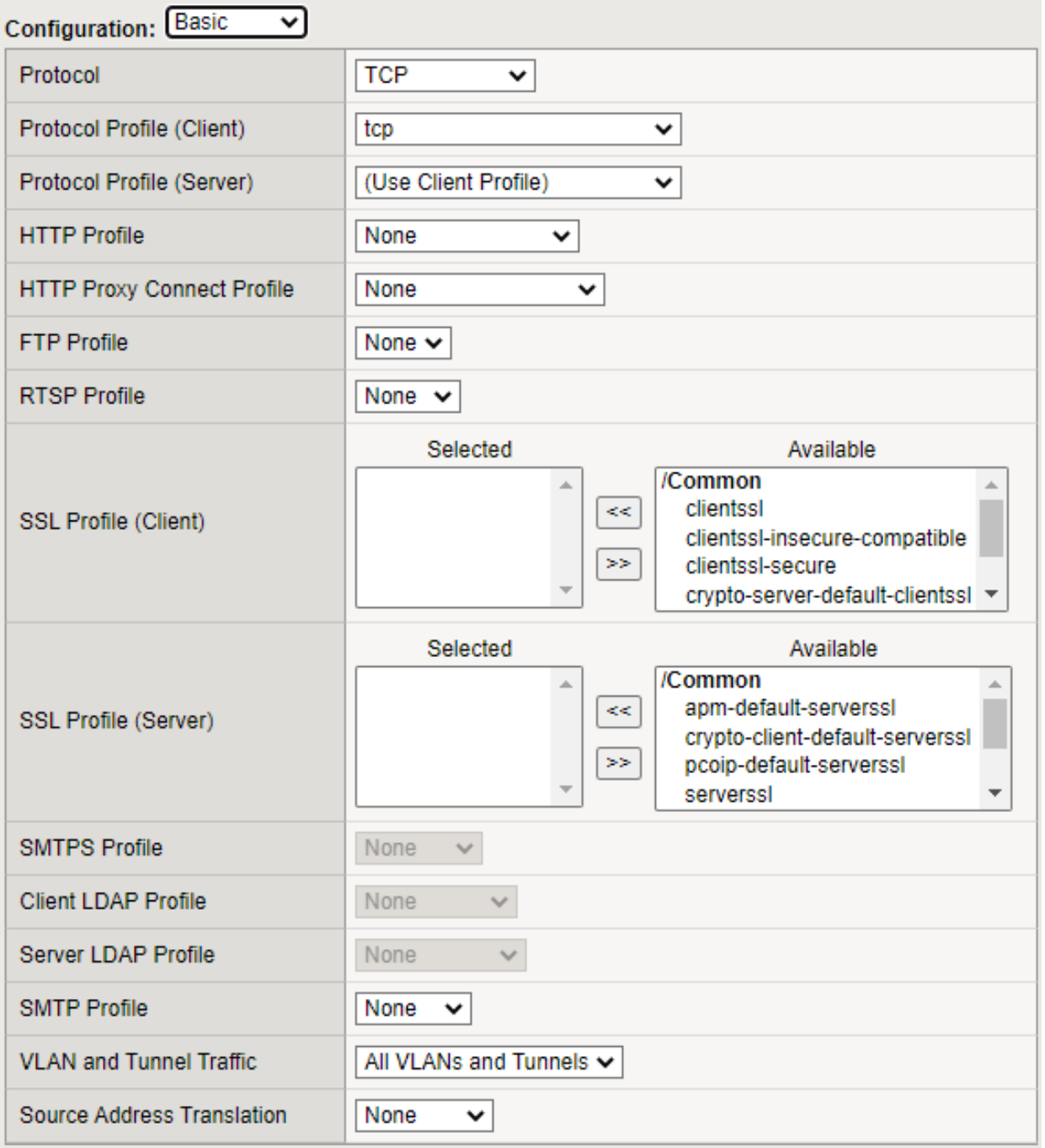
Profile Type	Description
Services profiles	
HTTP	Defines the behavior of HTTP traffic.
FTP	Defines the behavior of FTP traffic.
Persistence profiles	
Cookie	Implements session persistence using HTTP cookies.
Destination Address Affinity	Implements session persistence based on the destination IP address specified in the header of a client request. Also known as sticky persistence.
Hash	Implements session persistence in a way similar to universal persistence, except that the BIG-IP system uses a hash for finding a persistence entry.
Microsoft® Remote Desktop	Implements session persistence for Microsoft® Remote Desktop Protocol sessions.
SIP	Implements session persistence for connections using Session Initiation Protocol Call-ID.
Source Address Affinity	Implements session persistence based on the source IP address specified in the header of a client request. Also known as simple persistence.
SSL	Implements session persistence for non-terminated SSL sessions, using the session ID.
Universal	Implements session persistence using the BIG-IP system's Universal Inspection Engine (UIE).

 ©2026 F5

Misconfigured/Missing Profiles

Common mistakes/things to think about:

- The Protocol profile limits traffic to that protocol
  - i.e. Using the TCP profile, you can not ping through a virtual
- If looking into L4, L7, (ie HTTP), the appropriate protocol profile is needed
- SSL Profile requirements
  - HTTPS virtual, HTTPS pool members, where no HTTP profile is required, does NOT have to have SSL profiles, basically L4
  - SSL Offload, virtual HTTP, pool members HTTP will require a SSL Profile (Client)
  - HTTPS virtual, HTTPS pool members, where you need to look into the HTTP header (ie. Cookie persistence) and/or data require BOTH an SSL Profile (Client) and an SSL Profile (Server)



 ©2026 F5

Lab 1

 ©2026 F5

F5CAB2 : BIG-IP Administration Data Plane Concepts

Determine expected traffic behavior based on configuration

- Determine the egress source IP based on configuration
- Consider the packet and/or virtual server processing order (wildcard vips)
- Identify traffic diverted due to status of traffic objects (vs, pool, pool member)
- Identify when connection/rate limits are reached
- Identify traffic diverted due to persistence

 ©2026 F5

Determine the egress source IP based on configuration

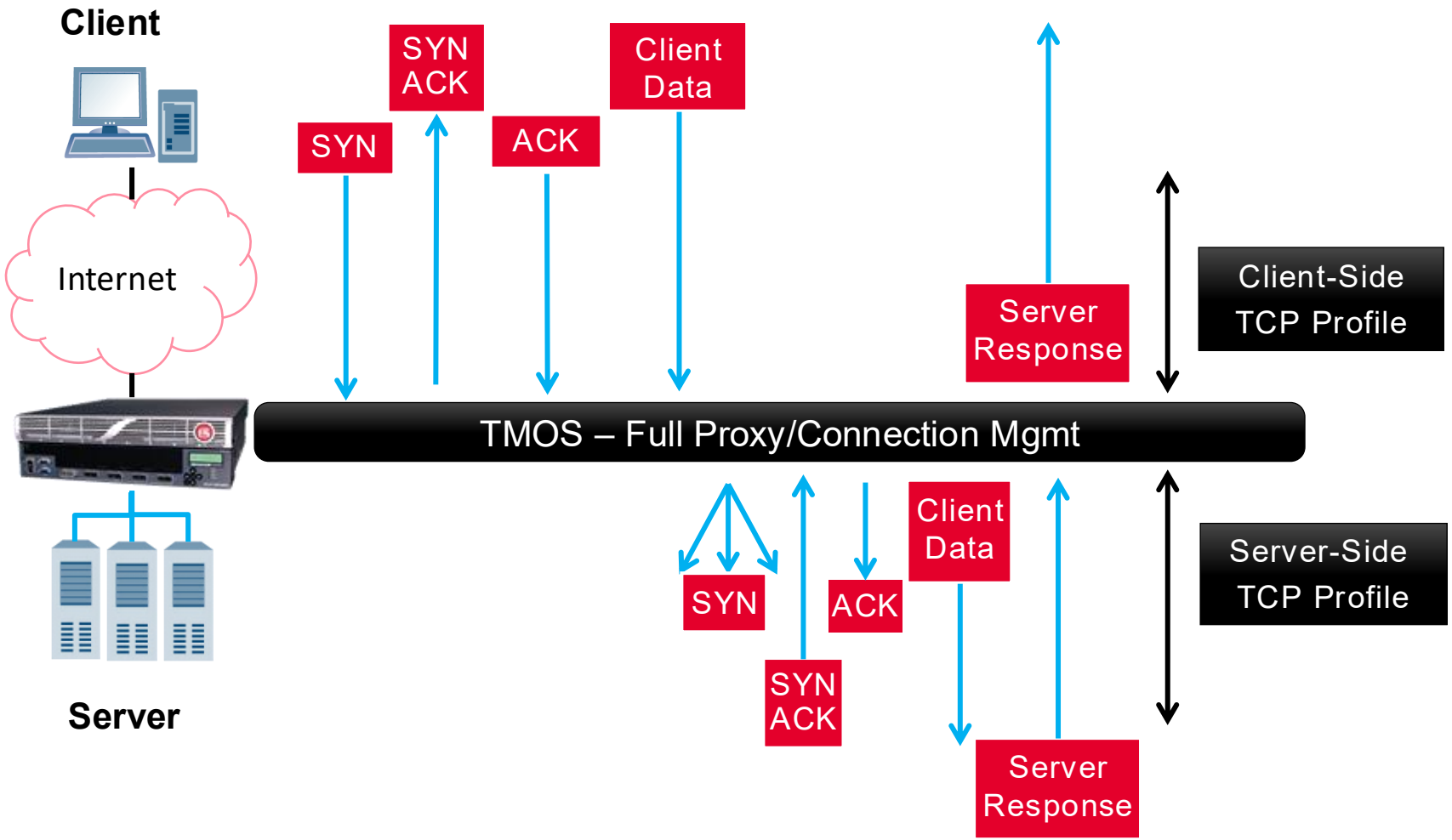
Traffic Flow through the BIG-IP

- TMOS is a full proxy architecture
- **Routed mode** (recommended)
  - Servers are on an internal network behind the BIG-IP
  - The BIG-IP is the default gateway for the servers
- **Secure Network Address Translation (SNAT) Mode**
  - The BIG-IP translates the original source IP, to an IP address owned by the BIG-IP
  - Allows a BIG-IP to be inserted into existing networks without changing the existing IP address structure
  - Can be used to create One-Armed/Single-Network mode

 ©2026 F5

TMOS – Full proxy Architecture

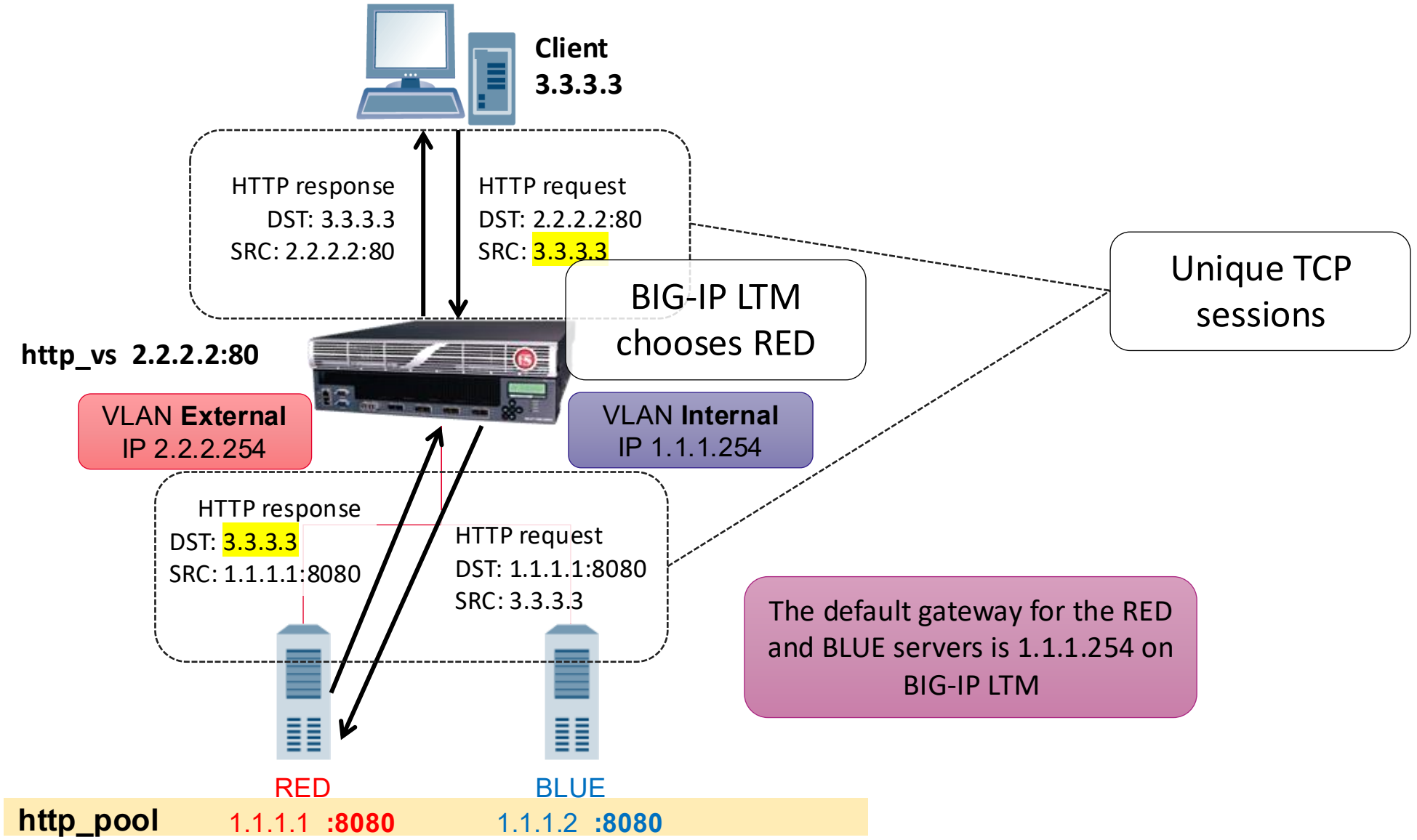
- Remember there are always two connections to a transaction.
- The BIG-IP connection table contains information about all the sessions currently established on the BIG-IP system.
  - Can be displayed via TMSH
  - Shows client-side/server side connection pairs



 ©2026 F5

Traffic flow through BIG-IP when BIG-IP is the default gateway

Routed mode



 ©2026 F5

NATs and SNATs

NATs and SNATs Comparison

- You can create **NAT**s on a BIG-IP
  - NAT is an address translation object to translate one IP address in a packet header to another IP address.
    - Consists of a one-to-one mapping of a public IP address to an internal private class IP address.
    - All ports are open
- Much more common and important are **SNAT**s, **understanding how SNATs work is key**.
- A *secure network address translation (**SNAT**)* is a BIG-IP Local Traffic Manager™ feature that translates the source IP address within a connection to a BIG-IP system IP address that you define. The destination node then uses that new source address as its destination address when responding to the request.
  - Can map multiple original addresses to a single translation address
  - Only the source can use the translation to establish connections
  - Only supports TCP and UDP by default
    - This makes SNATs more secure than NATs

 ©2026 F5

SNATs – How they are used

Manual Chapter : NATS and SNATs

- **When the default gateway of the server node is not the BIG-IP system**.
  - This is a very common scenario.
- **When clients and servers are on the same network.**
  - For example, web servers talking to applications or databases
- **SNATs for server-initiated (outbound) connections.**
  - Allow servers to access outside resources safely.

 ©2026 F5

SNAT Automap and Self IP Selection

K7336: The SNAT Automap and self IP address selection

- SNAT Automap uses the Self-IPs already assigned to the BIG-IP VLANs for translation.
  - SNATs are almost always assigned at the virtual server level
- SNAT Automap selects a translation address from the available self IP address in the following order of preference:

1. **Floating self IP addresses on the egress VLAN**
2. Floating self IP addresses on different VLANs
3. **Non-floating self IP addresses on the egress VLAN**
4. Non-floating self IP addresses on different VLANs

 ©2026 F5

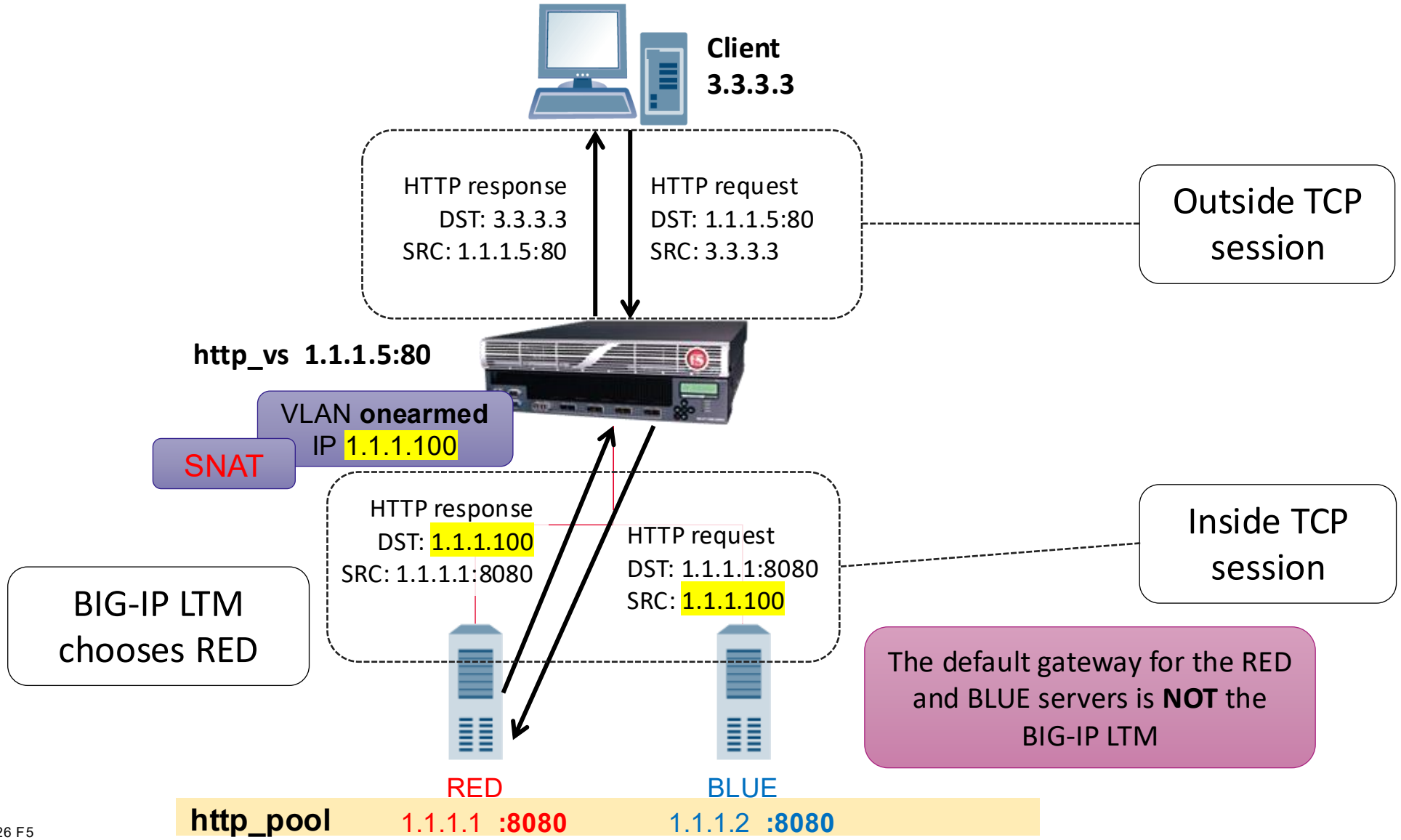
SNAT Pools

RECOMMENDED READING: K7820: Overview of SNAT features

- SNAT uses ports to separate client connections
  - More than roughly 65000 concurrent connections will exhaust the ports of a single SNAT'd IP
    - Note: BIG-IP Cluster Multiprocessing (CMP) can cause this limit to be exceeded, but always plan using this as the maximum
  - This is also known as port overload
  - Once the ports are exhausted connections will be dropped.
    - K8246: How the BIG-IP system handles SNAT port exhaustion
- SNAT Pools must be used if the concurrent connections will exceed this limit.
  - You will need enough IPs in the pool to handle the maximum number of concurrent connections.
- An additional benefit of SNAT pools is that they failover seamlessly if SNAT mirroring is selected.
  - SNAT mirroring mirrors the SNAT IP address and port utilized to the next active device in the cluster.

 ©2026 F5

Traffic flow through BIG-IP when SNATs are used



 ©2026 F5

Consider the packet and/or virtual server processing order (wildcard VIPs)

K9038: The order of precedence for local traffic object listeners

Packet Processing Priority

1. Existing connection in connection table
2. AFM/Packet filter rule
3. Virtual server
4. SNAT
5. NAT
6. Self-IP
7. Drop

 ©2026 F5

Virtual Server Order of Precedence

K14800: Order of precedence for virtual server matching

- Understand how a virtual server processes a request
  - Precedence is from most specific to least specific
- The BIG-IP system uses an algorithm that places virtual server precedence in the following order:
  - Destination address
    - Which virtual address (IP) is most specific?
  - Source address
    - Is the source address permitted to access the virtual address?
  - Service port
    - What is the most specific port match?

Order	Destination	Source	Service port
1	<host address>	<host address>	<port>
2	<host address>	<host address>	*
3	<host address>	<network address>	<port>
4	<host address>	<network address>	*
5	<host address>	*	<port>
6	<host address>	*	*
7	<network address>	<host address>	<port>
8	<network address>	<host address>	*
9	<network address>	<network address>	<port>
10	<network address>	<network address>	*
11	<network address>	*	<port>
12	<network address>	*	*
13	*	<host address>	<port>
14	*	<host address>	*
15	*	<network address>	<port>
16	*	<network address>	*
17	*	*	<port>
18	*	*	*

 ©2026 F5

Identify traffic diverted due to status of traffic objects (VS, pool, pool member)

BIG-IP Object State and Status

How traffic is processed is affected by the state and status of an object.

- States are:
  - Enabled
  - Disabled
- Status is based on monitor responses and object hierarchy
  - The virtual server status is determined by the status of the pool
  - The pool status is determined by the status of pool members
  - A pool member is determined by the status of the node
    - Node is an IP address

 ©2026 F5

Load Balancing Components (Brief review)

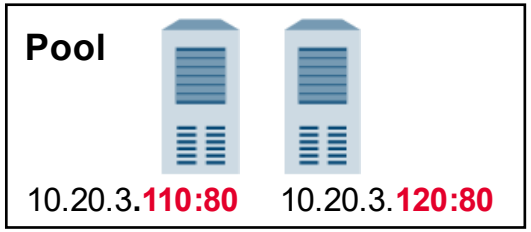
- **Node**
  - IP address of the server supporting applications

Node 10.20.3.110

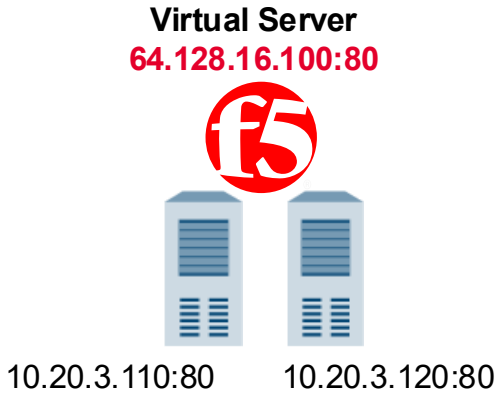
- **Pool Member**
  - A pool member is the IP Address:Port combination to access an application on the node (server)
  - Pool members are combined to form pools of applications
  - Since a single server may host multiple applications, a single node (server) may be a part of multiple pools

Pool Member
10.20.3.110**:80 (http)**

- **Pool**
  - A pool is a group of pool members supporting a particular application
  - Each pool has its own characteristics, such as monitor(s) and load balancing method



- **Virtual Server**
  - Is the IP Address:Port combination that represents a pool to the client side
  - Is a combination of a **virtual** IP address and **virtual** port
  - Access is limited to the defined port only
  - Multiple virtual servers can use the same servers or pools



 ©2026 F5

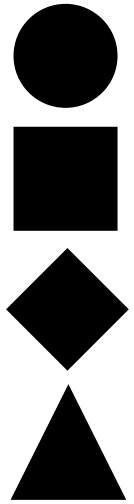
Monitor Status Reporting

Status	Status Definition	
	Node	• Most recent monitor successful
	Pool Member	• Most recent monitor successful
	Pool	• **<u>At least one</u> pool member is available**
	Virtual Server	• Associated pool is available
	Node	• No associated monitor (or timeout of first check not reached and not successful)
	Pool Member	• No associated monitor (or timeout of first check not reached and not successful)
	Pool	• All pool members are unknown/unmonitored (blue)
	Virtual Server	• Associated pool is unknown/unmonitored (blue)
	Node	• Most recent monitor failed (no successful checks within timeout period)
	Pool Member	• Most recent monitor failed (no successful checks within timeout period)
	Pool	• All members are offline and no members are available
	Virtual Server	• Associated pool is offline and no members available

 ©2026 F5

Other Statuses and State

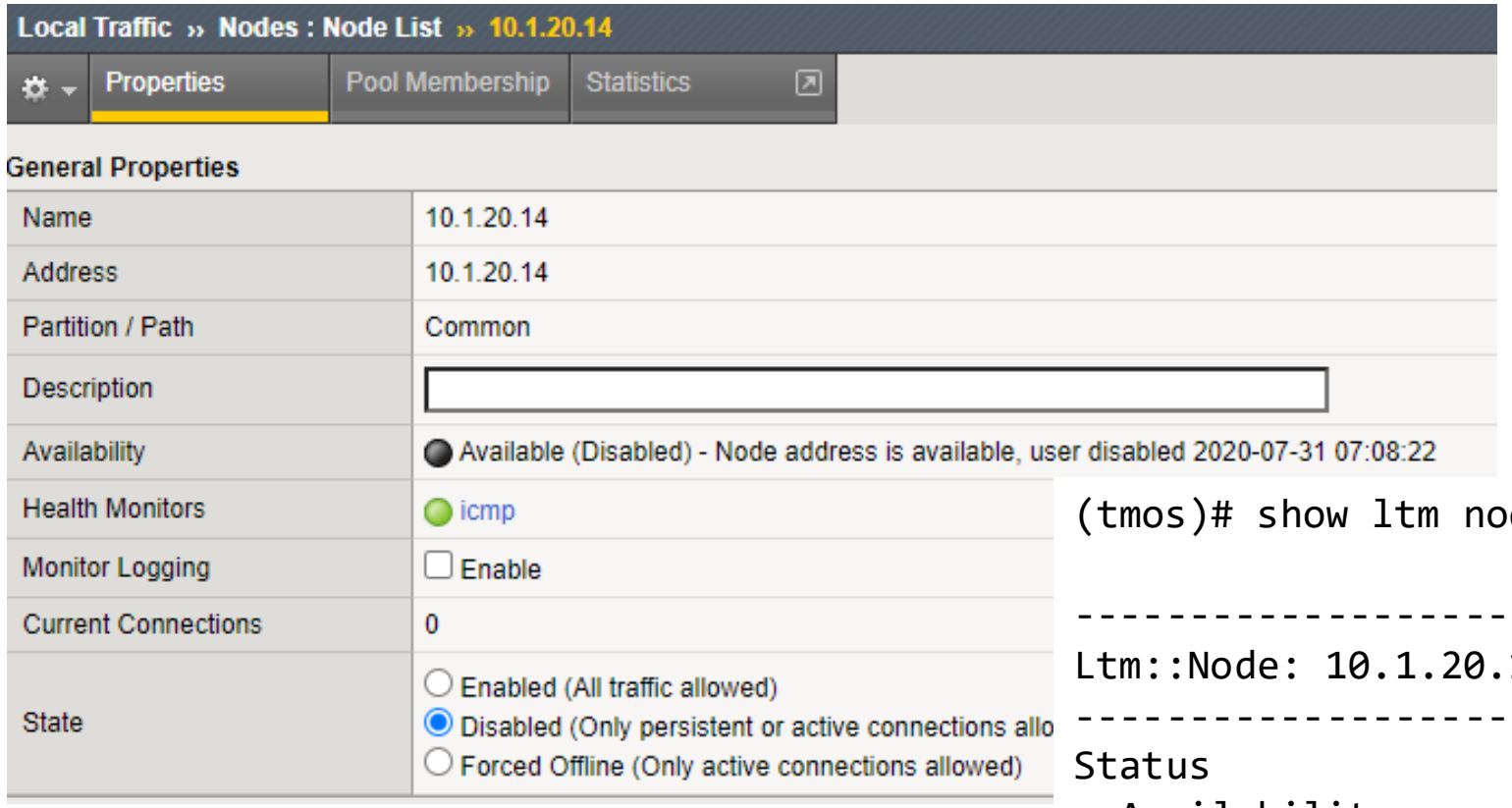
- **Currently Unavailable**
  - The virtual server or all its resources have reached a connection limit that has been set by the administrator
  - A pool member has reached a connection limit that has been set by the administrator
  - The object has no further capacity for traffic until the current connections fall below the connection limit settings.



- **Disabled / Forced Offline**
  - The object has administratively been marked down and will not process traffic
  - The status icon will be a shape that represents the current monitor status of the object but will always be colored **black**.
  - A grey status shape would mean the child object has been disabled.
    - If you disable a node, the pool member associated with the node would go grey

 ©2026 F5

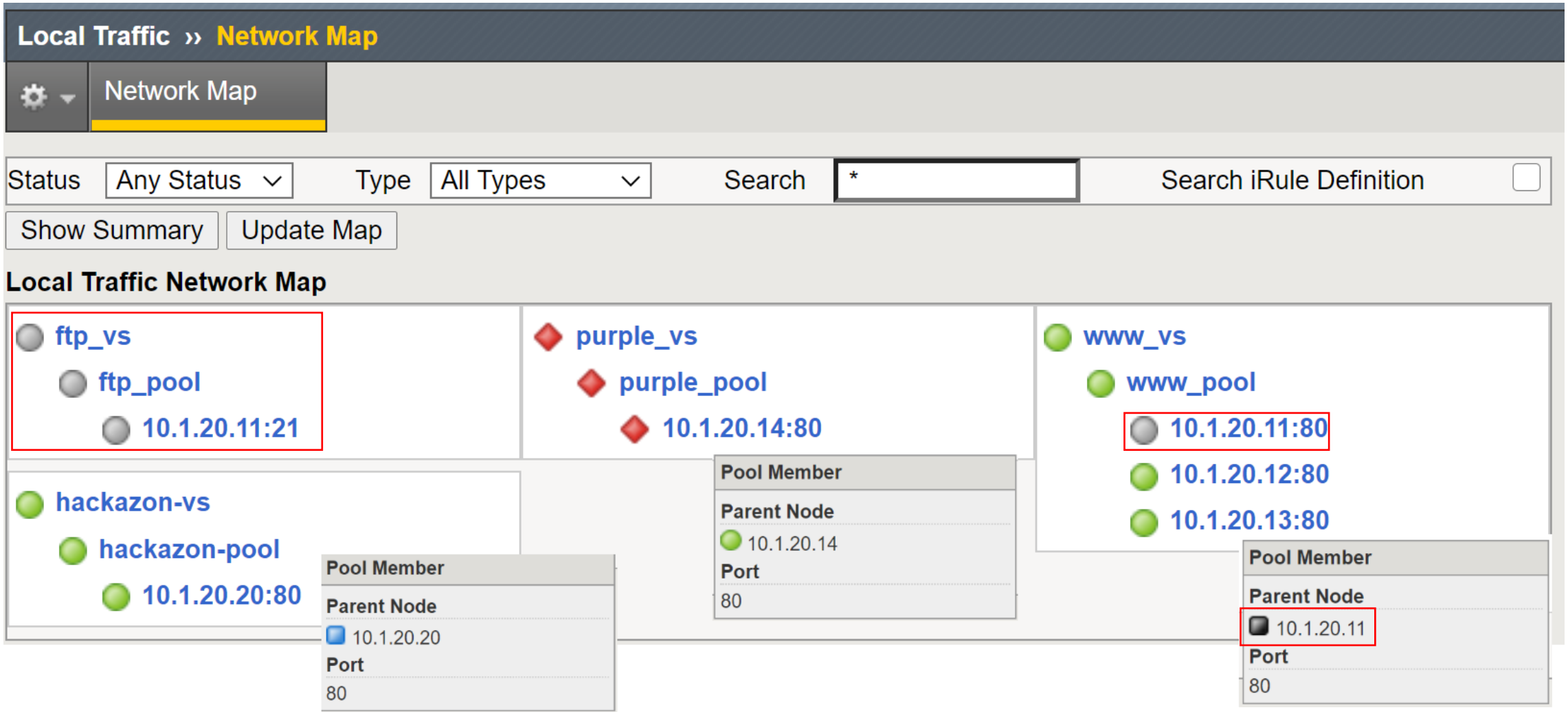
Status and State




```
(tmos)# show ltm node 10.1.20.14

-----------------------------------------
Ltm::Node: 10.1.20.14 (10.1.20.14)
-----------------------------------------
Status
  Availability   : available
  State          : disabled
  Reason         : Node address is available, user disabled
  Monitor        : icmp
  Monitor Status : up
  Session Status : user-disabled
```

 ©2026 F5

Status and State – Network Map



 ©2026 F5

Identify traffic diverted due to persistence

Manual Chapter : Session Persistence Profiles

- Directs a client back to the same server after the initial load balancing decision has been made
  - Is required for stateful applications
  - May skew load balancing statistics
- The persistence profile is assigned at the virtual server level.
- Persistence methods you should know
  - Source Address Affinity (aka Simple) Persistence (Based on source IP and network mask)
  - Cookie Persistence (Recommended for HTTP)
- Other persistence methods
  - SSL Session ID, Session Initiated Protocol (SIP), MSRDP
  - Universal Persistence
    - iRules can create persistence records based on anything in the client's request, such as, jsessionid, username, etc.

 ©2026 F5

Persistence Settings

Manual Chapter : Session Persistence Profiles

Match Across Services

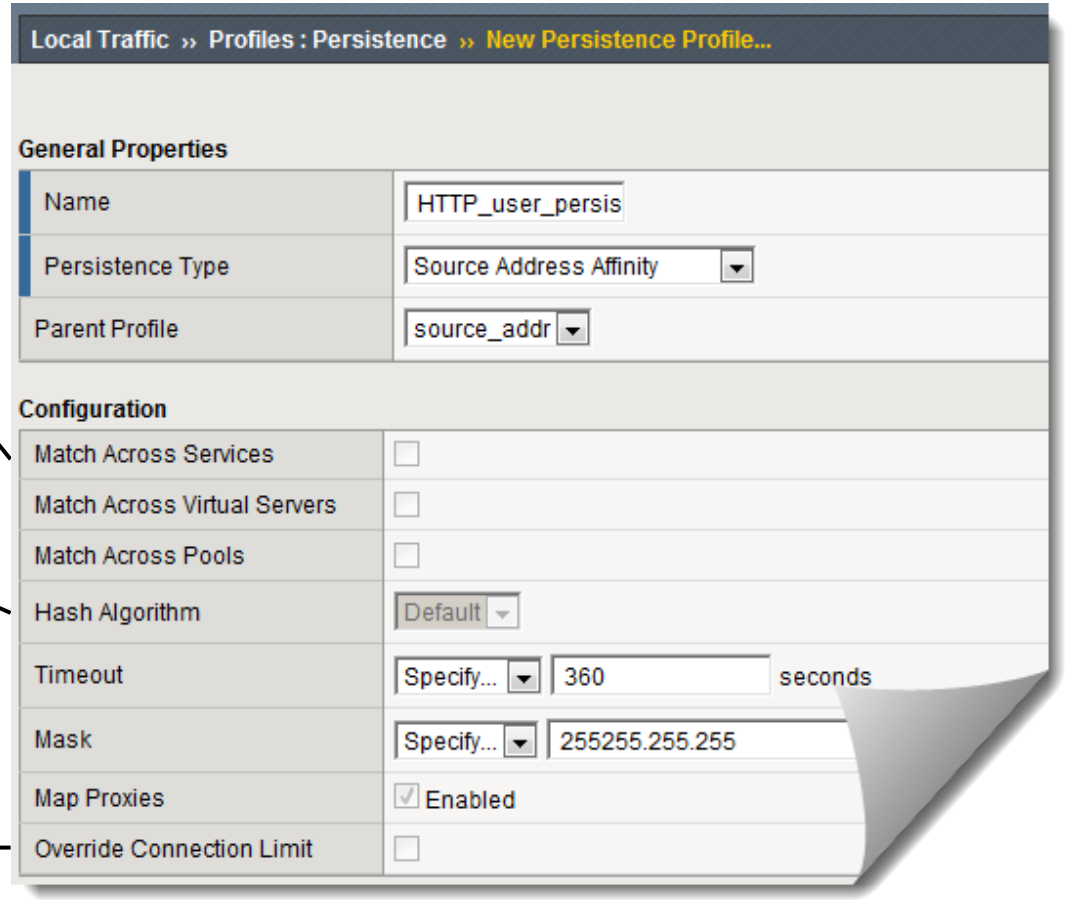
- When enabled, specifies that all persistent connections from a client IP address that go to the same virtual IP address also go to the same pool member

Timeout

- Specifies the duration of the persistence entries
- Resets on a new connection

Override Connection Limit

- Allows new connections to be established when the connection limit is reached, if there is an existing persistence record

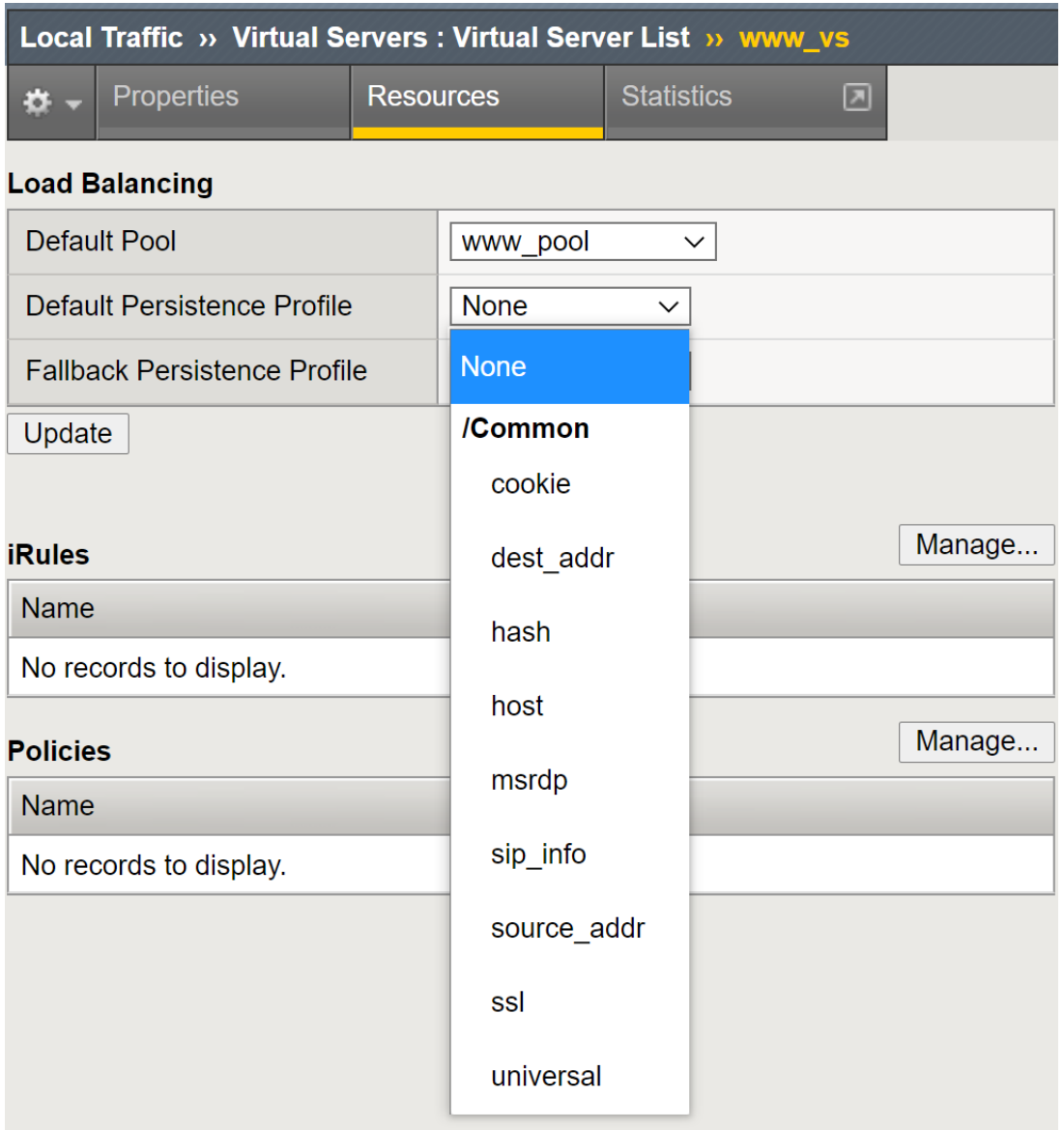


 ©2026 F5

Persistence Methods

[Manual: Session Persistence Profiles]

- Configured under Resources tab in a Virtual Server
- Fallback persistence
  - If there is not a persistence record from the Default Persistence Profile
  - Check if a persistence record was created by the fallback and use that record
- Fallback example:
  - If users don't allow cookies fallback to source persistence.



 ©2026 F5

F5CAB3 : BIG-IP Administration Data Plane Configuration

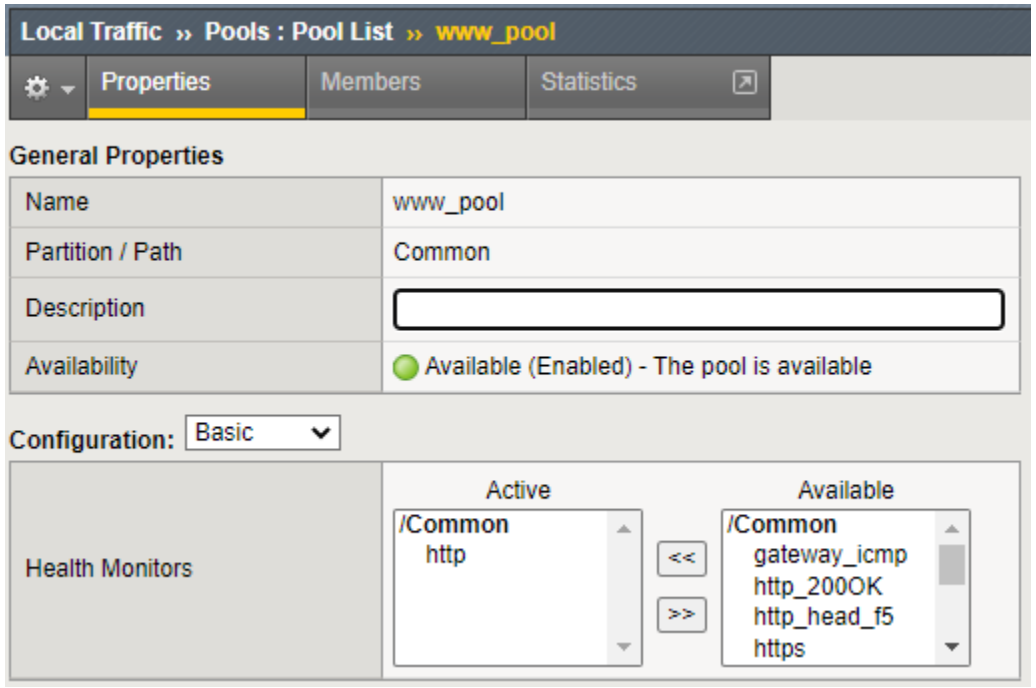
Apply procedural concepts required to modify and manage pools

- Determine configured health monitor
- Determine the load balancing method for a pool
- Determine pool member service port configuration
- Determine the active nodes in a priority group configuration
- Apply appropriate health monitor
- Apply load balancing method for a pool
- Apply pool member service port configuration

 ©2026 F5

Determine configured health monitor

Manual : BIG-IP Local Traffic Manager: Monitors Reference

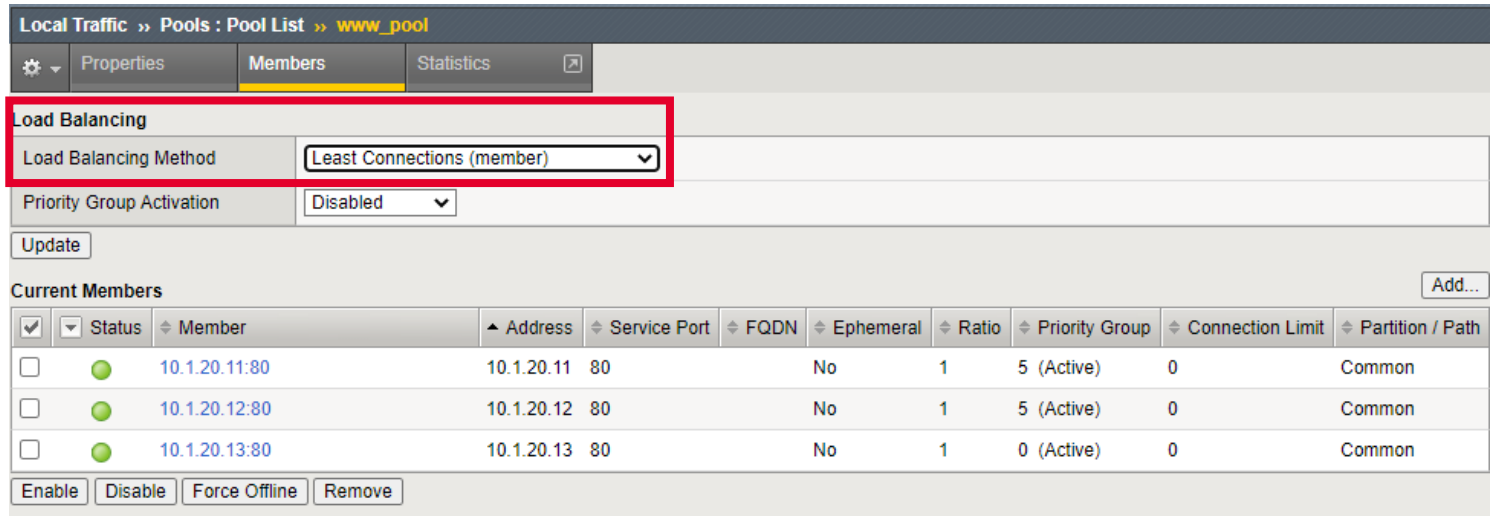



```
(tmos)# list ltm pool www_pool
ltm pool www_pool {
    members {
        10.1.20.11:http {
            address 10.1.20.11
            session monitor-enabled
            state up
        }
        10.1.20.12:http {
            address 10.1.20.12
            session monitor-enabled
            state up
        }
        10.1.20.13:http {
            address 10.1.20.13
            session monitor-enabled
            state up
        }
    }
    monitor http
}
```

 ©2026 F5

Determine the load balancing method for a pool

[Manual Chapter : About Pools]




```
(tmos)# list ltm pool www_pool
ltm pool www_pool {
    load-balancing-mode least-connections-member
    members {
        10.1.20.11:http {
            address 10.1.20.11
            priority-group 5
            session monitor-enabled
            state up
        }
        10.1.20.12:http {
            address 10.1.20.12
            priority-group 5
            session monitor-enabled
            state up
        }
        10.1.20.13:http {
            address 10.1.20.13
            session monitor-enabled
            state up
        }
    }
    monitor http
}
```

 ©2026 F5

Load Balancing methods

K6406: Overview of Least Connections, Fastest, Observed, and Predictive pool member load balancing

- A load balancing method is an algorithm used to determine which pool member to send traffic to
  - **Load balancing is connection based**
- Static load balancing methods distribute connections in a fixed manner
  - Round Robin (RR)
  - Ratio (Weighted Round Robin)
    - Distributes in a RR fashion for members/nodes whose ratio has not been met
- Dynamic load balancing looks at one or more factors, the most common method is:
  - Least Connections
    - Fewest L4 connections when load balancing decision is being made
    - Recommended when servers have similar capabilities
    - Very commonly used

 ©2026 F5

Load Balancing a Service (Member)

Client 18.200.150.10

Internet

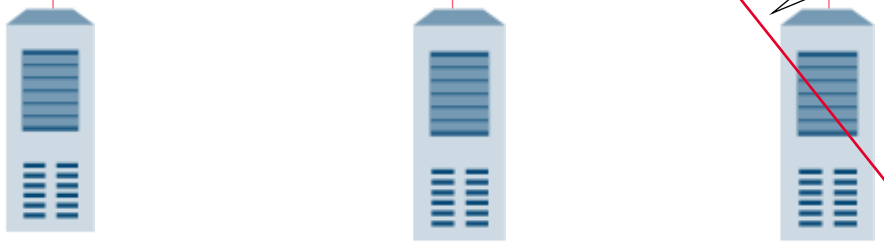
In this example, the HTTP pool is configured with the **Least Connections (member)** method

http_vs 10.2.2.100:80

BIG-IP LTM directs the request to the pool member with the least number of connections

With each new client request, BIG-IP LTM verifies which **pool member** within the pool has the fewest active connections

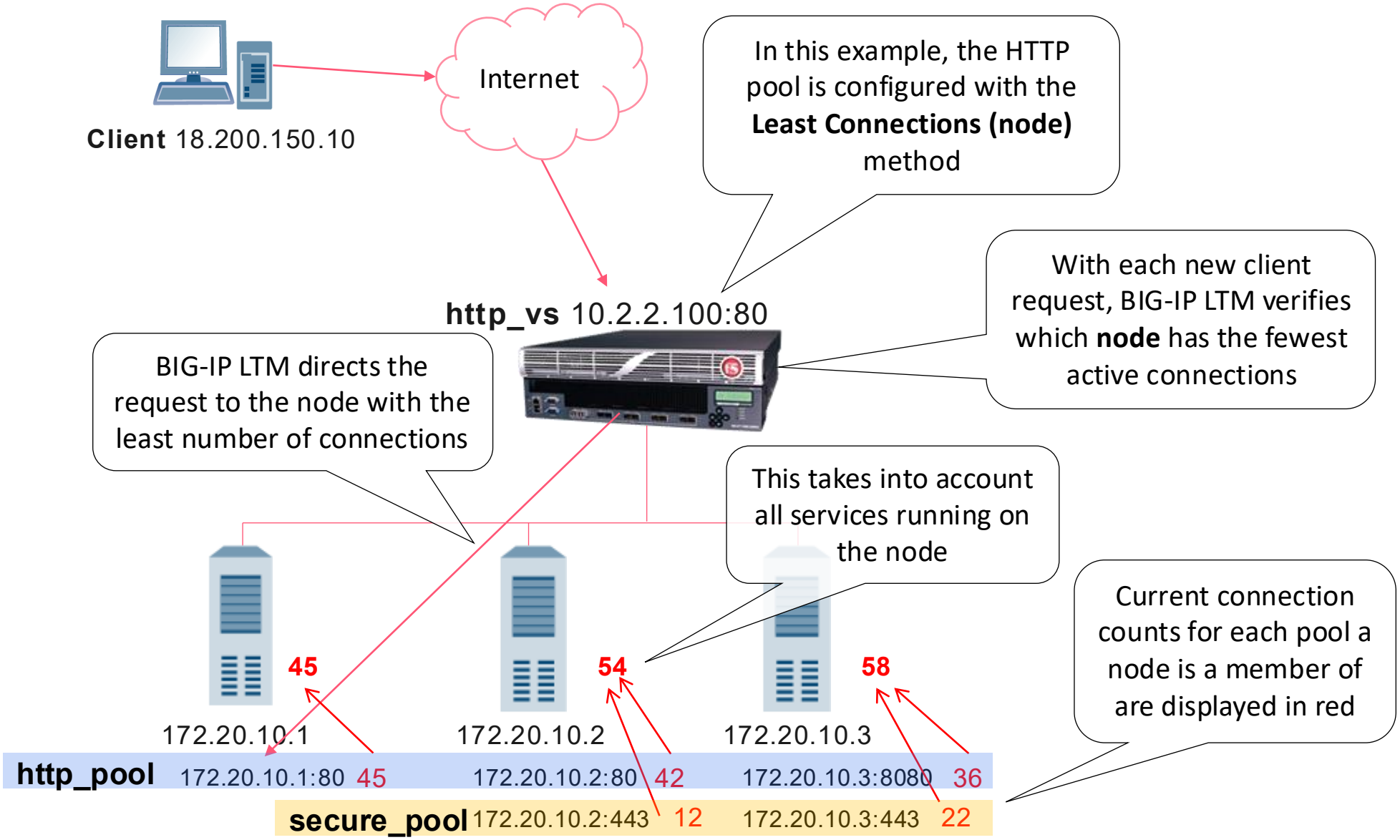
Current connection counts for each pool member are displayed in red



	172.20.10.1	172.20.10.2	172.20.10.3
http_pool	172.20.10.1:80 45	172.20.10.2:80 42	172.20.10.3:8080 36
secure_pool		172.20.10.2:443 12	172.20.10.3:443 22

 ©2026 F5

Load Balancing an IP Address (Node)



 ©2026 F5

F5CAB5: BIG-IP Administration Support and Troubleshooting

Identify the reason a pool is not working as expected

- Identify the current configured state of the pool/pool member
- Identify the current availability status of the pool/pool member
- Identify the reason a pool member has been marked down by health monitors
- Identify a pool member not in the active priority group

 ©2026 F5

Identify the current configured state/status of the pool/pool member

[Manual Chapter : About Pools]

Statistics » Module Statistics : Local Traffic » Pools

Traffic Summary | DNS | Local Traffic | Subscriber Management | Network | Memory | System

Display Options

Statistics Type	Pools
Data Format	Normalized
Auto Refresh	Disabled Refresh

* Search

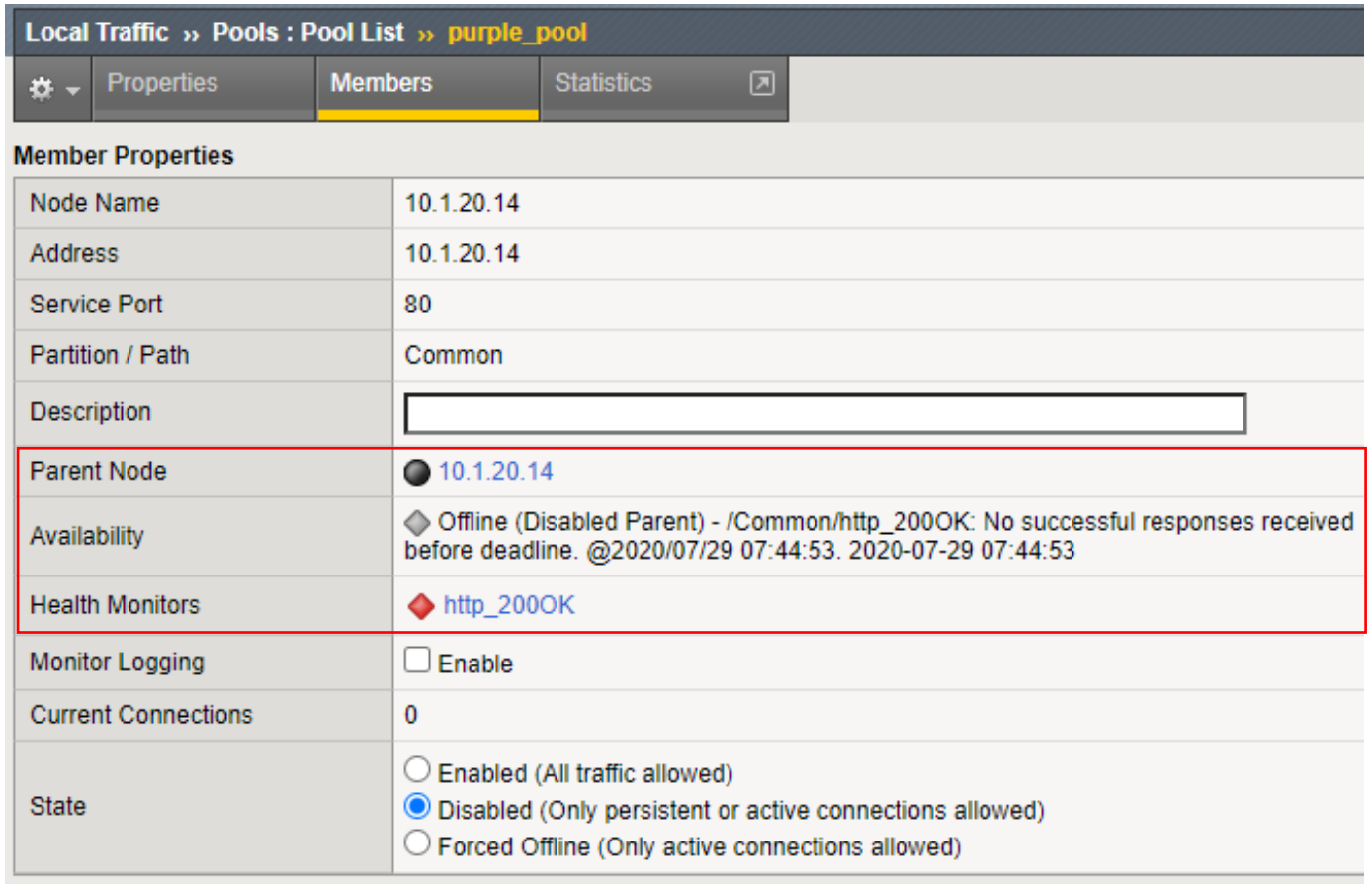
					Bits		Packets		Connections			Requests	Request Queue	
Status	Pool	Pool Member	Partition / Path		In	Out	In	Out	Current	Maximum	Total	Total	Depth	Maximum Age
	ftp_pool		Common		27.9K	34.0K	77	71	0	1	1	0	0	0
		10.1.20.11:21	Common		27.9K	34.0K	77	71	0	1	1	0	0	0
	hackazon-pool		Common		818.2K	11.4M	864	1.1K	0	9	28	0	0	0
		10.1.20.20:80	Common		818.2K	11.4M	864	1.1K	0	9	28	0	0	0
	purple_pool		Common		0	0	0	0	0	0	0	0	0	0
		10.1.20.14:80	Common		0	0	0	0	0	0	0	0	0	0
	www_pool		Common		182.7K	1.8M	208	253	0	7	14	0	0	0
		10.1.20.11:80	Common		66.6K	376.3K	72	78	0	4	8	0	0	0
		10.1.20.12:80	Common		116.1K	1.4M	136	175	0	3	6	0	0	0
		10.1.20.13:80	Common		0	0	0	0	0	0	0	0	0	0

Reset

 ©2026 F5

Identify the current configured state/status of the pool/pool member

[Manual Chapter : About Pools]



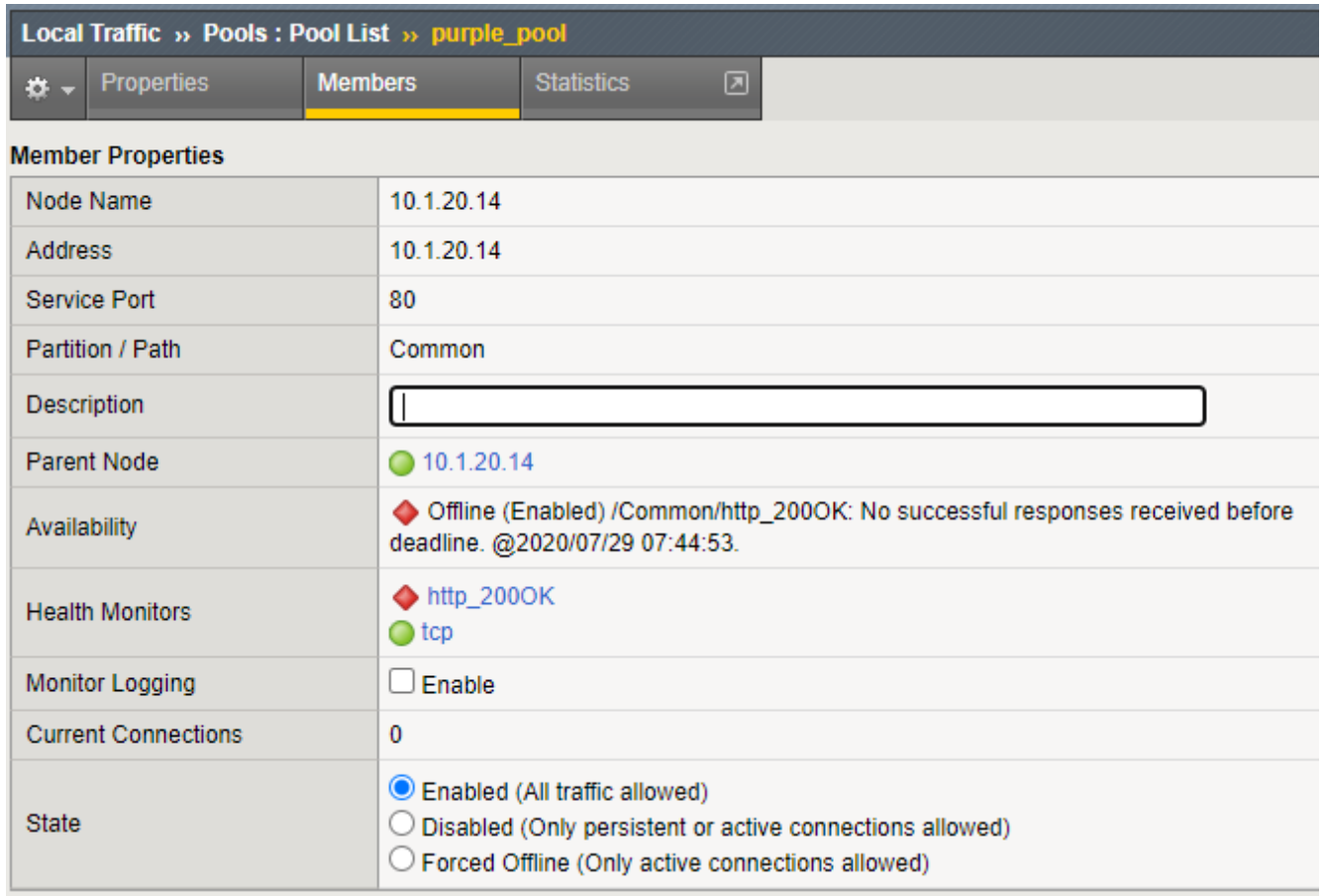

```
(tmos)# show ltm pool purple_pool members
-----------------------------------------------
Ltm::Pool: purple_pool
-----------------------------------------------
Status
  Availability : offline
  State        : enabled
  Reason       : The children pool member(s) are down
  Monitor      : http_200OK
  Minimum Active Members : 0
  Current Active Members : 0
       Available Members : 0
       Total Members : 1
          Total Requests : 0
        Current Sessions : 0
<cut>
  -----------------------------------------------
  | Ltm::Pool Member: 10.1.20.14:80
  -----------------------------------------------
  | Status
  |   Availability    : offline
  |   State           : disabled-by-parent
  |   Reason          : http_200OK: No successful
responses received before deadline. @2020/07/29 07:44:53.
  |   Monitor         : http_200OK (pool monitor)
  |   Monitor Status  : down
  |   Session Status  : addr-disabled
  |   Pool Name       : purple_pool
  |   IP Address      : 10.1.20.14
```

 ©2026 F5

Identify the reason a pool member has been marked down by health monitors

Manual Chapter : About Pools

- There are numerous reasons a pool member may be marked down.
  - Misconfigured monitor
  - Wrong monitor
  - Wrong port
  - Bad network path to servers
- IMPORTANT: Monitors are sourced from the **base** self IP on the outbound VLAN the BIG-IP uses to send traffic to the pool member being monitored.

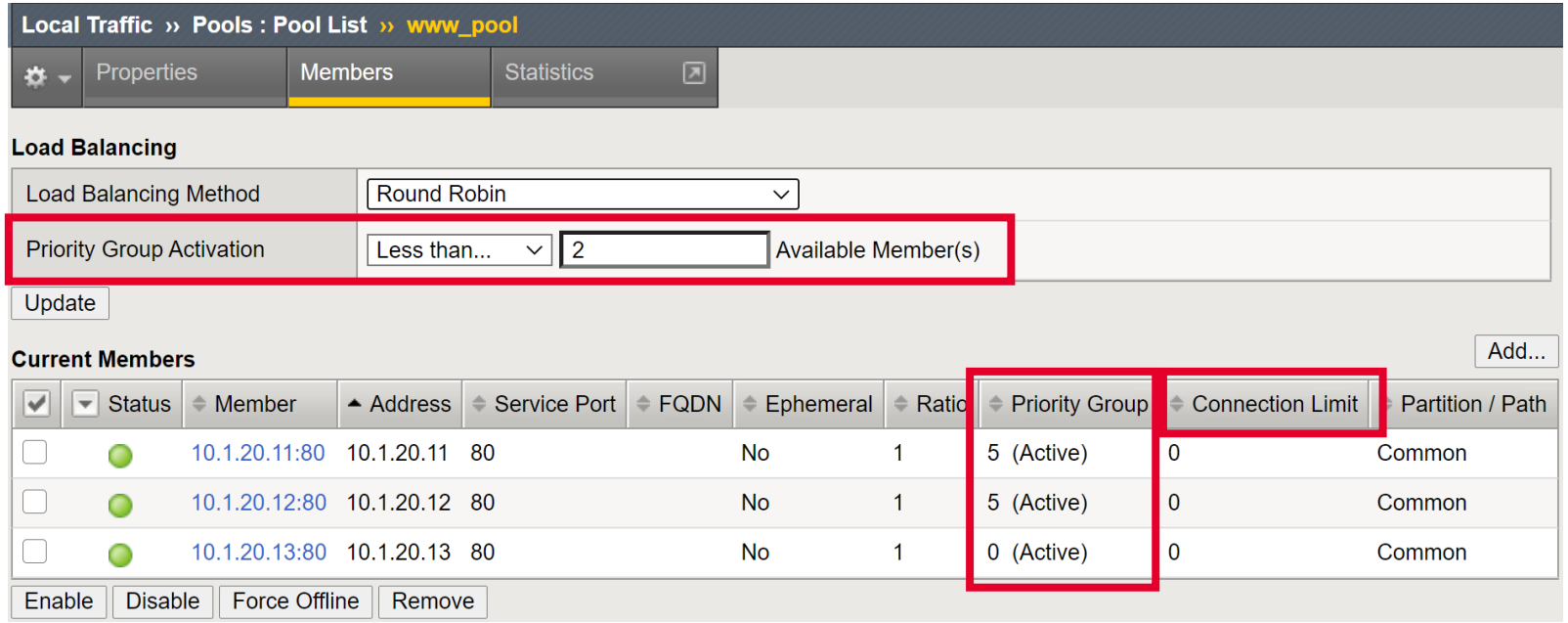


 ©2026 F5

Identify a pool member not in the active priority group

Priority Group Activation

- Priority Group Activation load balancing
  - Allows pool members to be used only if preferred pool members are unavailable.
  - Each pool member is assigned a priority
  - Connections are sent to the highest priority pool members first.
  - A minimum number of available members are assigned

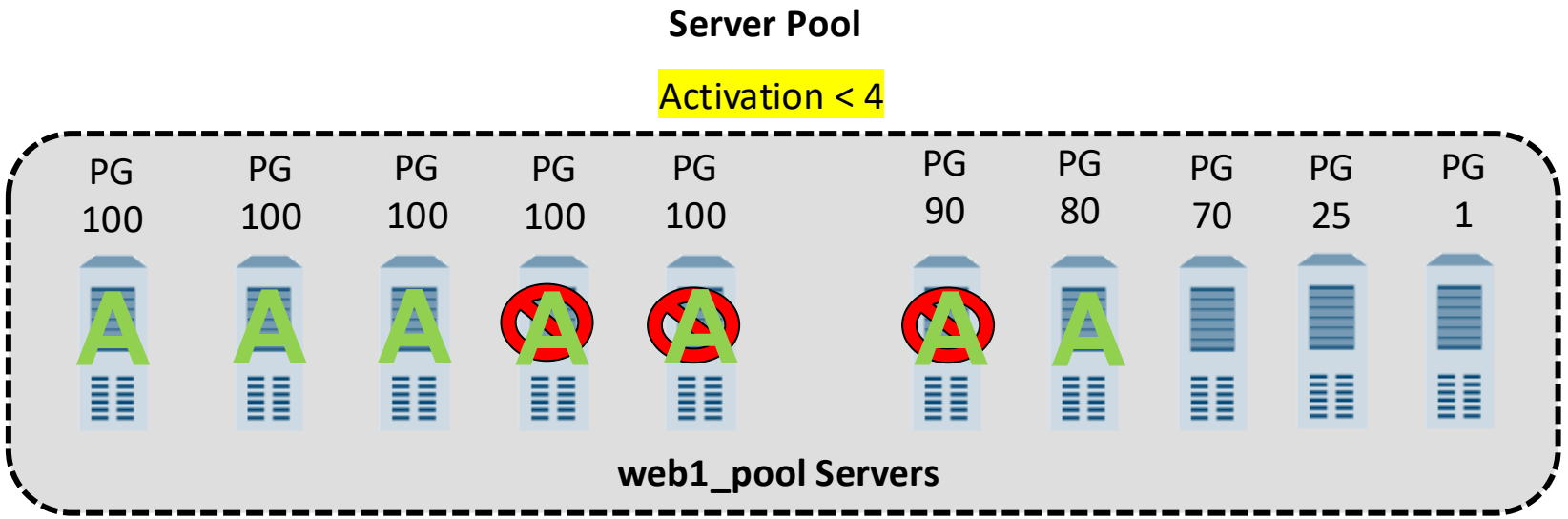


 ©2026 F5

Identify a pool member not in the active priority group

Priority Group Activation

- Priority Group Activation is a failure mechanism
  - Can dynamically pull in new members into the pool
  - Pulls lower priority groups into higher priority groups
  - Pulls in all members of a priority group together



 ©2026 F5

F5CAB5: BIG-IP Administration Support and Troubleshooting

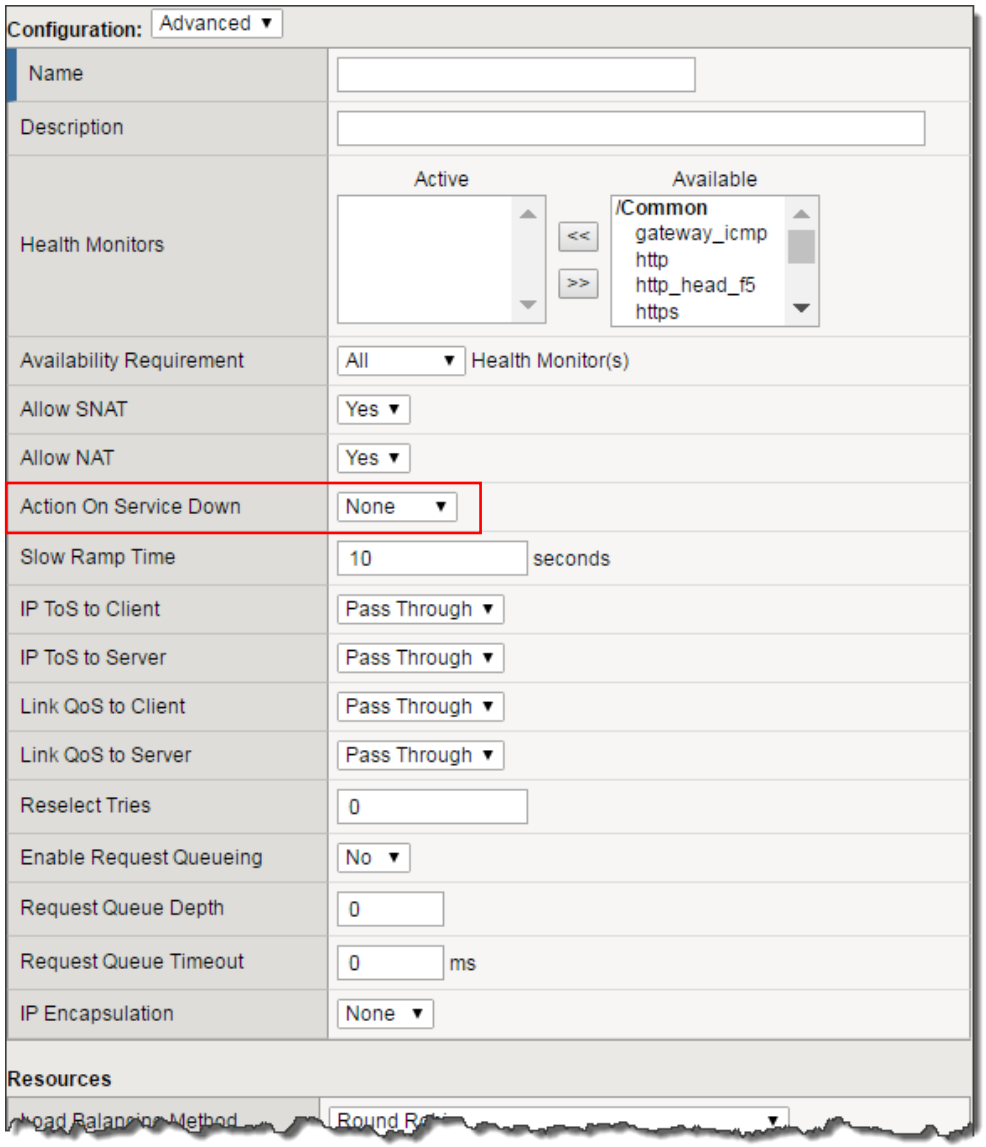
Identify the reason load balancing is not working as expected

- Identify current availability status
- Identify misconfigurations (incorrect health checks, action on service down, etc.)
- Consider persistence, priority group activation, rate/connection limits

 ©2026 F5

Action on Service Down

- Action on Service Down
  - **None** – RST to client after idle timeout reached (Default)
  - **Reject** – sent RST to active client connections
  - **Drop** – silently remove the connection
  - **Reselect** – move connection to alternate pool member
- Slow Ramp Time
  - Set less traffic to newly established pool member



 ©2026 F5

Enabling/Disabling Nodes and Pool Members

State determines how persistence and connections are handled

Pool Member State	Interaction with Pool Member
Enabled All Traffic Allowed	Existing Connection – Maintained New Persistence Records – Can be Created New Connections – Can be Created
Disabled (Members or Nodes) Only persistent or active connections allowed.	Existing Connection – Maintained New Persistence Records – Not Created New Connections – Can be Created *only* for Client with an Existing Persistence record
Forced Offline (Members or Nodes) Only active connections allowed.	Existing Connection – Maintained New Persistence Records – Not Created New Connections – Not Created

 ©2026 F5

Consider persistence, priority group activation, rate/connection limits

REVIEW

- Persistence
  - Check records
  - Object state
  - Understand the difference in behavior of
    - Pools and Nodes which are Disabled or Forced Offline
    - Persistence Override Connection limits

 ©2026 F5

Review

Is there something wrong with this pool?

If all members are up why aren't all members taking traffic?

If **node1** fails, which members will take traffic?

If all members are up, but you see traffic statistics on node3 and node4 what does that tell you?

Local Traffic » Pools : Pool List » pool1

Properties | Members | Statistics

Load Balancing

Load Balancing Method	Round Robin
Priority Group Activation	Less than... 2 Available Member(s)

Update

Current Members Add...

Status	Member	Address	Ratio	Priority Group	Connection Limit	Partition / Path
	node1:80	10.128.20.11	1	10 (Active)	0	Common
	node2:80	10.128.20.12	1	10 (Active)	0	Common
	node3:80	10.128.20.13	1	5 (Active)	0	Common
	node4:80	10.128.20.14	1	5 (Active)	0	Common
	node5:80	10.128.20.15	1	1 (Active)	0	Common

Statistics » Module Statistics : Local Traffic

Traffic Summary | Local Traffic | Network | Memory

Display Options

Statistics Type	Pools
Data Format	Normalized
Auto Refresh	Disabled Refresh

* Search

			Bits		Packets		Connections			Requests	Request Queue	
Status	Pool/Member	Partition / Path	In	Out	In	Out	Current	Maximum	Total	Total	Depth	Maximum Age
	pool1	Common	250.2K	2.5M	324	367	0	10	42	37	0	0
	-- node1:80	Common	126.5K	1.4M	168	195	0	6	21	18	0	0
	-- node2:80	Common	123.6K	1.1M	156	172	0	4	21	19	0	0
	-- node3:80	Common	0	0	0	0	0	0	0	0	0	0
	-- node4:80	Common	0	0	0	0	0	0	0	0	0	0
	-- node5:80	Common	0	0	0	0	0	0	0	0	0	0

 ©2026 F5

Load Balancing

Load Balancing Method	Round Robin
Priority Group Activation	Disabled

Update

Current Members Add...

Status	Member	Address	Service Port	FQDN	Ephemeral	Ratio	Priority Group	Connection Limit	Partition / Path
	10.1.20.11:80	10.1.20.11	80		No	5	10 (Active)	0	Common
	10.1.20.12:80	10.1.20.12	80		No	1	10 (Active)	0	Common
	10.1.20.13:80	10.1.20.13	80		No	1	5 (Active)	0	Common

Enable Disable Force Offline Remove

Given the configuration what pool member will take the most connections?

Load Balancing

Load Balancing Method	Round Robin
Priority Group Activation	Less than... 3 Available Member(s)

Update

Current Members Add...

Status	Member	Address	Service Port	FQDN	Ephemeral	Ratio	Priority Group	Connection Limit	Partition / Path
	10.1.20.11:80	10.1.20.11	80		No	5	10 (Active)	0	Common
	10.1.20.12:80	10.1.20.12	80		No	1	10 (Active)	0	Common
	10.1.20.13:80	10.1.20.13	80		No	1	5 (Active)	0	Common

Enable Disable Force Offline Remove

Given the configuration which pool members will process traffic?

 ©2026 F5

Load Balancing

Load Balancing Method	Round Robin
Priority Group Activation	Disabled

Update

Current Members Add...

Status	Member	Address	Service Port	FQDN	Ephemeral	Ratio	Priority Group	Connection Limit	Partition / Path
	10.1.20.11:80	10.1.20.11	80		No	5	10 (Active)	0	Common
	10.1.20.12:80	10.1.20.12	80		No	1	10 (Active)	0	Common
	10.1.20.13:80	10.1.20.13	80		No	1	5 (Active)	0	Common

Enable Disable Force Offline Remove

You have disabled 10.1.20.11:80, but the pool member continues to receive new connections. What does this tell you?

Load Balancing

Load Balancing Method	Ratio (member)
Ignore Persisted Weight	☐
Priority Group Activation	Less than... 1 Available Member(s)

Update

Current Members Add...

Status	Member	Address	Service Port	FQDN	Ephemeral	Ratio	Priority Group	Connection Limit	Partition / Path
	10.1.20.11:80	10.1.20.11	80		No	5	10 (Active)	0	Common
	10.1.20.12:80	10.1.20.12	80		No	1	10 (Active)	0	Common
	10.1.20.13:80	10.1.20.13	80		No	1	5 (Active)	0	Common

Enable Disable Force Offline Remove

Given the configuration what pool member will take the most connection?

 ©2026 F5

Lab 2

 ©2026 F5

F5CAB3 : BIG-IP Administration Data Plane Configuration

Apply procedural concepts required to modify and manage virtual servers

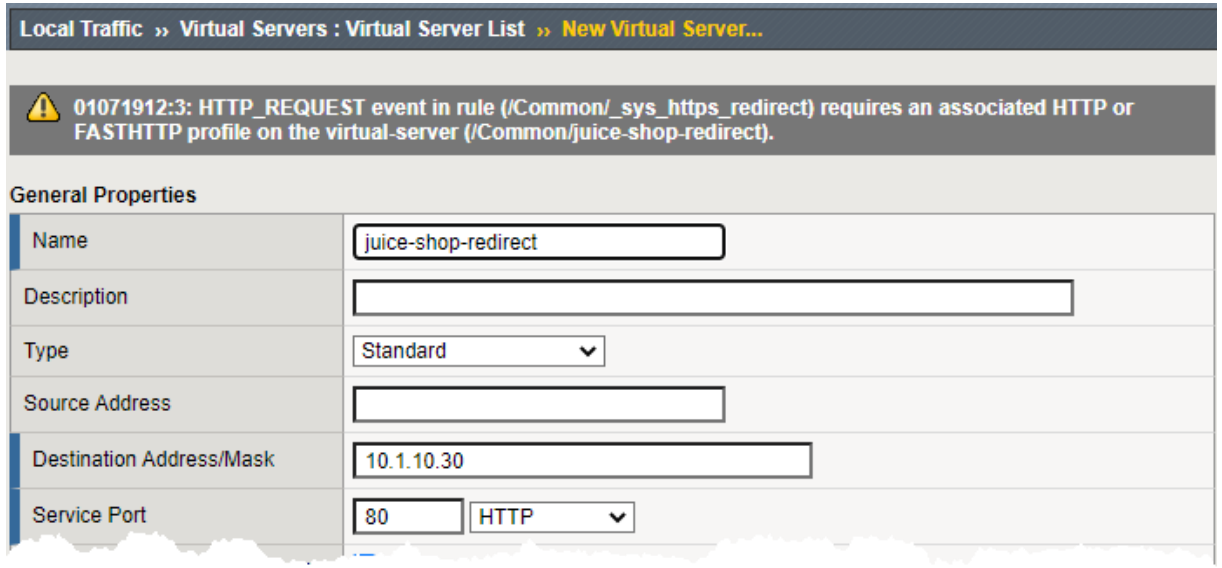
- Apply appropriate protocol specific profile
- Apply appropriate persistence profile
- Apply appropriate HTTPS encryption profile
- Identify iApp configured objects
- Report use of iRules
- Show default pool configuration

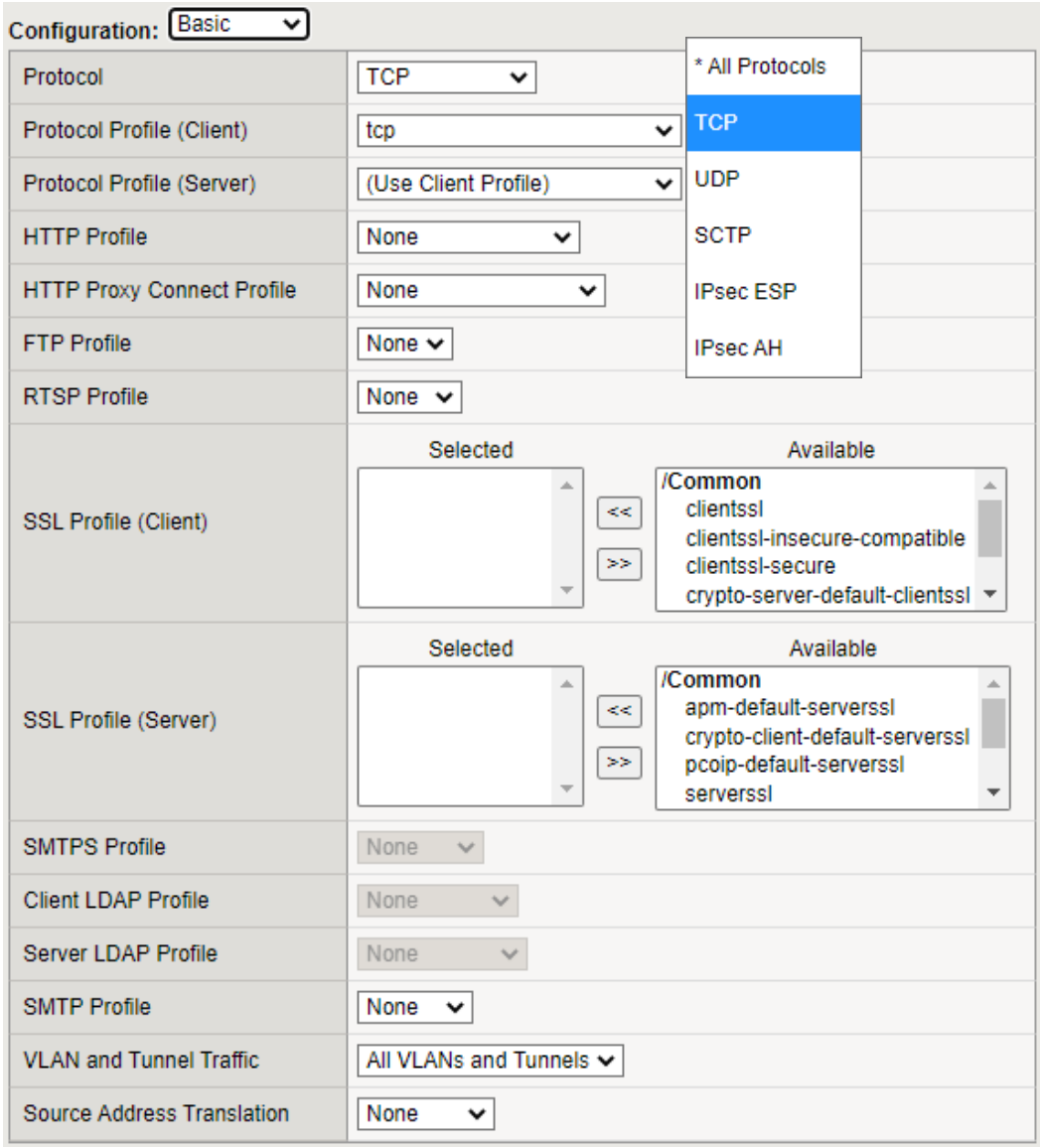
 ©2026 F5

Apply appropriate protocol specific profile

MANUAL CHAPTER: VIRTUAL SERVERS

- All virtual servers must have a Protocol profile assigned
- If looking beyond L4 information is required, then the appropriate L7 profile needs to be assigned.
  - For example, FTP profile for FTP applications
  - For example, HTTP profile if the cookie or other information needs to be viewed or manipulated.





 ©2026 F5

Apply appropriate HTTPS encryption profile

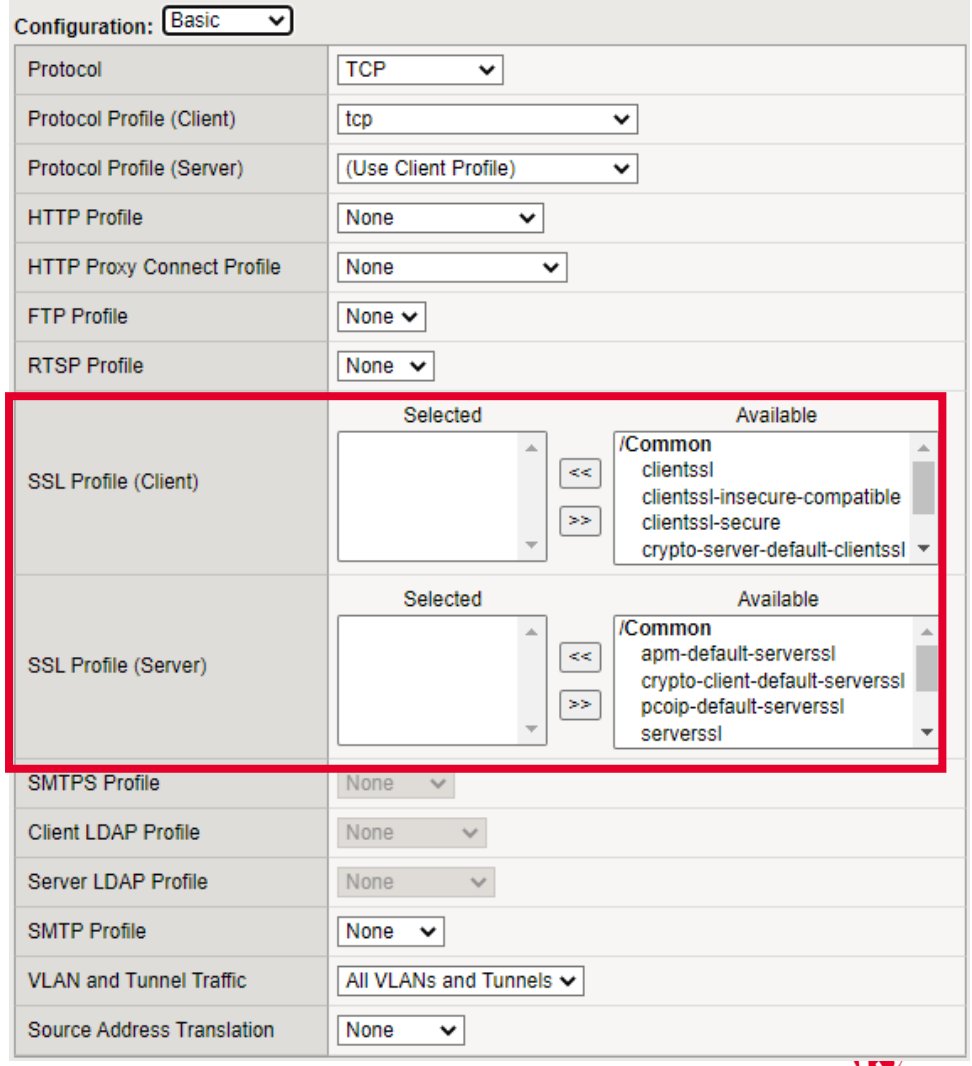
K14783: Overview of the Client SSL profile

K14806: Overview of the Server SSL profile

- SSL Profile requirements
  - SSL Client-Side profile, with the appropriate cert & key for SSL offload
  - SSL Server-Side profile, if the pool members service HTTPS traffic
- An HTTP profile is NOT required.

Local Traffic » Profiles : SSL : Client

Local Traffic » Profiles : SSL : Server



 ©2026 F5

Identify iApp configured objects

Local Traffic » Virtual Servers : Virtual Server List

Virtual Server List | Virtual Address List | Statistics

Search | Create...

Status	Name	Description	Application	Destination	Service Port	Type	Resources	Partition / Path
	created_with_iapp_vs		created_with_iapp	10.1.10.120	80 (HTTP)	Standard	Edit...	Common/created_with_iapp.app
	ftp_vs			10.1.10.100	21 (FTP)	Standard	Edit...	Common
	purple_vs			10.1.10.105	80 (HTTP)	Standard	Edit...	Common
	www_vs			10.1.10.100	80 (HTTP)	Standard	Edit...	Common

iApps » Application Services : Applications » created_with_iapp

Properties | Reconfigure | Components | Analytics

Application Service: Basic

Application Service	created_with_iapp
Partition / Path	Common/created_with_iapp.app
Description	
Template	f5.http

Update | Delete

 ©2026 F5

Identify iApp configured objects

iApps » Application Services : Applications » created_with_iapp

Properties | Reconfigure | Components | Analytics

Name	Availability	Type
BIG-IP		
created_with_iapp		Application Service
created_with_iapp_vs	Available	Virtual Server
created_with_iapp_pool	Available	Pool
http		Monitor
10.1.20.11:80	Available	Pool Member
10.1.20.11	Unknown	Node
10.1.20.12:80	Available	Pool Member
10.1.20.12	Unknown	Node
created_with_iapp_source-addr-persistence		Profile
10.1.10.120		Virtual Address
created_with_iapp_cookie-persistence		Virtual Server Persistence Profile
created_with_iapp_http		Profile
created_with_iapp_f5-tcp-lan		Profile
created_with_iapp_f5-tcp-wan		Profile
created_with_iapp_oneconnect		Profile
created_with_iapp_optimized-caching		Profile
created_with_iapp_wan-optimized-compression		Profile

Enable | Disable | Force Offline | Refresh

 ©2026 F5

F5CAB5: BIG-IP Administration Support and Troubleshooting

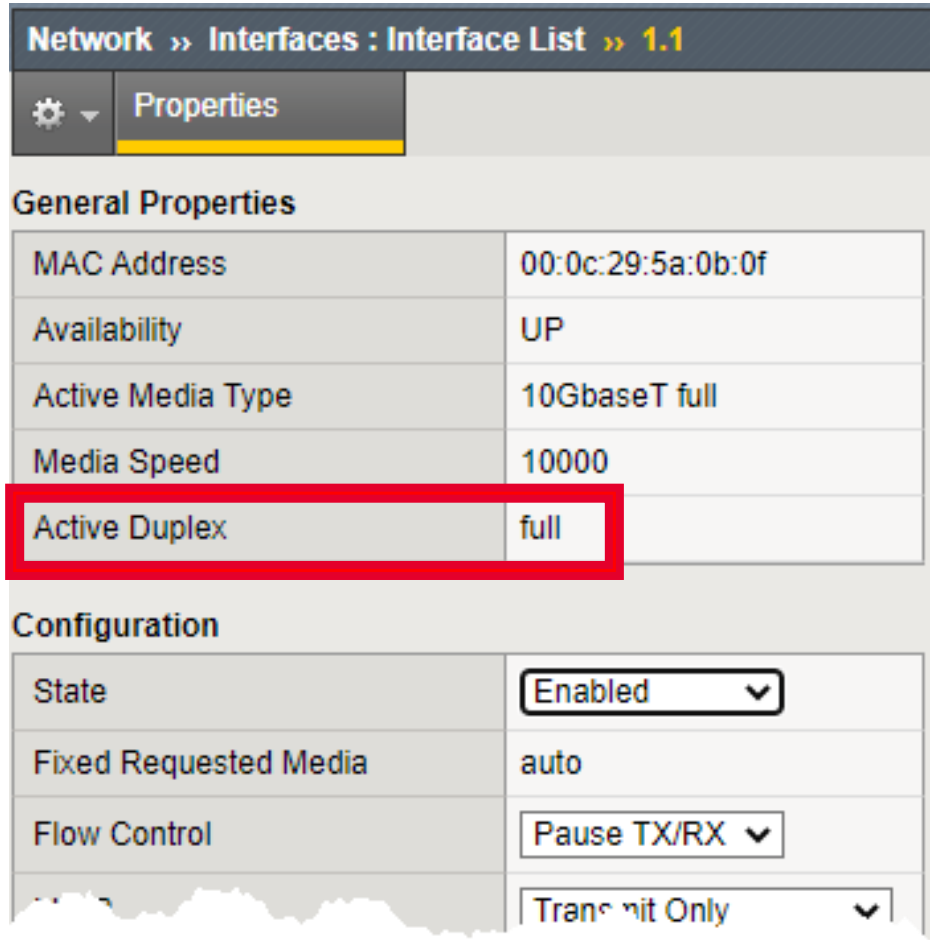
Identify network level performance issues

- Identify Speed and Duplex
- Distinguish TCP profiles (optimized profiles)
- Identify when a packet capture is needed within the context of a performance issue

 ©2026 F5

Identify Speed and Duplex

```
(tmos)# list net interface
net interface 1.1 {
    if-index 48
    mac-address 00:0c:29:5a:0b:0f
    media-active 10000T-FD
    media-fixed 10000T-FD
    media-max auto
}
net interface 1.2 {
    if-index 64
    mac-address 00:0c:29:5a:0b:19
    media-active 10000T-FD
    media-fixed 10000T-FD
    media-max auto
}
net interface 1.3 {
    if-index 80
    mac-address 00:0c:29:5a:0b:23
    media-fixed 10000T-FD
    media-max auto
}
net interface mgmt {
    if-index 32
    mac-address 00:0c:29:5a:0b:05
    media-active 100TX-FD
```



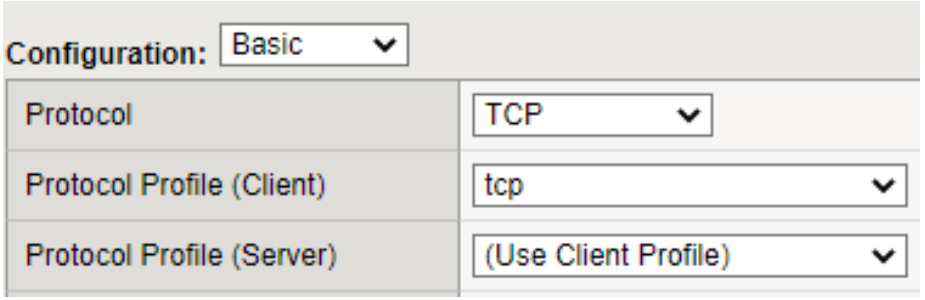
Be familiar with where things are in the GUI.

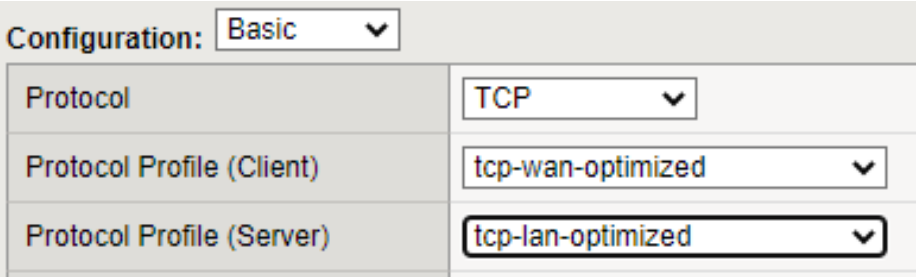
 ©2026 F5

Distinguish TCP profiles (optimized profiles)

Manual Chapter : Protocol Profiles

K50411377: Overview of the TCP profile (16.x and later)





- tcp-lan-optimized and f5-tcp-lan profiles
  - pre-configured profiles for LAN-based or interactive traffic
- tcp-wan-optimized and f5-tcp-wan profiles
  - pre-configured profile types for traffic over a WAN link
- tcp-mobile-optimized profile
  - pre-configured with default values set to give better performance to service providers' 3G and 4G customers.
- mptcp-mobile-optimized profile (Multipath TCP)
  - pre-configured profile type for use in reverse proxy and enterprise environments for mobile applications that are front-ended by a BIG-IP system

 ©2026 F5

Identify when a packet capture is needed within the context of a performance issue

K411: Overview of packet tracing with the tcpdump utility

- BIG-IP is a full proxy. Two *separate* tcpdumps (one on each side of the proxy) are often needed.
  - Can by done by opening two SSH sessions, or running the dumps in background (&)
  - Note - be very careful running tcpdumps in the background! (fg brings to foreground)
- When a tcpdump is required, always make it as specific as possible
  - Limit it to the appropriate interfaces/VLANs and hosts/ports
  - -i 0.0 captures on all ints except mgmt

system# tcpdump –i external –eX host 10.10.10.10 and port 80

system# tcpdump –i (1.1, f5_trunk1, external, 0.0) –eX –w /var/tmp/dump.cap

 ©2026 F5

Troubleshooting Tools

- Curl Utility - http://curl.haxx.se/
  - ***curl*** *is a command line tool for transferring data with URL syntax, supporting DICT, FILE, FTP, FTPS, Gopher, HTTP, HTTPS, IMAP, IMAPS, LDAP, LDAPS, POP3, POP3S, RTMP, RTSP, SCP, SFTP, SMTP, SMTPS, Telnet and TFTP.*
  - *It is supported on BIG-IP and is great for troubleshooting connectivity and monitors*

curl http://www.mysitename.com

curl http://10.128.20.11

```
[root@bigip249] config # curl -i 10.128.20.11
HTTP/1.1 200 OK
Date: Wed, 06 Aug 2014 20:05:13 GMT
Server: Apache/2.2.22 (Ubuntu)
X-Powered-By: PHP/5.4.9-4ubuntu2.2
Vary: Accept-Encoding
Content-Length: 3819
Connection: close
Content-Type: text/html

<html>
<head>
<TITLE>Using virtual server 10.128.20.11 and pool member 10.128.20.11 (Node
#1)</TITLE>
<meta http-equiv="Content-Type" content="text/html; charset=us-ascii" />
 <script language="javascript">
    ……................
 </script>
```

 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

Apply procedural concepts required to manage the state of a high availability pair

- Report current active/standby failover state
- Show device trust status
- Execute force to standby procedure
- Execute force to offline procedure

 ©2026 F5

Before we begin: A little more on Device Service Clusters.

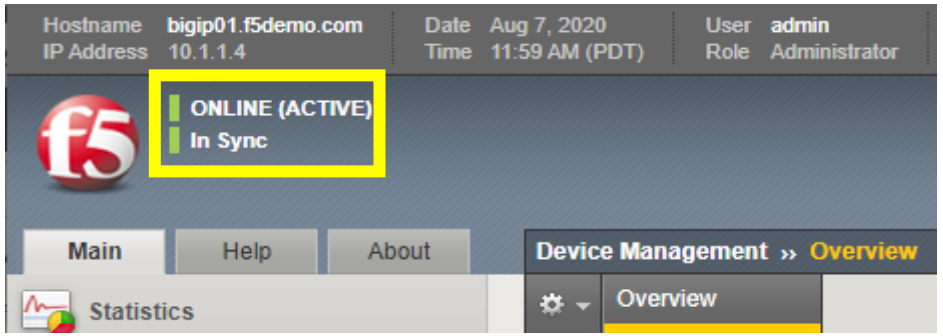
Manual : BIG-IP Device Service Clustering: Administration

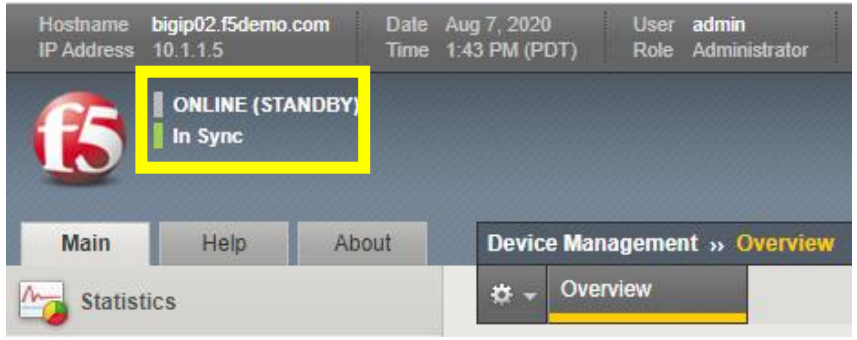
- For BIG-IPs to be combined into clusters for high availability, certain things must configured:
  - BIG-IPs must have a valid device certificate
  - On the device, IP addressing must be defined for failover
  - Devices must be place into a trust group
  - Devices in a trust group and then be place into a failover group

 ©2026 F5

Report current active/standby failover state

Manual : BIG-IP Device Service Clustering: Administration





[root@bigip01:**Active:**In Sync] config #

Active – there are one of more active traffic groups that can failover

Standby – there are no active traffic groups that can failover

[root@bigip02:**Standby:**In Sync] config #

Have a working knowledge of mirroring.

- SNAT
- Persistence
  - Only if persistence records are kept locally on the BIG-IP, not necessary for Cookie persistence.
- Connection Table
  - Only for long term connections, ie. FTP, resource intensive

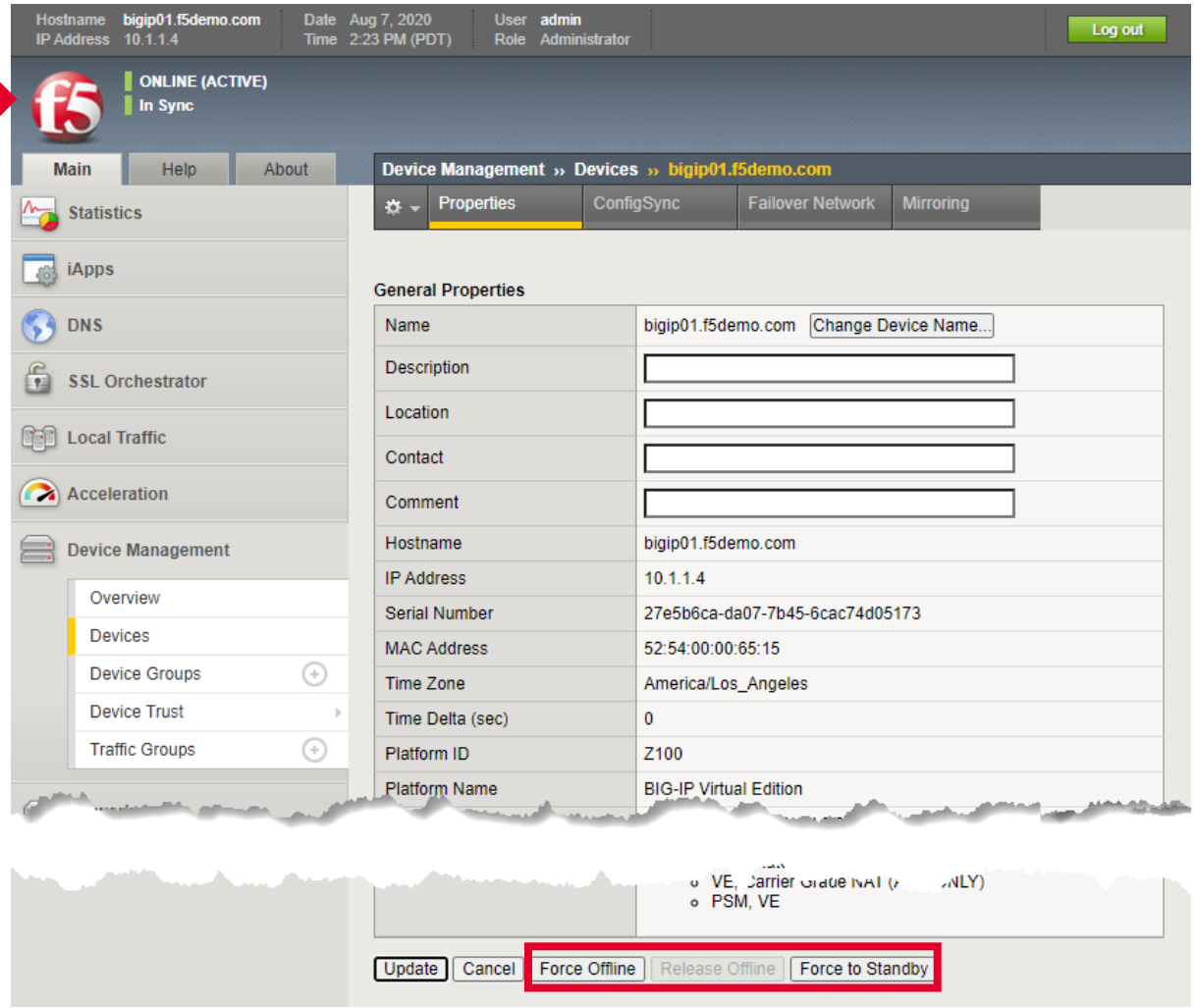
 ©2026 F5

Execute force to standby or offline procedure

Manual : BIG-IP Device Service Clustering: Administration

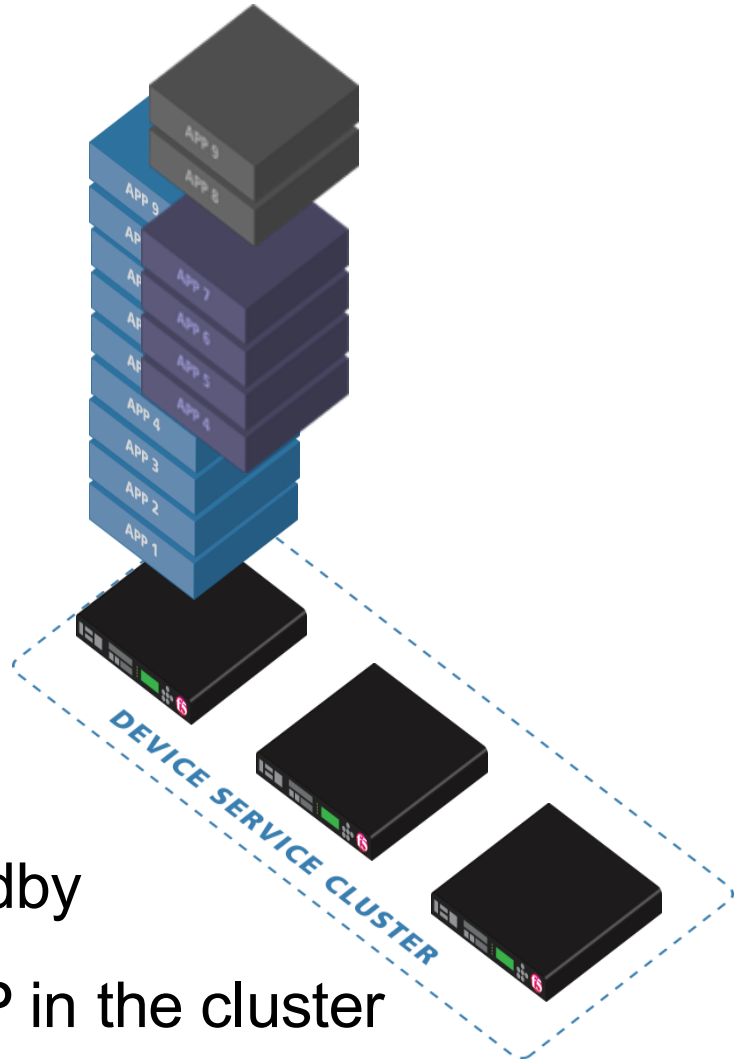
(tmos)# run sys failover

```
offline  Changes the status of a unit or cluster to
         Forced Offline. If persist or no-persist are
         not specified, the change in status will be
         persisted in-between system restarts.

online   Changes the status of a unit or cluster from
         Forced Offline to either Active or Standby,
         depending upon the status of the other unit
         or cluster in a redundant pair.

standby Specifies that the active unit or cluster
         fails over to a Standby state, causing the
         standby unit or cluster to become Active.
```



 ©2026 F5

Traffic Groups

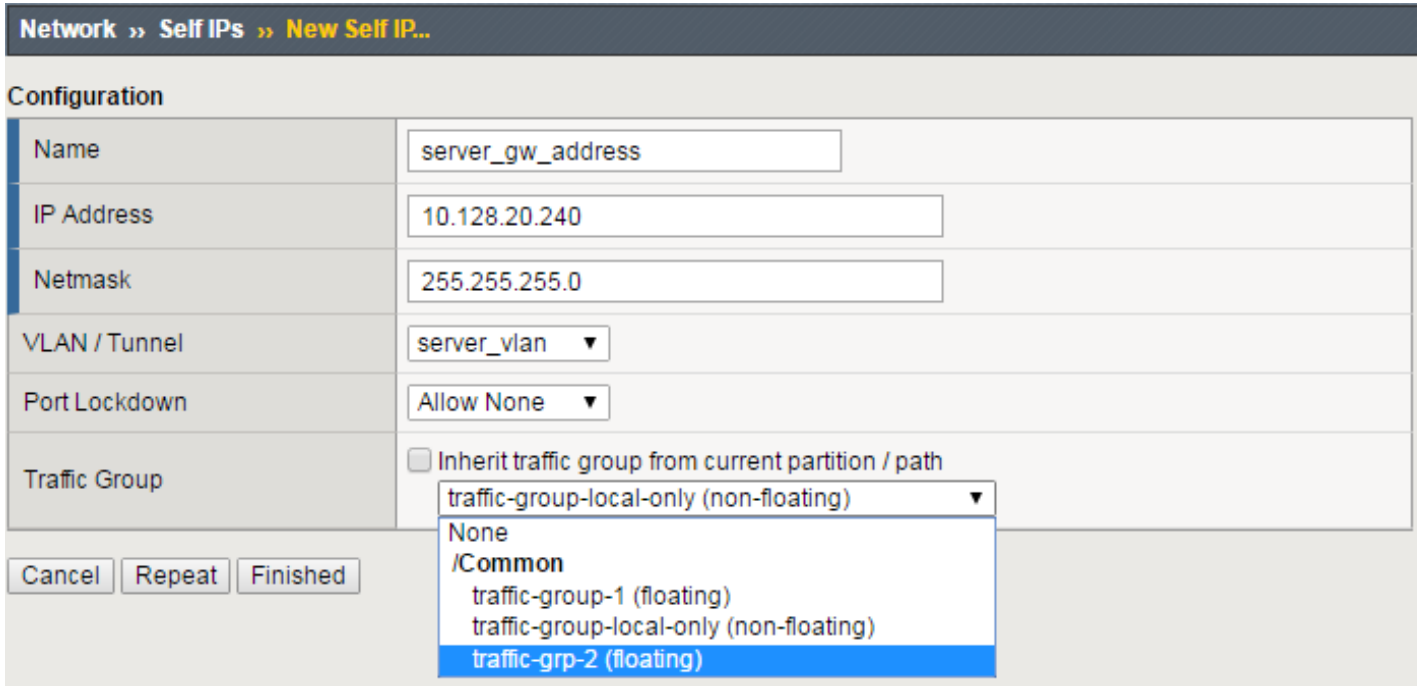


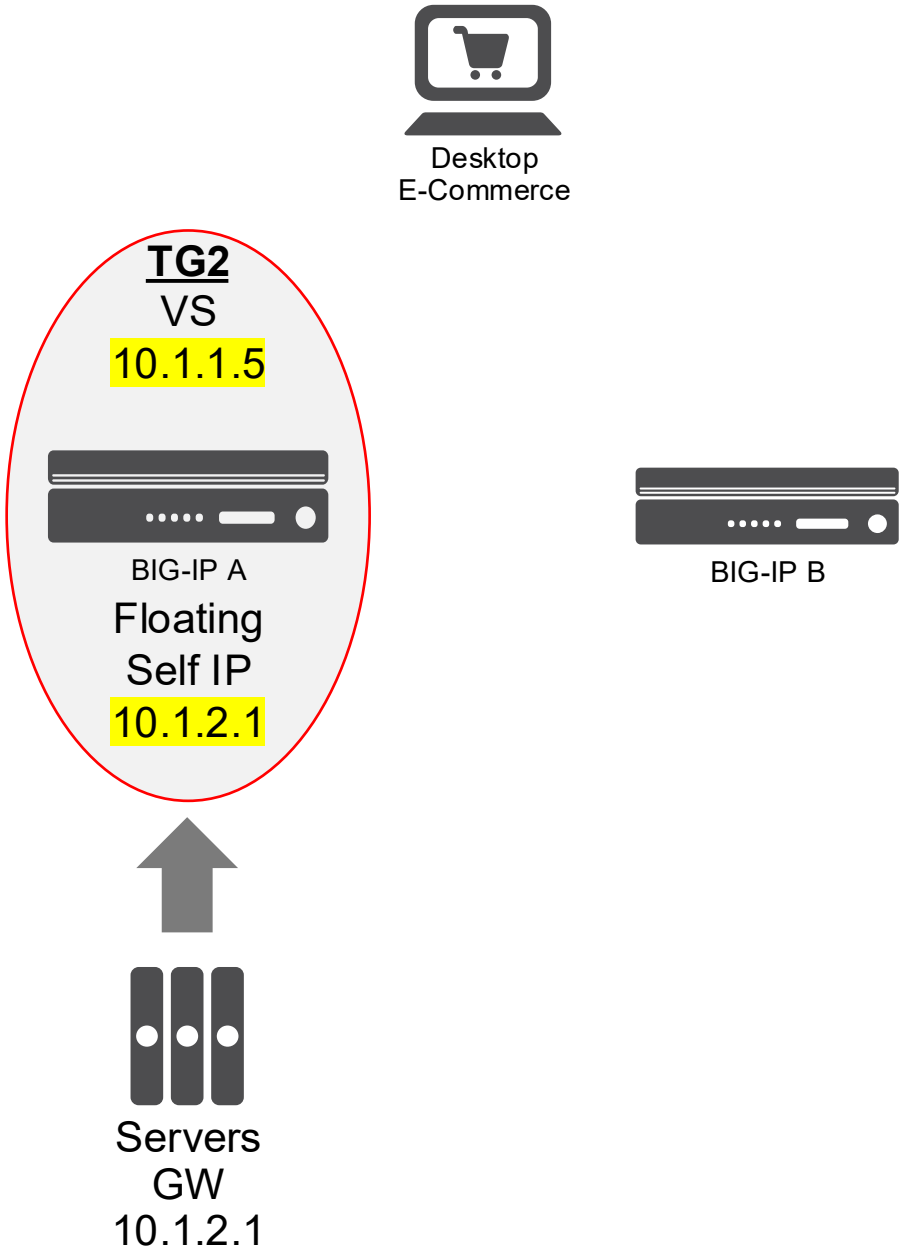
- A collection of listeners to failover
- Create traffic groups and assign applications to the group
- Activate traffic groups on cluster members
- If a cluster member has no active traffic groups it is in standby
- If a device fails, the traffic group migrates to another BIG-IP in the cluster

 ©2026 F5

The all important Floating Self IP

- Self IP addresses that need to move on failover to ensure application access
  - The server's default gateway is the BIG-IP





 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

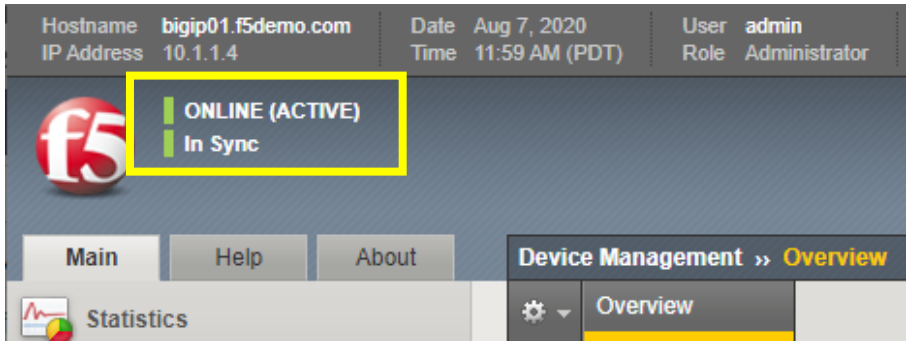
Explain config sync

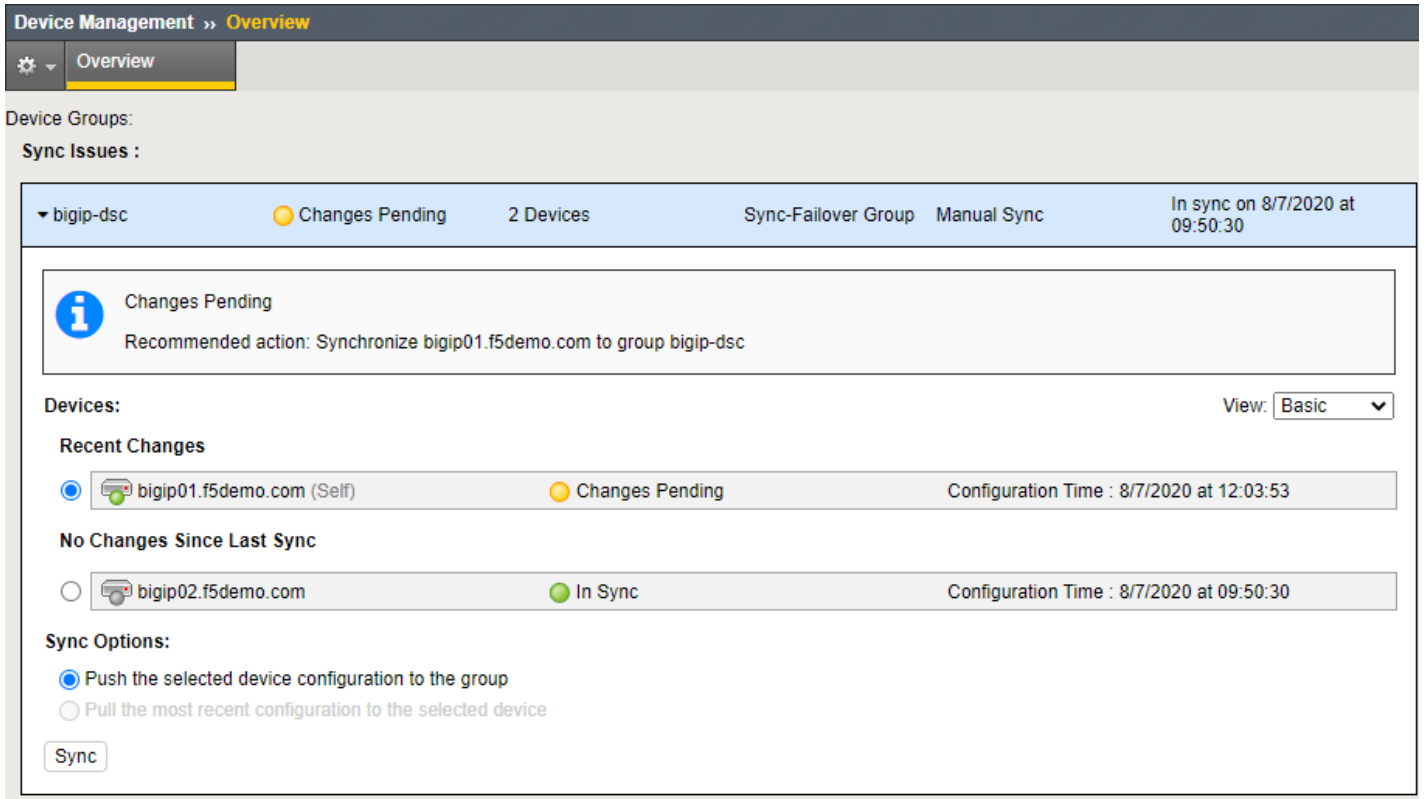
- Show config sync status
- Explain when a config sync is necessary
- Compare configuration timestamp
- Demonstrate config sync procedure
- Report errors which occur during config sync

 ©2026 F5

Show config sync status

[Manual Chapter : Managing Configuration Synchronization]





- By default, syncing a configuration is a manual process

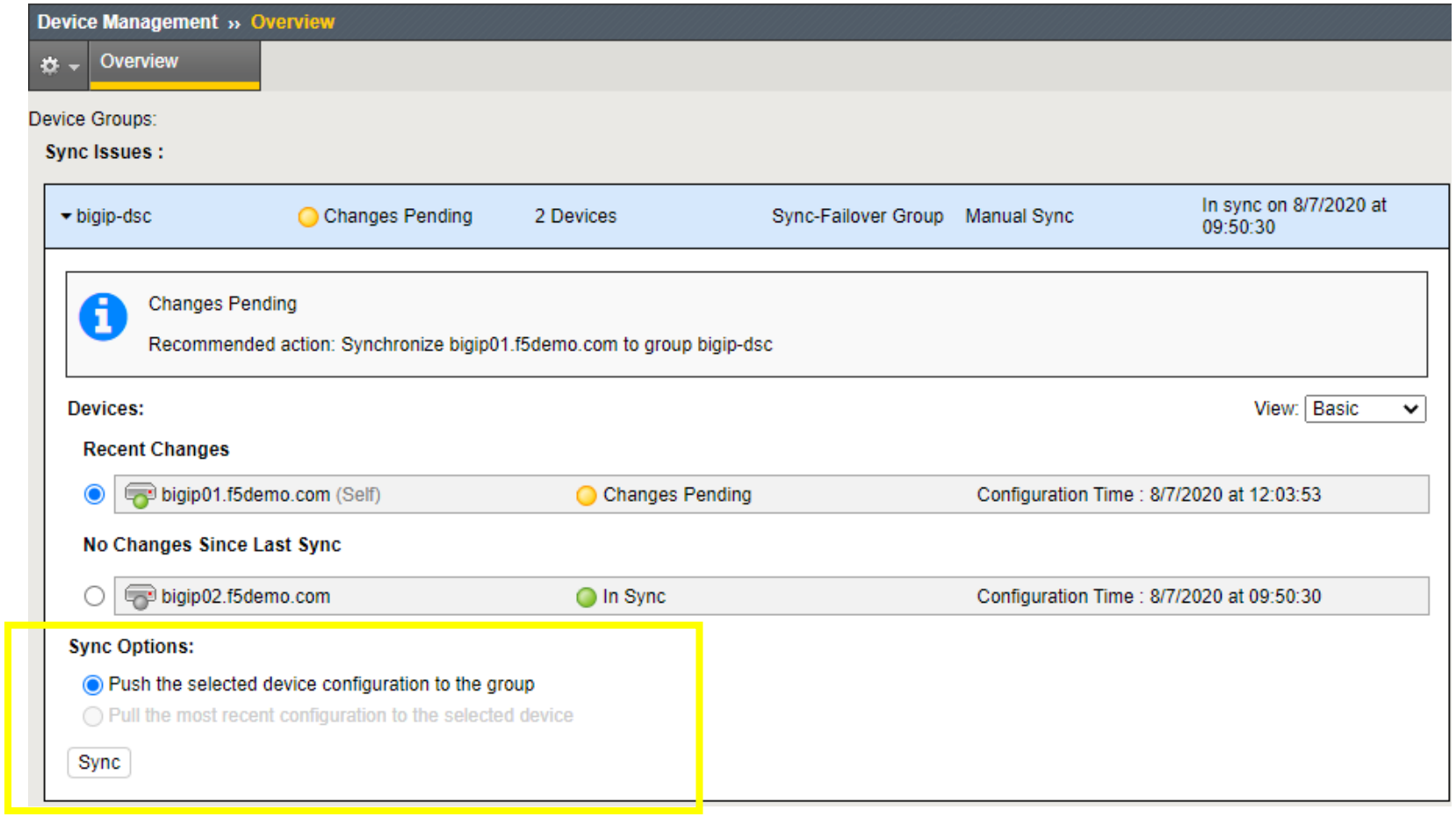
[root@bigip01:Active:**Changes Pending**] config #

 ©2026 F5

Demonstrate config sync procedure (GUI)

Manual Chapter : Managing Configuration Synchronization

- You can Push or Pull a configsync
  - You may want a pull if you make changes you regret



 ©2026 F5

Demonstrate config sync procedure (TMSH)

K14856: Performing a ConfigSync using tmsh

- F5 YouTube: Performing a ConfigSync using tmsh ~1min
- run /cm config-sync <sync_direction> <sync_group>
- <sync_direction>

```
 force-full-load-push    Sync configuration to the specified device group even if
                         the system would deem this unsafe. This may result in
                         loss of configuration on other devices.

  from-group             Sync configuration from specified device group.

  recover-sync           Resets the local device configuration and restores trust
                         domain, device, and device-group information to default
                         settings.

  to-group               Sync configuration to specified device group.
```

 ©2026 F5

Labs 3-6

 ©2026 F5

F5CAB1: BIG-IP Administration Install, Initial Configuration, and Upgrade

Identify which modules are licensed and/or provisioned

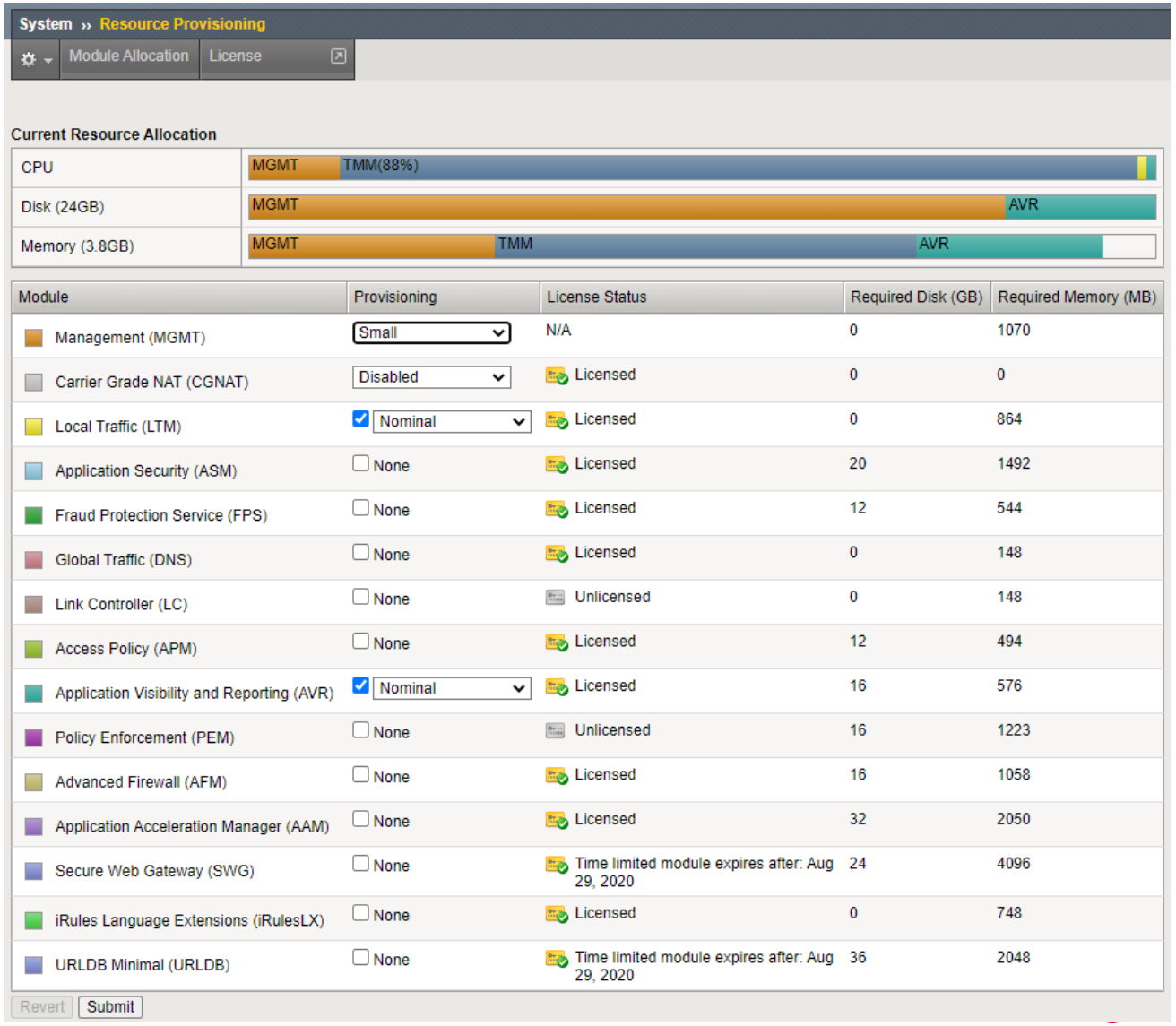
- Show provisioned modules
- Report modules which are licensed
- Report modules which are provisioned but not licensed
- Show resource utilization of provisioned modules

 ©2026 F5

Show provisioned modules

- The Resource Provisioning page
  - Shows licensed modules
  - Show subscriptions license and expiration
  - Show provisioned modules

A module must be Licensed and Provisioned to process traffic.



Module	Provisioning	License Status	Required Disk (GB)	Required Memory (MB)
Management (MGMT)	Small	N/A	0	1070
Carrier Grade NAT (CGNAT)	Disabled	Licensed	0	0
Local Traffic (LTM)	Nominal	Licensed	0	864
Application Security (ASM)	None	Licensed	20	1492
Fraud Protection Service (FPS)	None	Licensed	12	544
Global Traffic (DNS)	None	Licensed	0	148
Link Controller (LC)	None	Unlicensed	0	148
Access Policy (APM)	None	Licensed	12	494
Application Visibility and Reporting (AVR)	Nominal	Licensed	16	576
Policy Enforcement (PEM)	None	Unlicensed	16	1223
Advanced Firewall (AFM)	None	Licensed	16	1058
Application Acceleration Manager (AAM)	None	Licensed	32	2050
Secure Web Gateway (SWG)	None	Time limited module expires after: Aug 29, 2020	24	4096
iRules Language Extensions (iRulesLX)	None	Licensed	0	748
URLDB Minimal (URLDB)	None	Time limited module expires after: Aug 29, 2020	36	2048

 ©2026 F5

F5CAB1: BIG-IP Administration Install, Initial Configuration, and Upgrade

Explain the processes of licensing, license reactivation, and license modification

- Show where to license (activate.F5.com)
- Identify license issues
- Identify Service Check Date (upgrade)

 ©2026 F5

Identify Service Check Date (upgrade)
bigip.license

```
[root@bigip01:Active:Standalone] config # cat /config/bigip.license
#
#       Licensing Information
#
Licensed date :                     20260330
License start :                     20260329
License end :                       20260430
Service check date :                20260304
#
#       Platform Information
#
Registration Key :          L8490-48363-55503-18604-3374990
Licensed version :                 17.5.1
```

 ©2026 F5

Identify Service Check Date (upgrade)
CLI: show sys license

```
[root@bigip01:Active:Standalone] config # tmsh
root@(bigip01)(cfg-sync Standalone)(Active)(/Common)(tmos)# show sys license

Sys::License
Licensed Version                        17.5.1
Registration key                        L8490-48363-55503-18604-3374990
Licensed On                             2026/03/30
License Start Date                      2026/03/29
License End Date                        2026/04/30
Service Check Date                      2026/03/04
Platform ID                             Z100k
Daily Renewal Notification Days         5
Daily Renewal Notification Start Date   2026/04/25
Permitted Versions                      5.*.* - 18.*.*

Active Modules
  APM, Base, VE GBB (500 CCU) (M523127-2218598)
    Anti-Virus Checks
    Base Endpoint Security Checks
    Firewall Checks
    Machine Certificate Checks
    Network Access ...
```

 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

Explain authentication methods

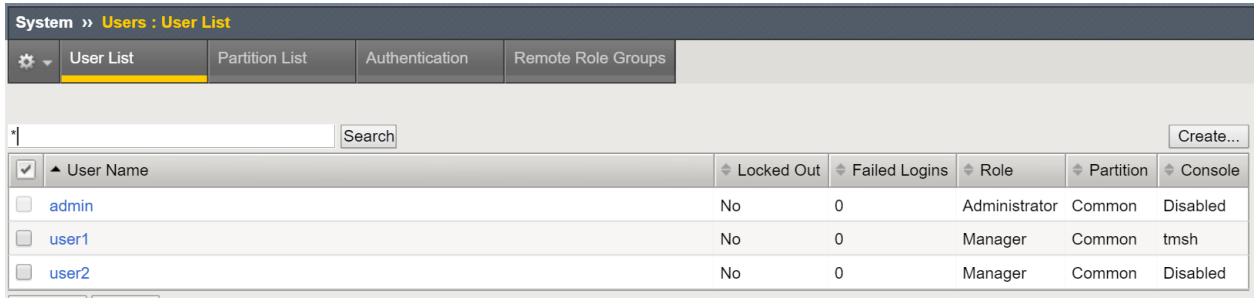
- Explain how to create a user
- Explain how to modify user properties
- Explain options for remote authentication provider
- Explain use of groups using remote authentication provider

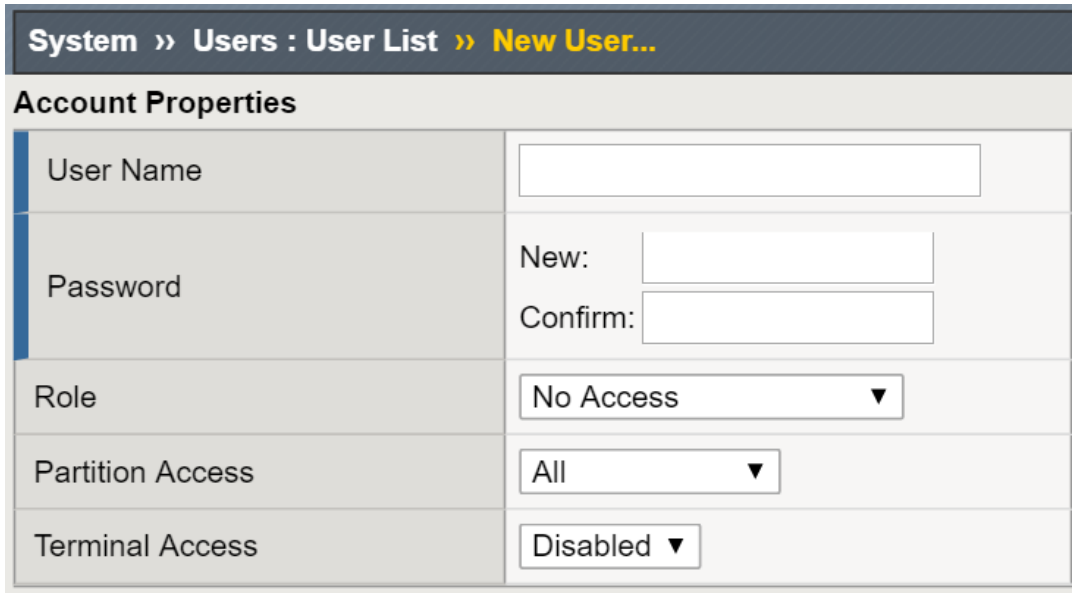
 ©2026 F5

Explain how to create a user

Manual : BIG-IP Systems: User Account Administration

- User and Password are required
- Assign a role
- Assign partition access
  - A user may be assigned to one partition or All partitions
- Assign the type of terminal access (Specify the type of CLI access)
  - Disabled
    - The user may access only the GUI interface
  - TMSH
    - Permits the user access to the TMOS CLI shell via SSH
  - Advanced Shell
    - Permits user access to the Linux prompt
    - Administrator and Resource Administrator only





 ©2026 F5

User Roles (most common)

Manual : BIG-IP Systems: User Account Administration

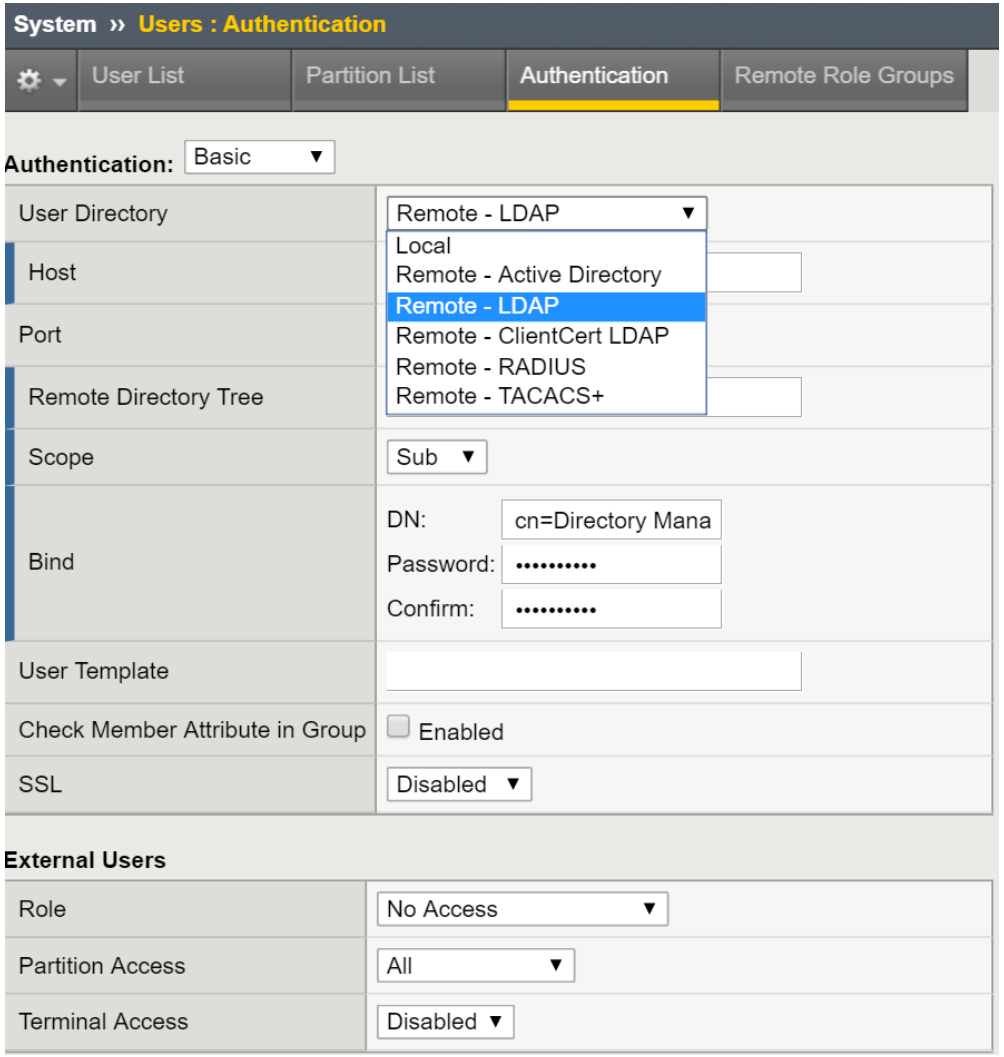
- **No Access**
  - Prevents users from accessing the system. Basically turns off the account without deleting the account.
- **Guest**
  - Grants users limited, view-only access to a specific set of objects.
- **Operator**
  - Grants users permission to enable or disable existing nodes and pool members. Cannot enable/disable virtual servers.
- **Application Editor**
  - Grants users permission to modify existing nodes, pools, pool members, and monitors.
- **Manager**
  - Permission to create, modify, and delete virtual servers, pools, pool members, nodes, custom profiles, custom monitors, and iRules.
- **Resource Administrator**
  - Grants users complete access to all objects on the system, except access to create/modify users (except for themselves)
- **Administrator**
  - Grants users complete access to all objects on the system.

 ©2026 F5

Explain options for remote authentication provider

Manual : BIG-IP Systems: User Account Administration

- Still will always need a least one admin local account
  - For config sync functionality
  - In case you lose access to authentication server
- Supports AD, LDAP, TACACS+ and RADIUS



 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

Apply procedural concepts required to create, manage, and restore a UCS archive

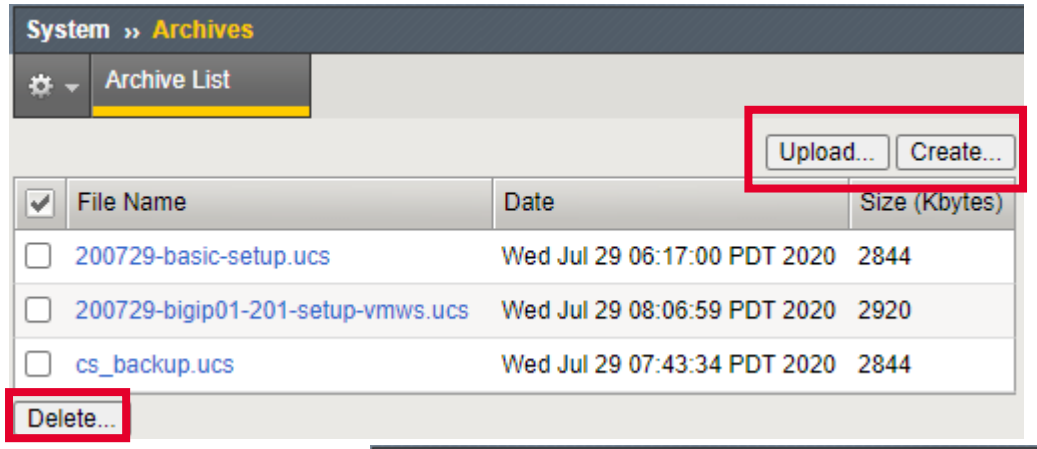
- Summarize the use case of a UCS backup
- Execute UCS backup procedure
- Execute UCS restore procedure
- Explain proper long-term storage of UCS backup file
- Explain the contents of the UCS file (private keys)

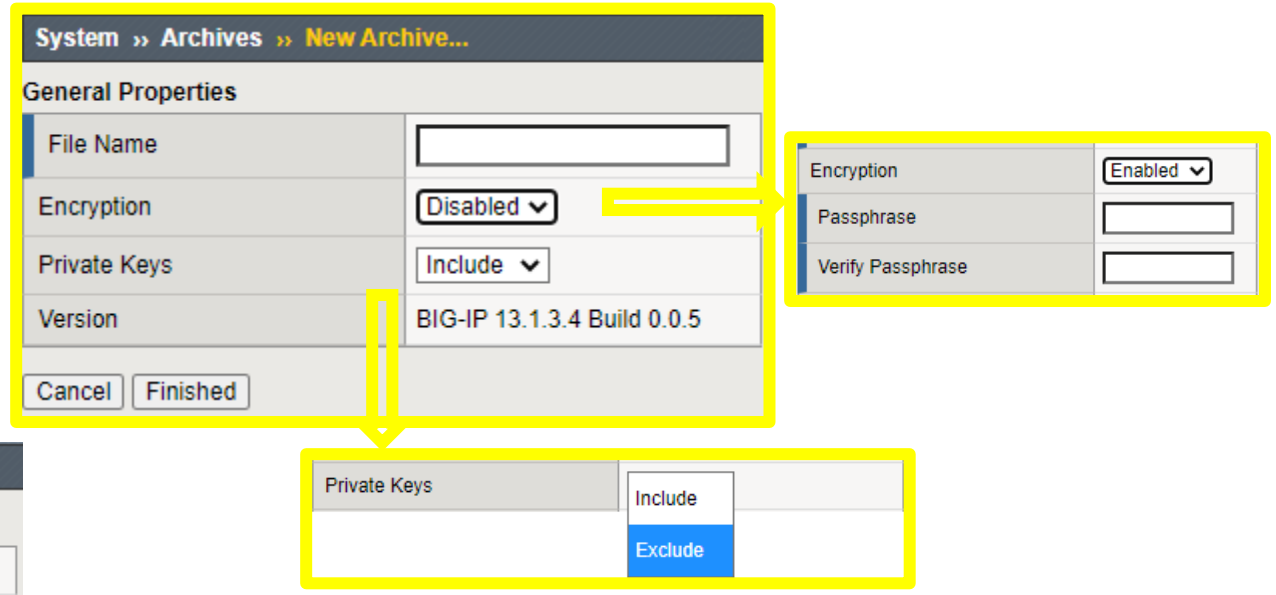
 ©2026 F5

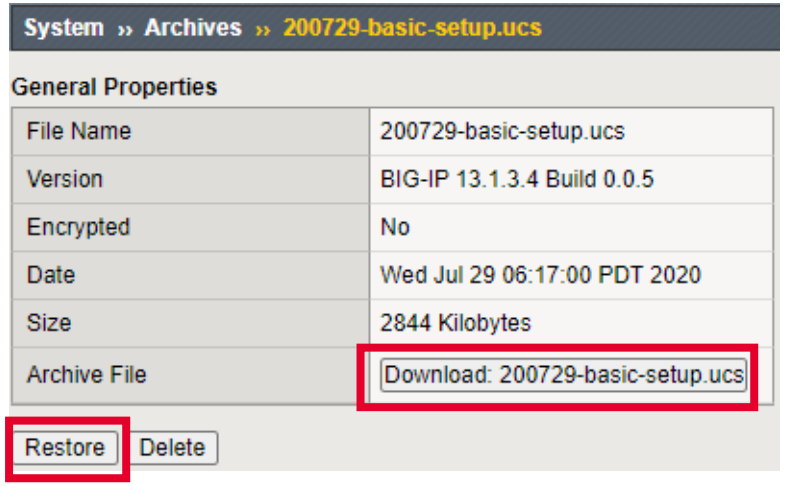
Execute UCS backup and restore procedure

K13132: BACKING UP AND RESTORING BIG-IP CONFIGURATION FILES WITH A UCS ARCHIVE

You can create, delete, restore, upload and download UCS archives from the GUI interface:







 ©2026 F5

Execute UCS backup and restore procedure

Manual Chapter : Archives

- You can also create, delete and restore UCS backups using TMSH, but TMSH has options the GUI doesn't.
  - Backup the BIG-IP: save sys ucs <ucs filename>
  - Restore the BIG-IP: load sys ucs <ucs filename>
- If you are restoring an RMA or migrating to a new platform you do NOT want to restore the license.
  - load sys ucs <filename> **no-license**
  - If you are migrating platforms, you may not want to restore the base configurations as interfaces may be different.
  - On the system you are restoring you would build the base first, interfaces, VLANs, self IPs, etc
  - load sys ucs **platform-migrate** <filename> **no-license**
- Other TMSH options
  - no-platform-check Bypass platform check.
  - passphrase Passphrase for (un)encrypting UCS.
  - reset-trust Reset device and trust domain certificates and keys when loading a UCS.

 ©2026 F5

F5CAB1: BIG-IP Administration Install, Initial Configuration, and Upgrade

Apply procedural concepts required to manage software images

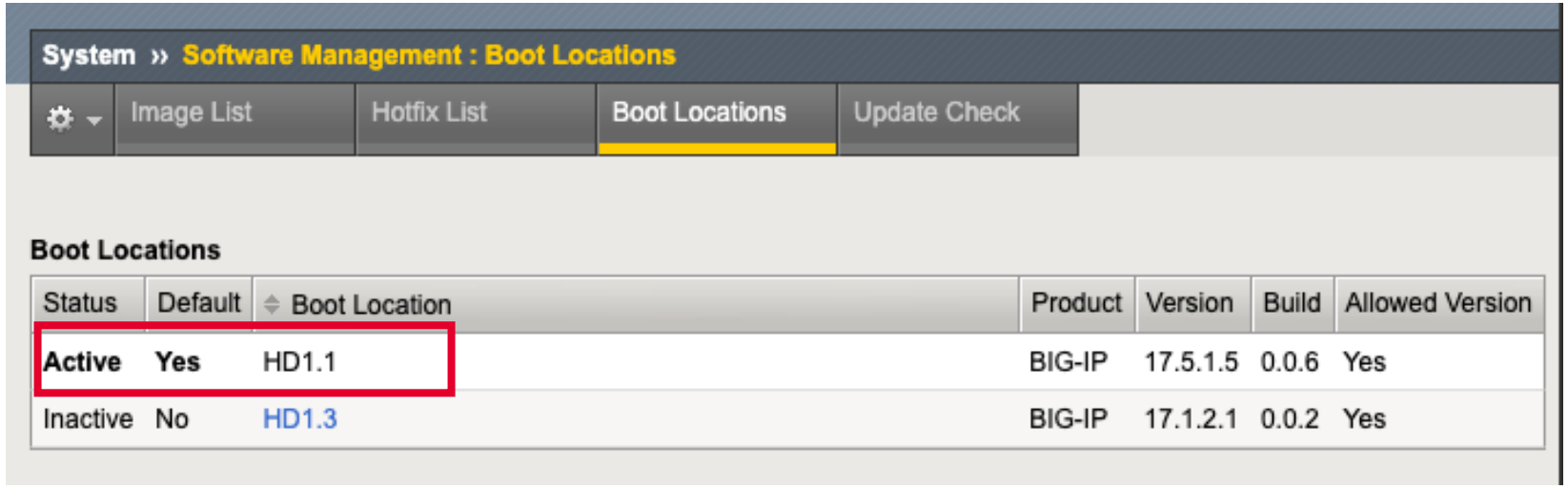
- Given an HA pair, describe the appropriate strategy for deploying a new software image
- Perform procedure to upload new software image
- Show currently configured boot location
- Demonstrate creating new volume for software images

 ©2026 F5

Show currently configured boot location

```
(tmos)# show sys software
-------------------------------------------------------------------
Sys::Software Status
Volume  Product   Version  Build  Active    Status  Allowed Version
-------------------------------------------------------------------
HD1.1    BIG-IP  17.5.1.5  0.0.6     yes  complete             yes
HD1.3    BIG-IP  17.1.2.1  0.0.2      no  complete             yes
```

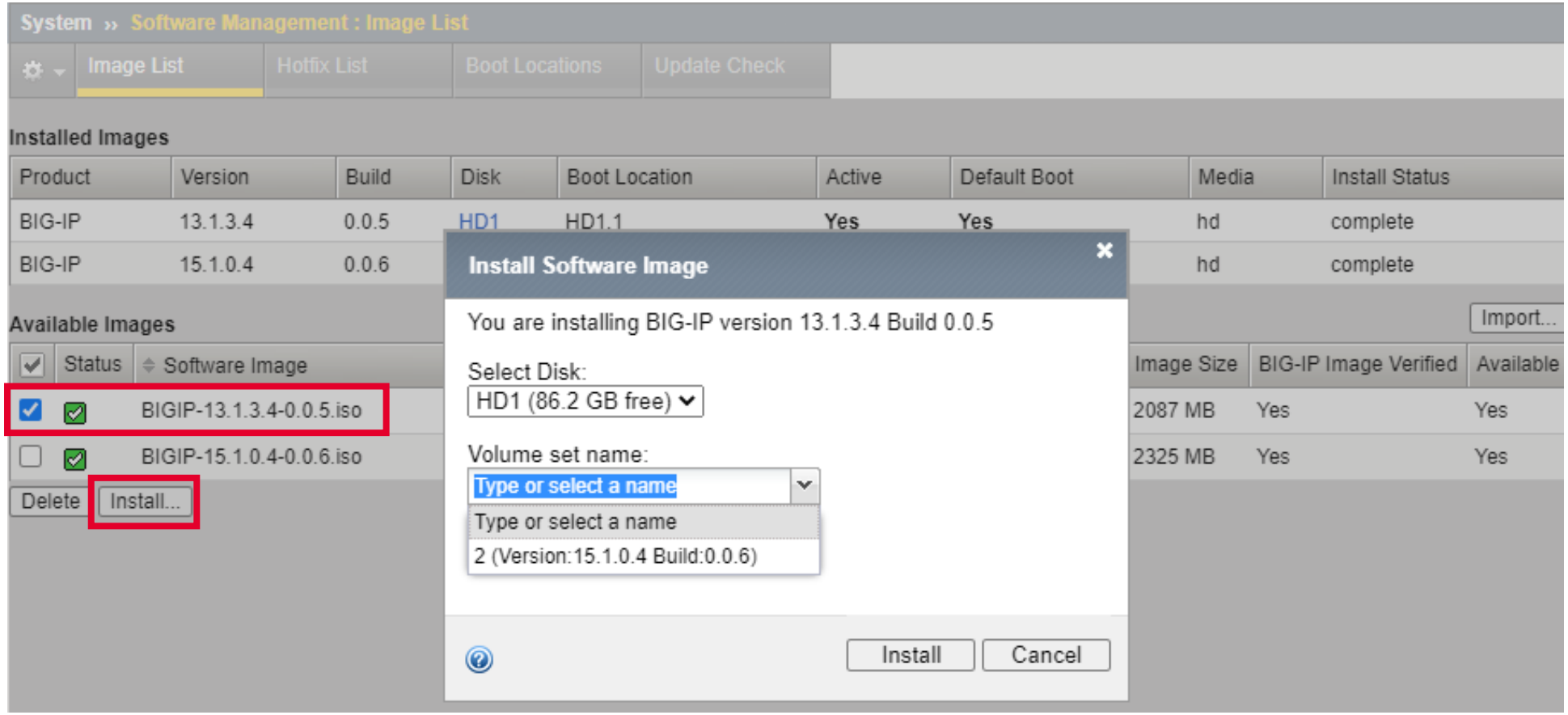
```
----------------------------
Sys::Software Update Check
----------------------------
  Check Enabled          true
  Phonehome Enabled      true
  Frequency            weekly
  Status              failure
  Errors                    8
```



 ©2026 F5

Demonstrate creating new volume for software images

install sys software image <iso> volume <name>



 ©2026 F5

F5CAB5: BIG-IP Administration Support and Troubleshooting

Determine resource utilization

- Distinguish between control plane and data plane resources
- Identify CPU statistics per virtual server
- Interpret Statistics for interfaces
- Determine Disk utilization and Memory utilization

 ©2026 F5

Distinguish between control plane and data plane resources

Overview of F5 BIG-IP TMOS Routing

- Control Plane
- Linux OS
  - Hardened CentOS
  - Use to boot HW/SW
  - Runs TMSH CLI and APIs
  - Runs Out-of-Band Management
    - By default, uses DHCP
    - IP address can be assigned manually
    - Unique IP subnet and default gateway

- Data Plane
- TMOS (Traffic Management OS)
  - aka TMM
  - Runs TMM switch interface
  - L3 Switching and Routing
    - VLANs, Self IPs, Routing for TMM
  - Pools and Virtual Servers
  - Monitors
  - basically all things related to Local Traffic Management and application security.

 ©2026 F5

Identify CPU statistics per virtual server

Statistics » Module Statistics : Local Traffic » Virtual Servers

Traffic Summary | DNS | Local Traffic | Subscriber Management | Network | Memory | System

Display Options

Statistics Type	Virtual Servers
Data Format	Normalized
Auto Refresh	Disabled Refresh

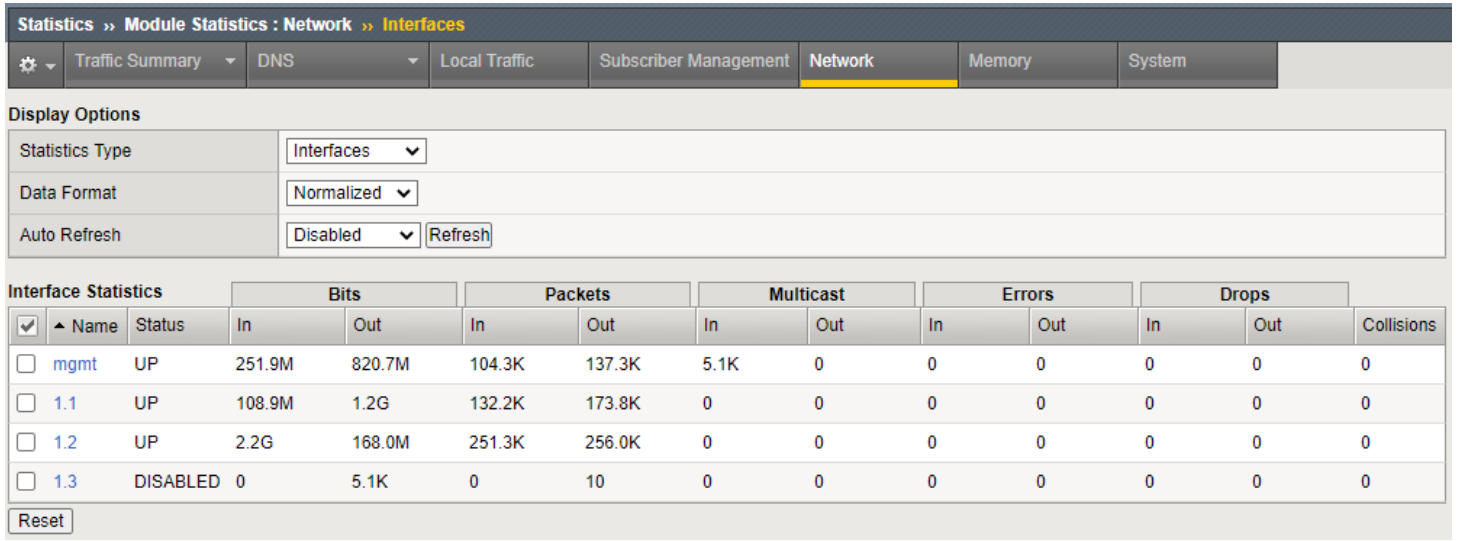
* Search

				Bits		Packets		Connections			Requests	CPU Utilization Avg.		
Status	Virtual Server	Partition / Path	Details	In	Out	In	Out	Current	Maximum	Total	Total	5 Sec.	1 Min.	5 Min.
	ftp_vs	Common	View...	41.7K	105.6K	91	107	1	2	10	0	0%	0%	0%
	hackazon-redirect	Common	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	hackazon-vs	Common	View...	1.6M	26.5M	2.9K	3.6K	3	7	31	0	0%	0%	0%
	purple_vs	Common	View...	0	0	0	0	0	0	0	0	0%	0%	0%
	www_vs	Common	View...	4.9M	40.5M	3.7K	5.5K	8	14	27	0	0%	0%	0%

Reset

 ©2026 F5

Interpret Statistics for interfaces



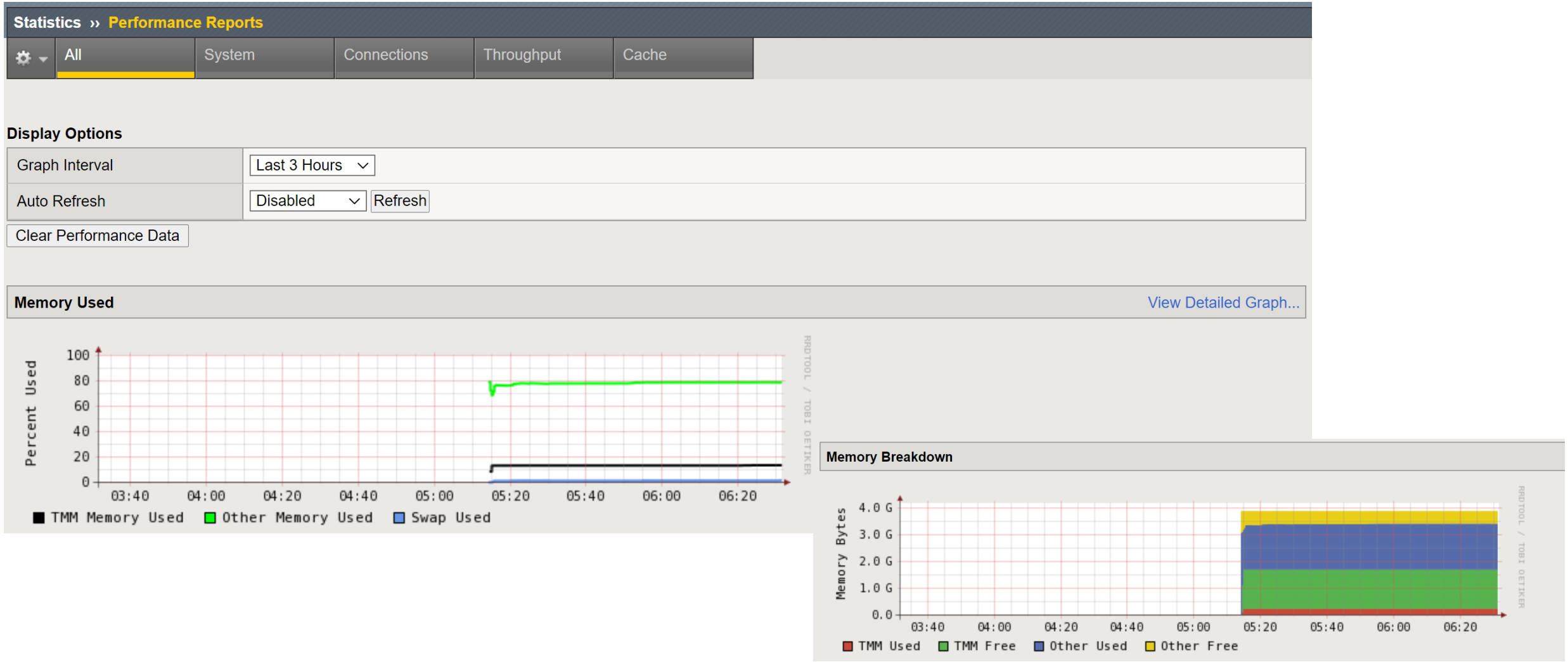
Name	Status	Bits In	Bits Out	Packets In	Packets Out	Multicast In	Multicast Out	Errors In	Errors Out	Drops In	Drops Out	Collisions
mgmt	UP	251.9M	820.7M	104.3K	137.3K	5.1K	0	0	0	0	0	0
1.1	UP	108.9M	1.2G	132.2K	173.8K	0	0	0	0	0	0	0
1.2	UP	2.2G	168.0M	251.3K	256.0K	0	0	0	0	0	0	0
1.3	DISABLED	0	5.1K	0	10	0	0	0	0	0	0	0

- Errors – number of packets containing errors
- Drops – number of packets drop for processing or packet errors
- Collisions – should only occur on half-duplex links (not common)

```
(tmos)# show net interface
-----------------------------------------------------------------
Net::Interface
Name     Status     Bits     Bits     Pkts     Pkts  Drops  Errs      Media
                      In      Out       In      Out
-----------------------------------------------------------------
1.1          up   111.4M     1.3G   136.1K   178.7K      0     0   10000T-FD
1.2          up     2.2G   170.3M   256.0K   260.3K      0     0   10000T-FD
1.3    disabled        0     5.1K        0       10      0     0        none
mgmt         up   254.3M   831.2M   105.4K   139.0K      0     0    100TX-FD
```

 ©2026 F5

Determine Disk utilization and Memory utilization



 ©2026 F5

Determine Disk utilization and Memory utilization

Statistics » Module Statistics : Memory

Traffic Summary | DNS | Local Traffic | Subscriber Management | Network | Memory | System

Display Options

Data Format	Normalized
Auto Refresh	Disabled Refresh

System Memory	Total	Used	Free	Percent Used
TMM	1.6G	232.7M	1.4G	13.5%
Other	2.1G	1.7G	471.9M	78.8%
Total	3.8G	1.9G	1.9G	50.3%
Swap	999.9M	14.2M	985.7M	1.4%

Memory Pool Name	Allocated	Max Allocated	Object Size
ADM Mitigation	0	0	1
ADM Statistics	0	0	1
APMD proxy	0	0	1
Application Family Name	2.0M	2.0M	1
Application filter	408.0K	408.0K	1
BIGTCP PKTSEG cache	0	0	48

 ©2026 F5

Determine Disk utilization and Memory utilization

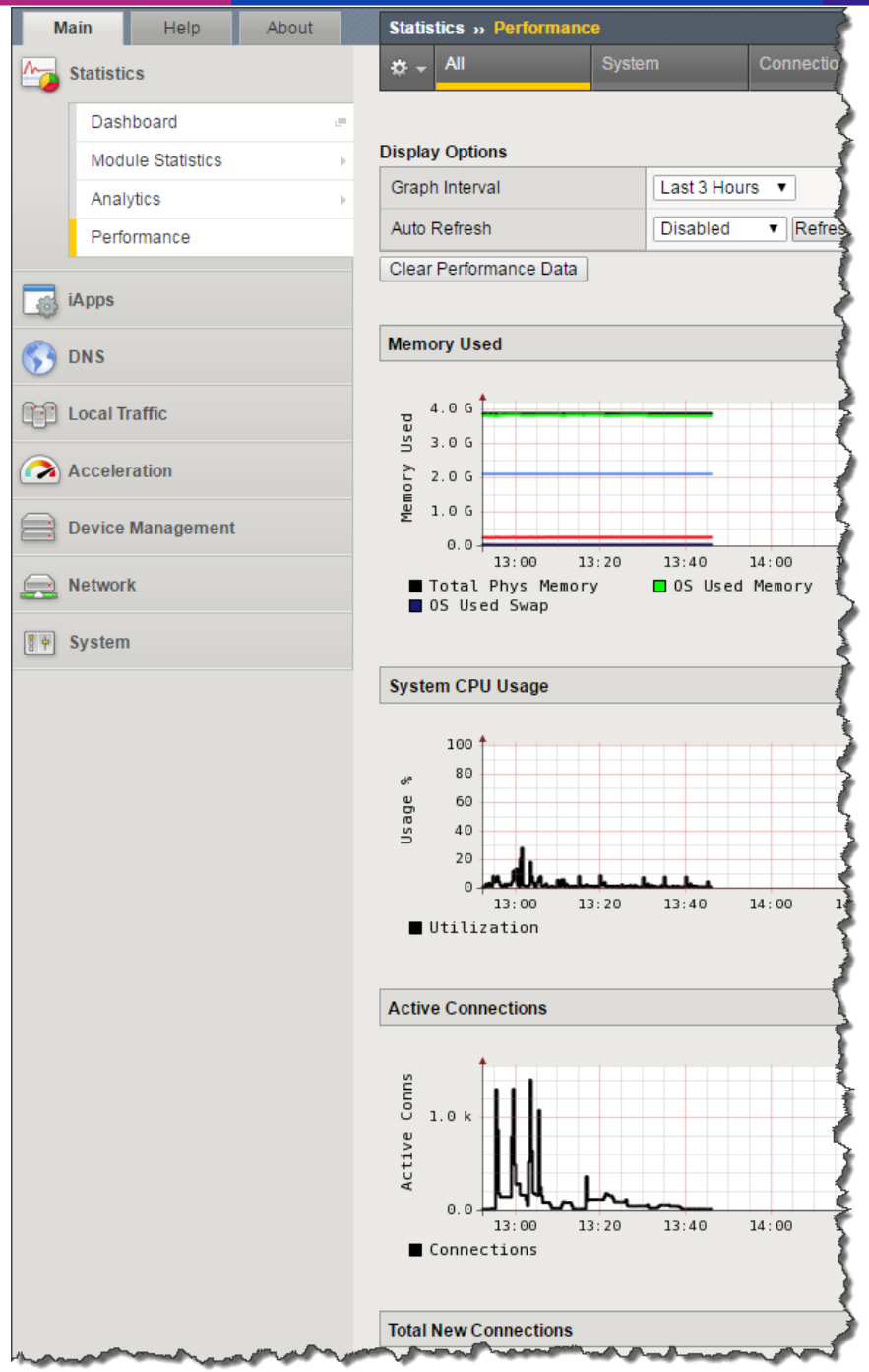
K14403: Maintaining disk space on the BIG-IP system

```
[root@bigip01:Active:Disconnected] config # df -h
Filesystem            Size  Used Avail Use% Mounted on
/dev/mapper/vg--db--vda-set.1.root
                      427M  274M  131M  68% /
none                  3.9G  2.3M  3.9G   1% /dev/shm
/dev/mapper/vg--db--vda-set.1._config
                      3.2G   87M  2.9G   3% /config
/dev/mapper/vg--db--vda-set.1._usr
                      4.0G  3.2G  655M  83% /usr
/dev/mapper/vg--db--vda-set.1._var
                      3.0G  792M  2.1G  28% /var
/dev/mapper/vg--db--vda-dat.share
                       20G  306M   19G   2% /shared
/dev/mapper/vg--db--vda-dat.log
                      2.9G  106M  2.7G   4% /var/log
/dev/mapper/vg--db--vda-dat.appdata
                       25G  190M   24G   1% /appdata
none                  3.9G   35M  3.9G   1% /shared/rrd.1.2
none                  3.9G   16M  3.9G   1% /var/tmstat
none                  3.9G  1.6M  3.9G   1% /var/run
prompt                4.0M   28K  4.0M   1% /var/prompt
none                  3.9G     0  3.9G   0% /var/loipc
```

 ©2026 F5

Performance Statistics

- On the Statistics >> Performance page you can find:
  - Memory Used
  - System CPU Usage
  - Active Connections and Total New Connections
  - Throughput - (bits) and (packets)
  - TMM Client-side and Server-side Throughput
  - HTTP Requests
  - RAM Cache Utilization
  - SSL Transactions
  - And more …….
- In TMSH, `show /sys performance all-stats`



 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

List which log files could be used to find events and/or hardware issues

- Identify use of /var/log/ltm, var/log/secure, /var/log/audit
- Identify severity log level of an event
- Identify event from a log message

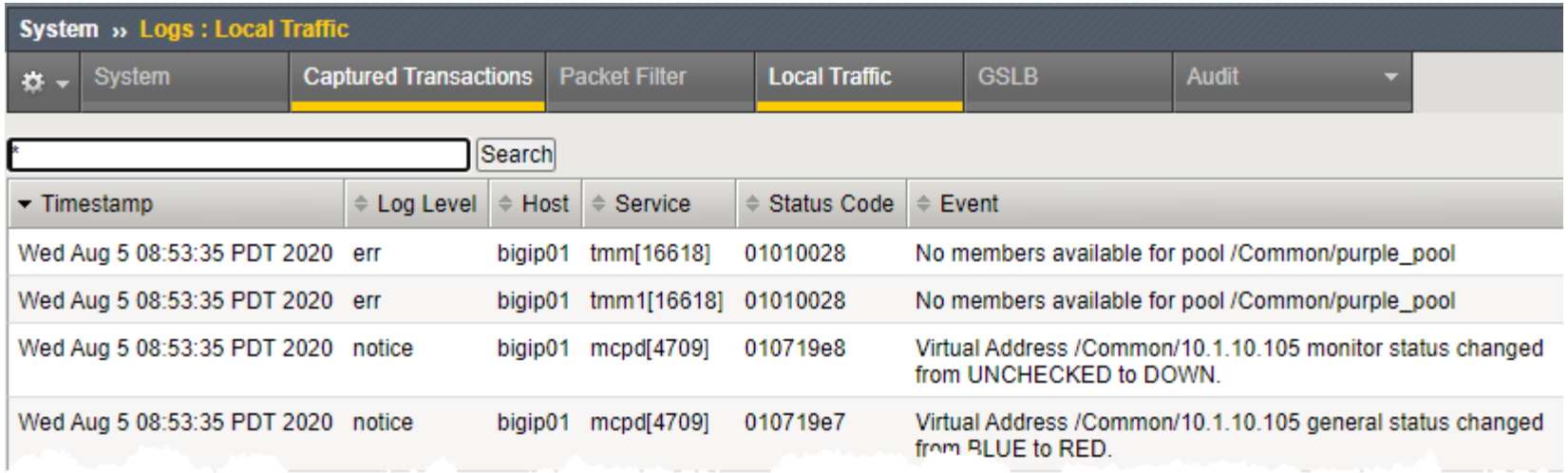
 ©2026 F5

Identify use of /var/log/ltm, var/log/secure, /var/log/audit

Manual Chapter : About Logging

K16197: Reviewing BIG-IP log files

- /var/log/ltm
  - The local traffic messages pertain specifically to the BIG-IP local traffic management events
  - Can be found in the GUI under System >> Logs >> Local Traffic
  - In TMSH: `show sys log ltm`
  - In bash: `cat /var/log/ltm`



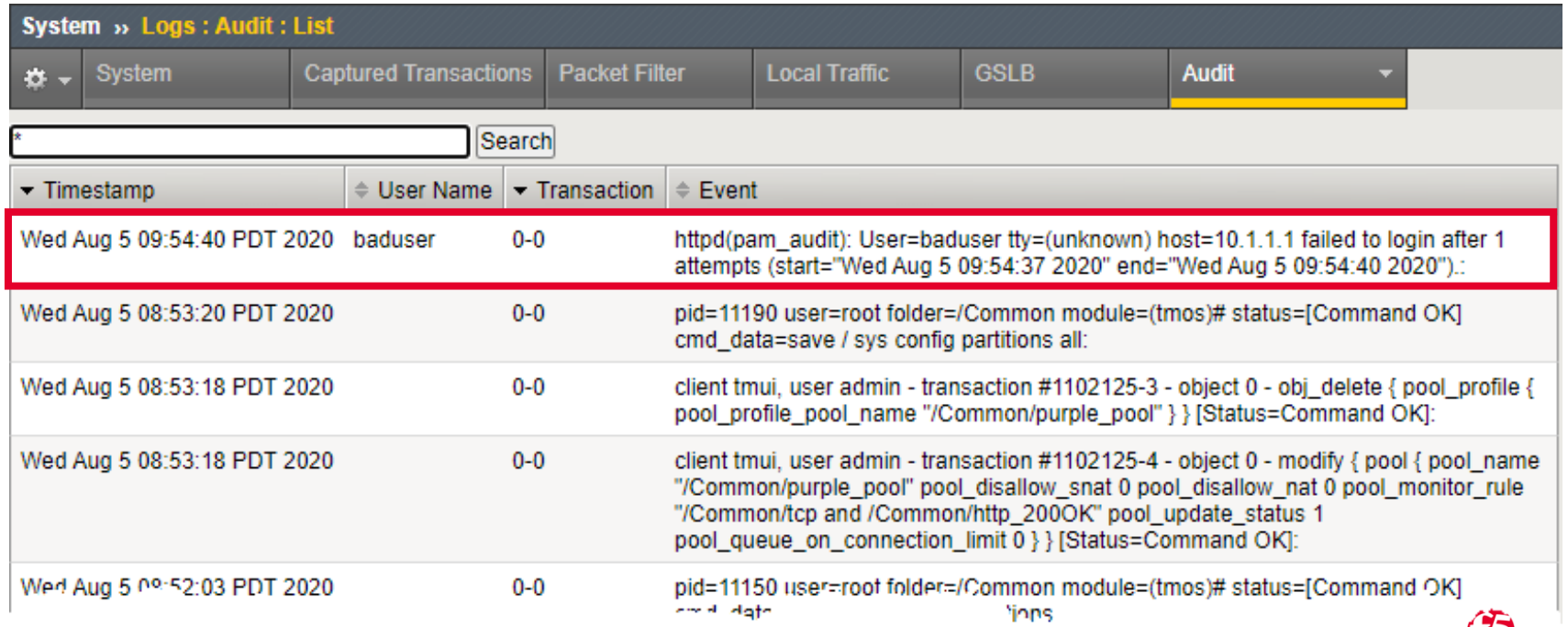
Timestamp	Log Level	Host	Service	Status Code	Event
Wed Aug 5 08:53:35 PDT 2020	err	bigip01	tmm[16618]	01010028	No members available for pool /Common/purple_pool
Wed Aug 5 08:53:35 PDT 2020	err	bigip01	tmm1[16618]	01010028	No members available for pool /Common/purple_pool
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	010719e8	Virtual Address /Common/10.1.10.105 monitor status changed from UNCHECKED to DOWN.
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	010719e7	Virtual Address /Common/10.1.10.105 general status changed from BLUE to RED.

 ©2026 F5

Identify use of /var/log/ltm, var/log/secure, /var/log/audit

Auditing User Access

- /var/log/secure
  - Log information related to authentication and authorization privileges.
  - Can be found in the GUI under System >> Logs >> Audit
  - In TMSH, `show sys log secure`
  - In Bash, `cat /var/log/secure`



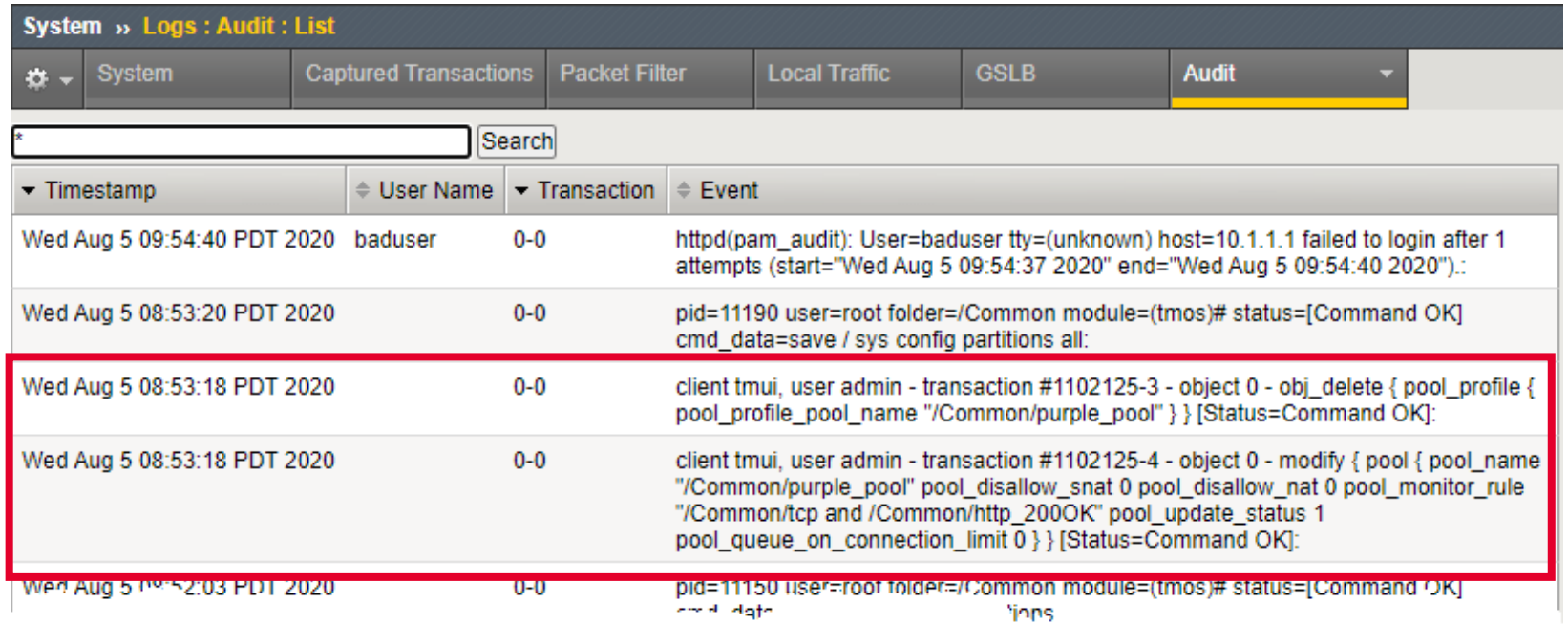
 ©2026 F5

Identify use of /var/log/ltm, var/log/secure, /var/log/audit

Manual Chapter : About Logging

K16197: Reviewing BIG-IP log files

- /var/log/audit
  - Log changes to the BIG-IP system configuration. Logging audit events is optional.
  - Can be found in the GUI under System >> Logs >> Audit
    - In TMSH, `show sys log audit`
    - In Bash, `cat /var/log/audit`



 ©2026 F5

Identify event from a log message

Local Traffic

Timestamp	Log Level	Host	Service	Status Code	Event	
Wed Aug 5 08:53:35 PDT 2020	err	bigip01	tmm[16618]	01010028	No members available for pool /Common/purple_pool	2
Wed Aug 5 08:53:35 PDT 2020	err	bigip01	tmm1[16618]	01010028	No members available for pool /Common/purple_pool	
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	010719e8	Virtual Address /Common/10.1.10.105 monitor status changed from UNCHECKED to DOWN.	3
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	010719e7	Virtual Address /Common/10.1.10.105 general status changed from BLUE to RED.	
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	01071682	SNMP_TRAP: Virtual /Common/purple_vs has become unavailable	
Wed Aug 5 08:53:35 PDT 2020	notice	bigip01	mcpd[4709]	01070638	Pool /Common/purple_pool member /Common/10.1.20.14:80 monitor status down. [/Common/tcp: up, /Common/http_200OK: down; last error: /Common/http_200OK: No successful responses received before deadline. @2020/07/29 07:44:53.] [was up for 0hr:1min:34sec]	1

Audit

Wed Aug 5 08:53:18 PDT 2020	0-0	client tmui, user admin - transaction #1102125-4 - object 0 - modify { pool { pool_name "/Common/purple_pool" pool_disallow_snat 0 pool_disallow_nat 0 pool_monitor_rule "/Common/tcp and /Common/http_200OK" pool_update_status 1 pool_queue_on_connection_limit 0 } } [Status=Command OK]:	4

 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

Identify configured system services

- Show proper configuration for: DNS, NTP, SNMP, syslog

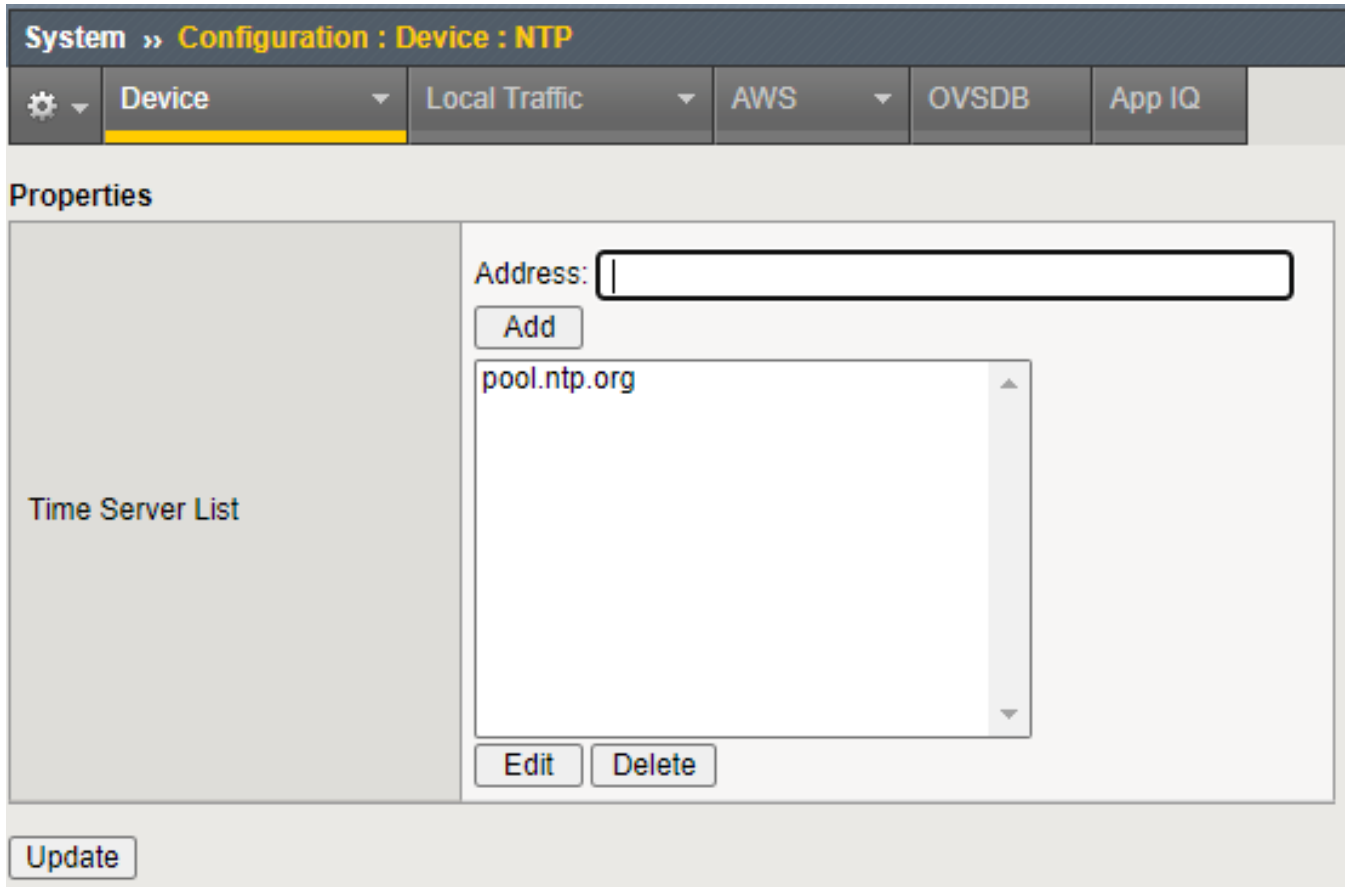
 ©2026 F5

Show proper configuration for: DNS, NTP, SNMP, syslog

Manual Chapter : General Configuration Properties

K13380: Configuring the BIG-IP system to use an NTP server from the command line

- NTP is essential for:
  - Device Service Clusters
  - Configsync
  - Logging

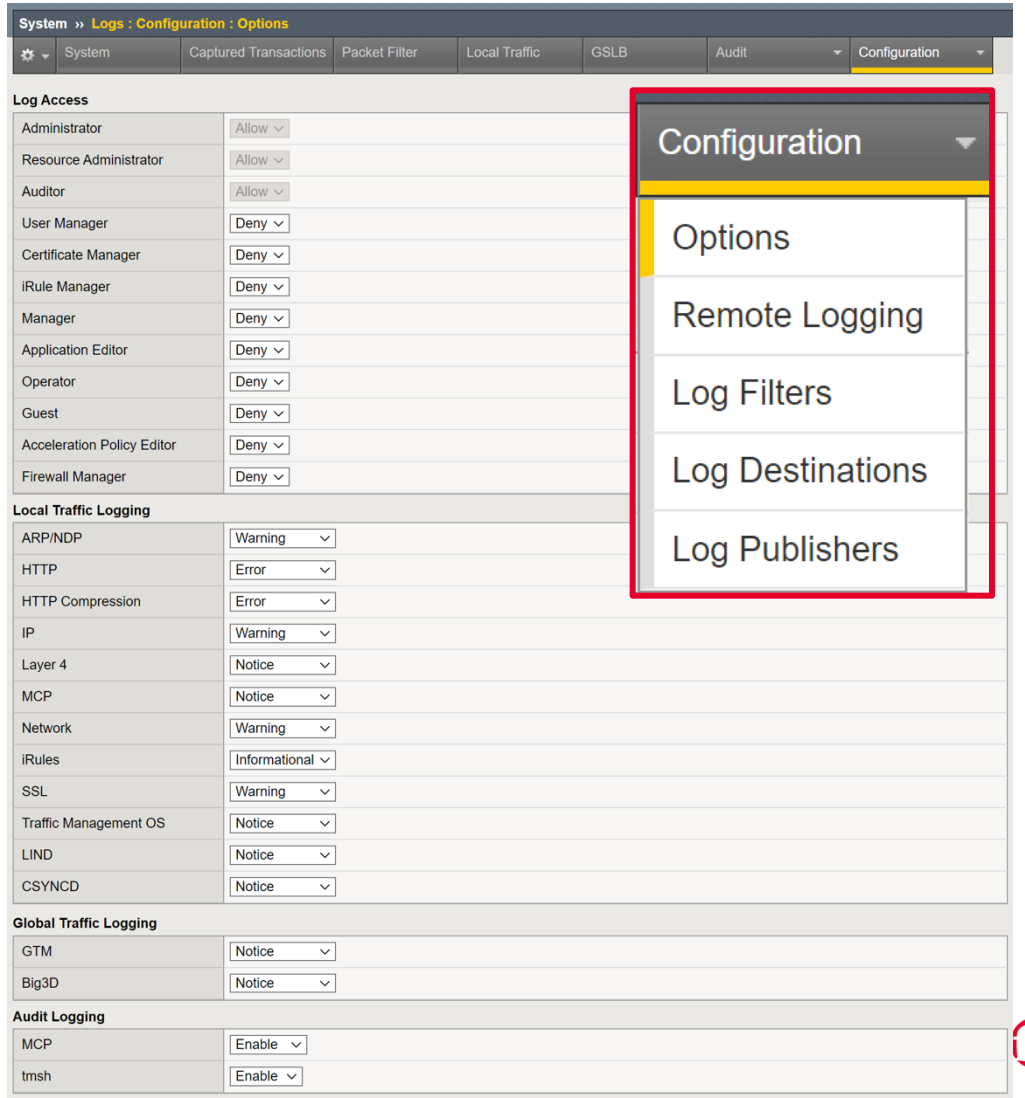


 ©2026 F5

Show proper configuration for: DNS, NTP, SNMP, syslog

Manual Chapter : About Logging

- Log Destinations
  - The High-Speed Logging (HSL) or Unformatted destination
  - Defines the protocol to use (UDP or TCP)
  - Defines the server pool the log message will go too
- The Formatted destination defines the format of the messages being sent
  - There are two parts to a Destination
    - Where a message is going : HSL Destination
    - What the message looks like: Formatted Destination
- Publisher
  - A Publisher is a collection of Formatted Destinations



 ©2026 F5

Tools for Testing – DNS, NTP, SNMP, SYSLOG

- DNS
  - You should know to use and interpret the results of the `dig` utility
- NTP
  - K10240: Verifying NTP peer server communications
- SNMP
  - There is a test snmp button on the configuration page
- Good old tcpdump
- Show services
  - `tmsh show service <service>` or `tmsh show service` (shows all services)
  - From the linux prompt: **bigstart status**
    - This will show you the status of the various daemons the BIG-IP uses.

 ©2026 F5

F5CAB4 : BIG-IP Administration Control Plane Administration

Identify and report current device status

- Interpret the LCD panel warning messages
- Use the dashboard to gauge the current running status of the system
- Review the Network Map in order to determine the status of objects
- Interpret current systems status via GUI or TMSH
- Interpret high availability and device trust status

 ©2026 F5

Interpret the LCD panel warning messages

K15521451: BIG-IP TMOS operations guide | Chapter 12: Log files and alerts

- /etc/alertd/alert.conf – contains the LCD error message

LCD Warning: Critical: 9d Blocking Dos Attack

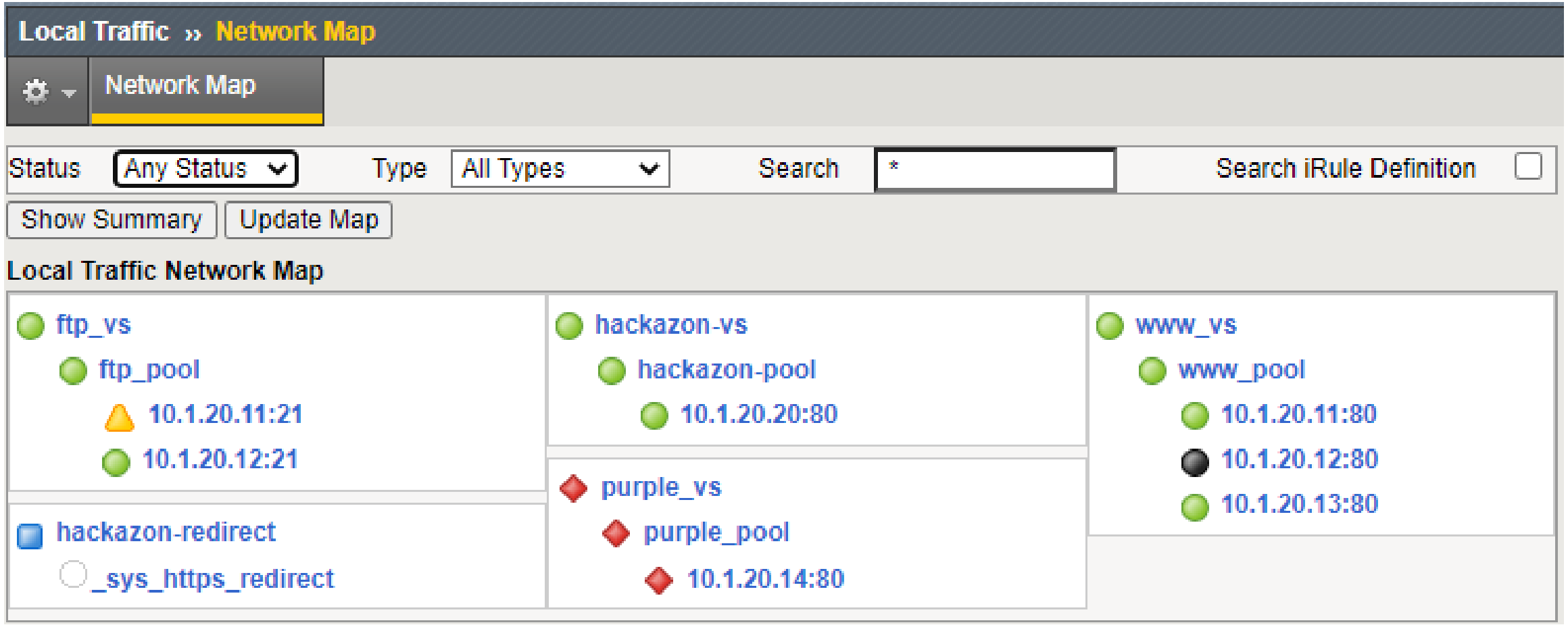
Local Traffic Log: sweeper_update: aggressive mode activated. 372313/438016 pages

https://support.f5.com/kb/en-us/products/big-ip_ltm/manuals/product/platform-b5000/2.html?sr=54998935

 ©2026 F5

Review the Network Map in order to determine the status of objects

REVIEW

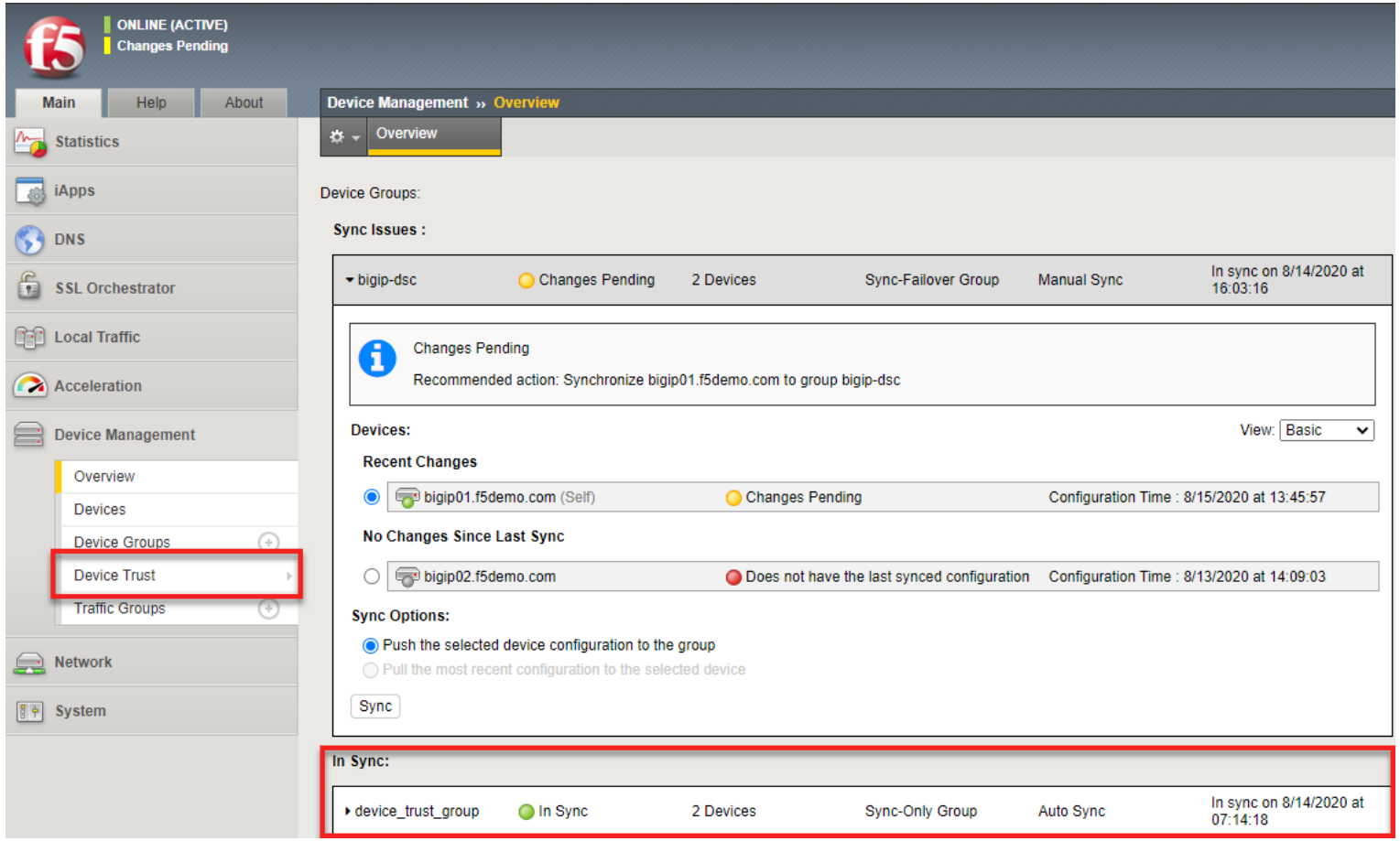


 ©2026 F5

Interpret high availability and device trust status

Manual : BIG-IP Device Service Clustering: Administration

- To create secure communications between BIG-IPs in a HA configuration (Device Service Cluster – DSC) they are place into a Device Trust Group:
  - BIG-IP exchanges device certificates
  - If a certificate expires the trust is broken
  - The device_trust_group must be in sync for configsync, mirroring and network failover to be available.
- More on HA later…



 ©2026 F5

F5CAB1: BIG-IP Administration Install, Initial Configuration, and Upgrade

Identify management connectivity configurations

- Identify the configured management-IP address
- Show remote connectivity to the BIG-IP Management interface
- Explain management IP connectivity issue
- Interpret port lockdown settings to Self-IP
- Identify HTTP/SSH access list to management-IP address

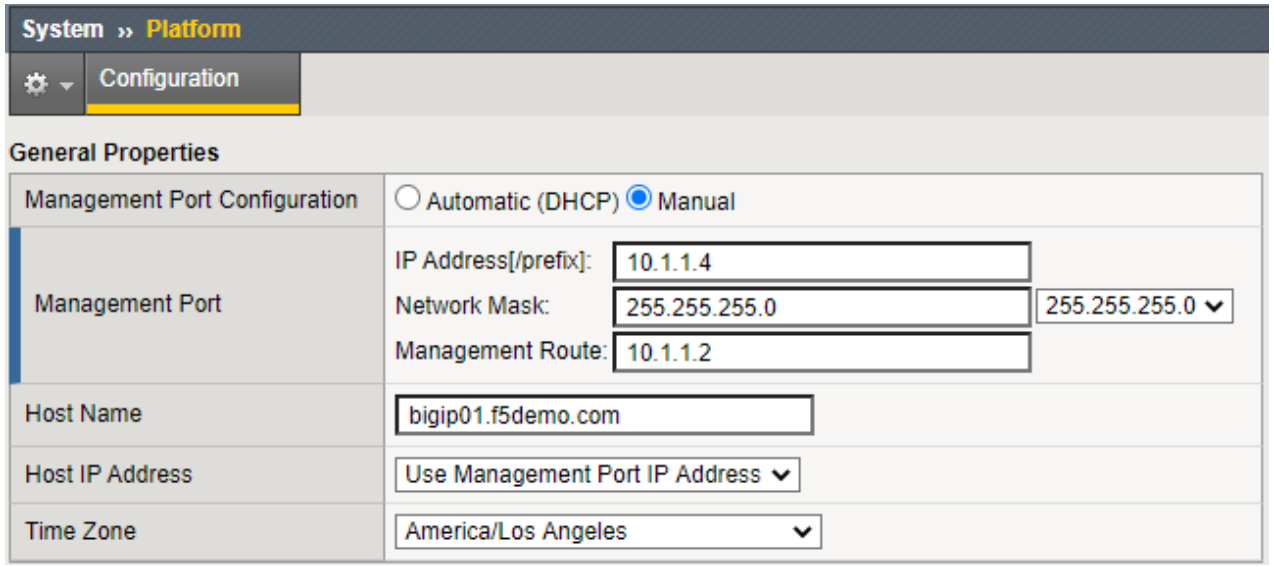
 ©2026 F5

Identify the configured management-IP address

K15040: Configuring and displaying the management IP address for the BIG-IP system

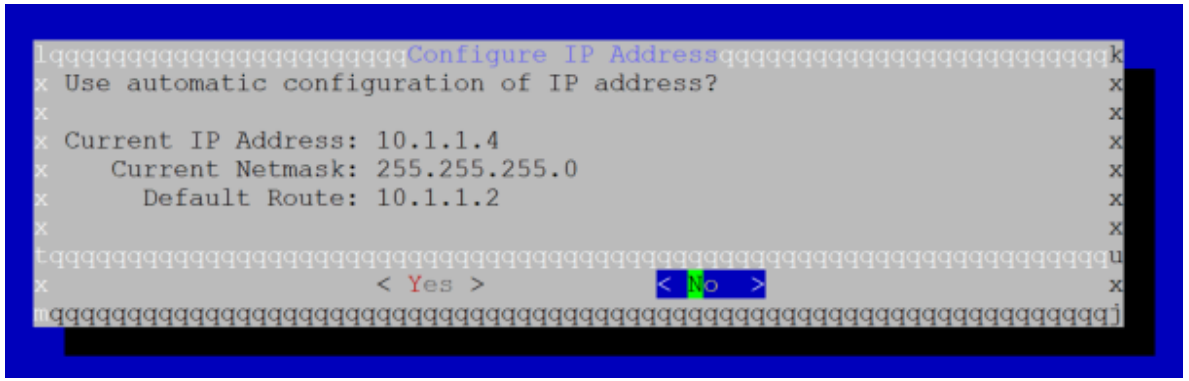
K7312: Overview of the management interface (port)

GUI



TMSH

```
tmos)# list sys management-ip
sys management-ip 10.1.1.4/24 {
    description configured-statically
}
```

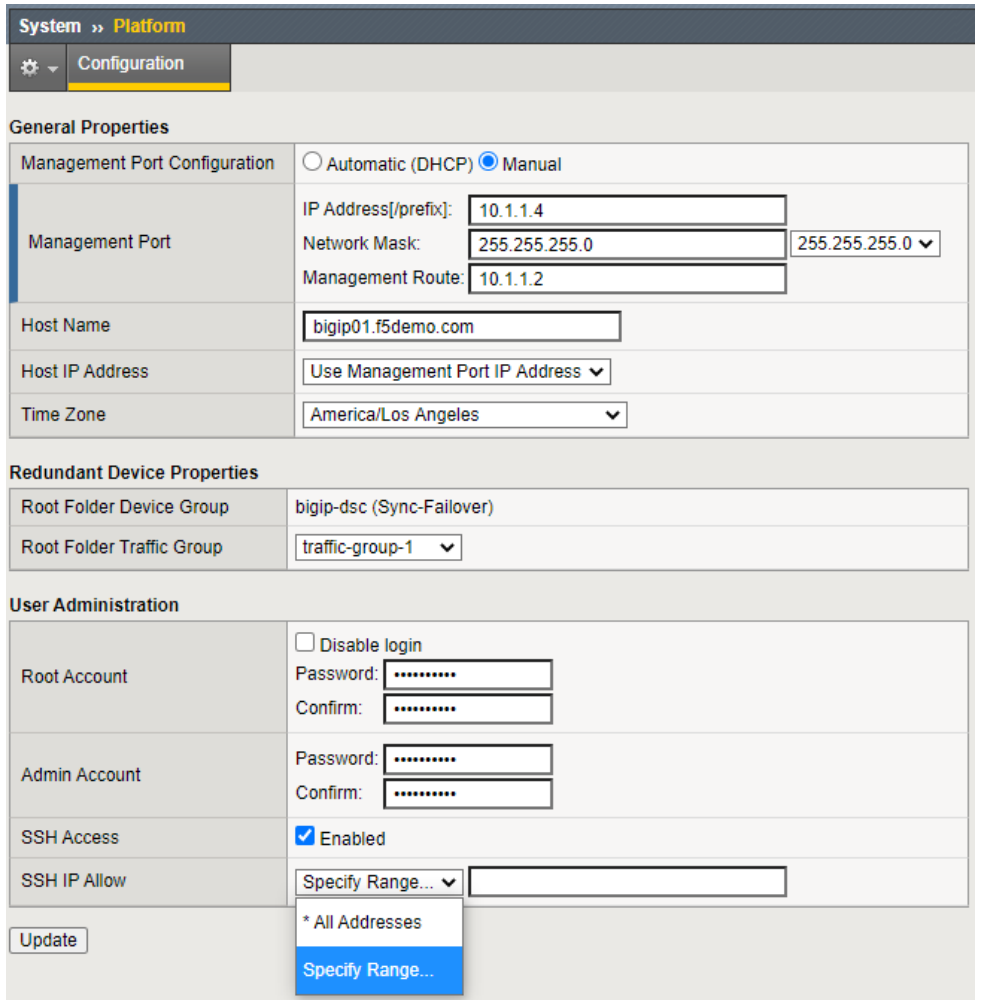


“config” utility at the linux prompt

 ©2026 F5

Identify SSH access list to management-IP address

K13309: Restricting access to the Configuration utility by source IP address



- To add to the allow list:
  - modify /sys sshd allow add { <IP address or IP address range> }
- To replace the list
  - modify /sys sshd replace-all-with {<IP address or IP address range>}
- Default is:

```
(tmos)# list sys sshd allow
sys sshd {
    allow { All }
}
```

- Save the change by entering the following command:
  - save /sys config

 ©2026 F5

Identify HTTP access list to management-IP address

K13309: Restricting access to the Configuration utility by source IP address

- To add to the allow list:
  - modify /sys httpd allow add { <IP address or IP address range> }
- To replace the list
  - modify /sys httpd replace-all-with {<IP address or IP address range>}
- Default is:

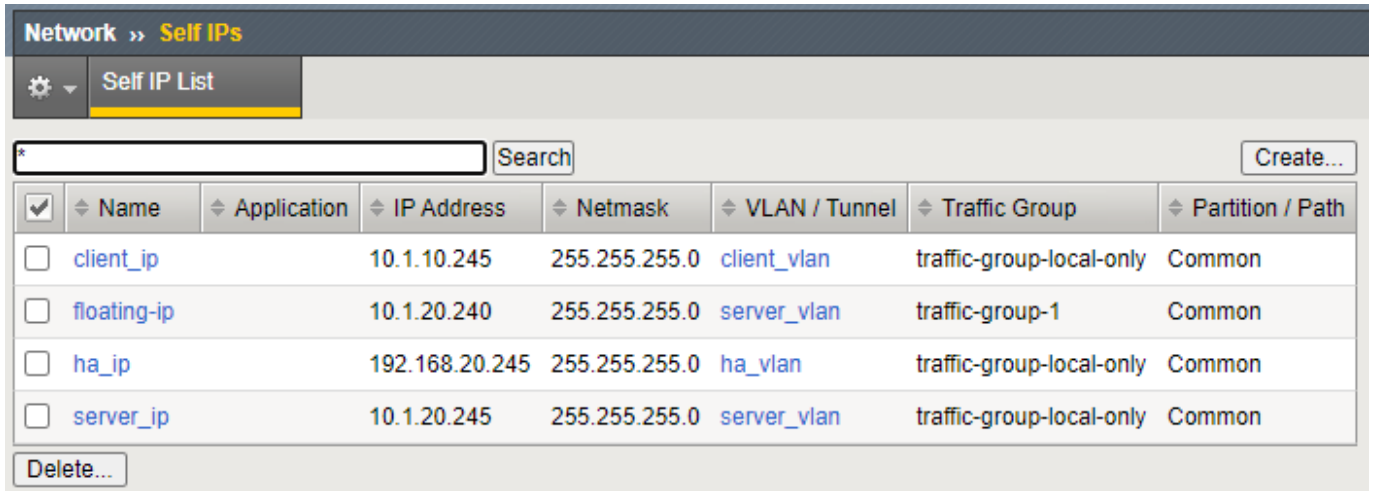
```
(tmos)# list sys httpd
allow
sys httpd {
    allow { All }
}
```

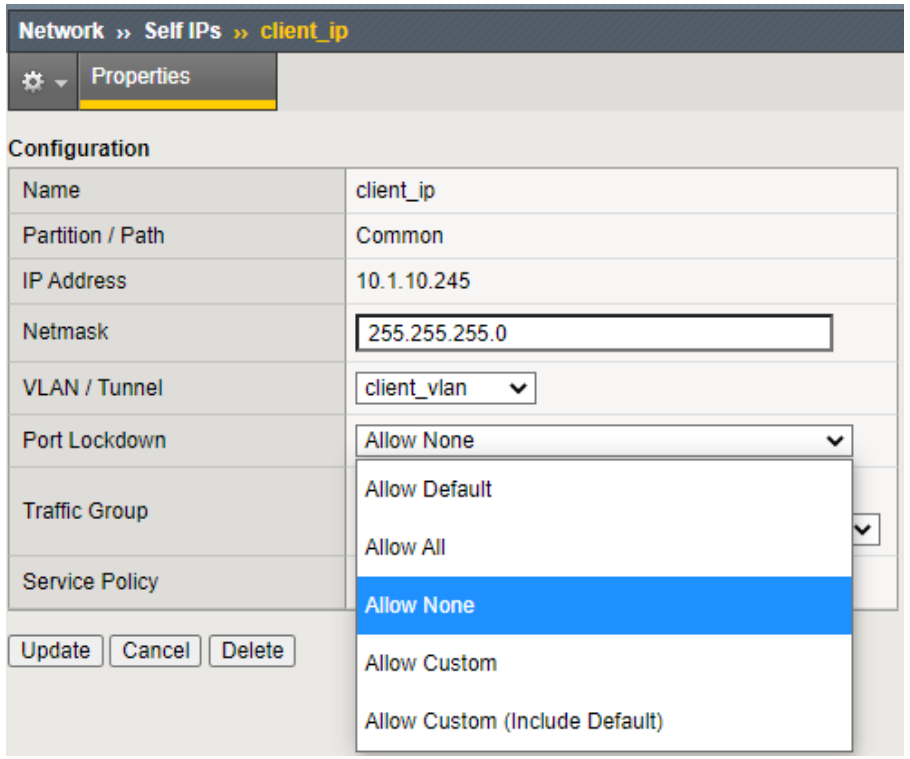
- Save the change by entering the following command:
  - save /sys config

 ©2026 F5

Interpret port lockdown settings to Self-IP

- Port Lockdown determines which ports a self IP address will respond to
  - By default Port Lockdown is none, and the self IP only responds to ICMP
- Port Lockdown settings can be modified to allow other traffic, such as, port 443 or 22 for management

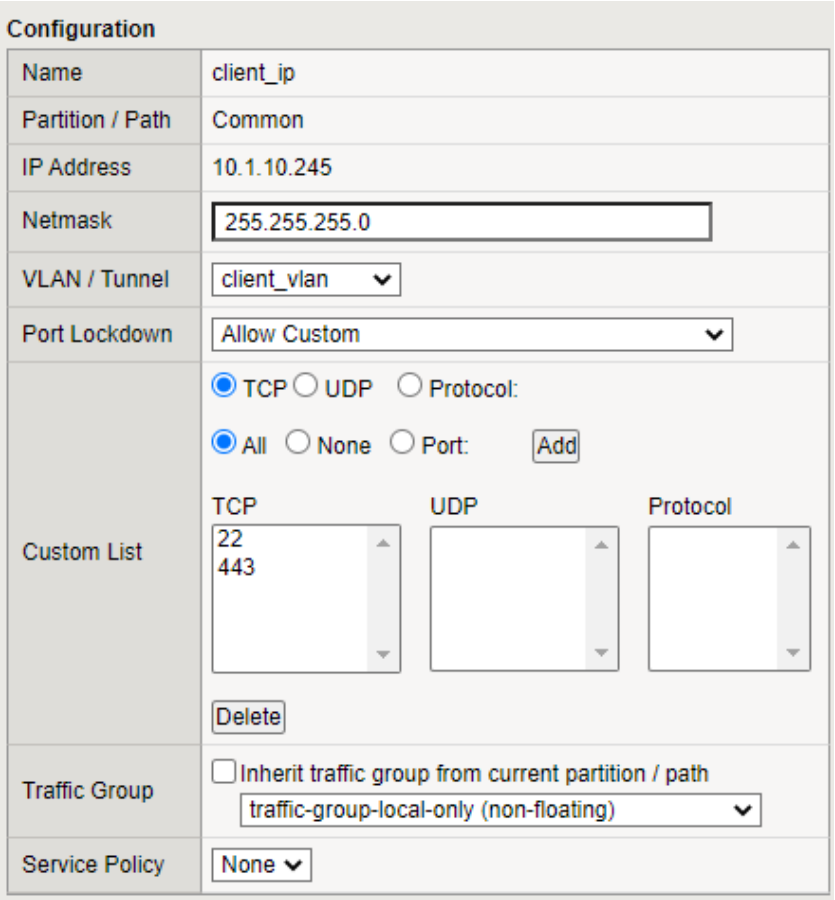




 ©2026 F5

Interpret port lockdown settings to Self-IP

- You can select “Allow Default” which opens the following:
  - ospf:any
  - tcp:domain (53)
  - tcp:f5-iquery (4353)
  - tcp:https (443)
  - tcp:snmp (161)
  - tcp:ssh (22)
  - udp:520
  - udp:cap (1026 - for network failover)
  - udp:domain (53)
  - udp:f5-iquery (4353)
  - udp:snmp (161)
- Or you can select custom ports to open




```
list net self
net self client_ip {
    address
10.1.10.245/24
    allow-service {
        tcp:ssh
        tcp:https
    }
```

 ©2026 F5

Explain management IP connectivity issue

- If using OOB Management
  - Is the IP, netmask and default gateway configured correctly
  - Is the interface up
    - At the Linux prompt: **ifconfig -a mgmt**
- If using a Self IP
  - Is the IP and netmask configured correctly
    - Are they routable
  - Are the appropriate ports open, 22 for SSH and/or 443 for the GUI interface
  - Are the any packet filters blocking traffic

 ©2026 F5

Labs 7-10

 ©2026 F5

Virtual Server Match Examples

Match the connections on the right to the virtual server configurations on the left

1. Destination IP 10.0.33.199:80 with IP source of 10.30.1.0/24
2. Destination IP 10.0.33.199:80 with network source of 0.0.0.0/0
3. Destination IP 10.0.33.199:* with network source 10.30.1.0/24
4. Destination IP 10.0.33.199:* with network source 0.0.0.0/0
5. Destination Net 10.0.33.0/24:443 with network source 0.0.0.0/0
6. Destination Net 10.0.33.0/24:* with network source 0.0.0.0/0
7. Destination Net 0.0.0.0/0:80 with network source 10.128.20.0/24
8. Destination Net 0.0.0.0/0:* with network source 0.0.0.0/0

Connect to	Source IP
10.1.33.199:80	10.30.1.120
10.0.33.199:80	10.30.2.120
10.0.33.199:443	17.64.223.120
10.0.33.196:443	10.30.1.120
74.125.21.106:80	10.128.20.100

 ©2026 F5